

APPLYING LANGUAGE MODELS TO PATIENT HEALTH RECORDS: ACRONYM
EXPANSION, LONG DOCUMENT CLASSIFICATION AND EXPLAINABLE PREDICTIONS

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ABSTRACT

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The health industry is experiencing a digital transformation, with Electronic Health Records (EHRs) becoming central repositories for an ever-growing volume of patient data. While EHR clinical notes offer rich, detailed insights into patient conditions, treatments and outcomes, extracting meaningful information from these notes remains a significant challenge due to their unstructured nature, widespread occurrence of acronyms and medical jargon, and varying writing styles. This dissertation addresses three challenges in applying Machine Learning (ML) and Natural Language Processing (NLP) to clinical text, specifically focusing on (1) understanding medical acronyms in context, (2) building models that can analyze multi-modal data (structured and unstructured patient EHR data) that includes lengthy clinical notes to study a stigmatized condition, namely opioid prescribing patterns and opioid use disorder (OUD) risk, and (3) developing explainable models that utilize clinical notes for complex diagnoses such as dementia.

This thesis has three main contributions. The first contribution introduces CLASSE GATOR, a novel system for disambiguating medical acronyms using a distantly supervised approach. Our method leverages medical research papers for contextual learning and eliminates the need for expensive manual annotation, achieving an average accuracy of 63% on Mimic-III clinical notes. We dramatically reduced the cost for clinical annotations of these acronyms. The second contribution is the development of a novel deep learning architecture that leverages structured EHR data (e.g., demographics, billing code diagnosis data) and unstructured EHR data (i.e., clinical notes) for predicting both a stigmatized health disorder, namely opioid prescription likelihood and (OUD)

diagnosis from EHR data, achieving F1 scores of 0.88 and 0.82 respectively. The model uniquely handles multiple data types and variable-length clinical notes through a combination of feed-forward layers, transformers, and a Hierarchical Attention Model built on ClinicalBERT. Finally, we present a new approach to improving interpretability in clinical prediction models using Concept Bottleneck Models (CBMs). By leveraging the Oxford Textbook of Medicine and GPT-4, we extract 254 clinically relevant features for dementia and demonstrate superior performance in dementia type prediction (0.72 accuracy) compared to baseline models (0.64 for an ngram-Logistic Regression baseline and 0.48 for a GPT-4 baseline) while maintaining interpretability.

Across all contributions, this dissertation emphasizes scalable, interpretable methods that minimize reliance on manual data curation, aiming to create computational tools that are both effective and deployable across diverse healthcare settings.

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CHAPTER 1

Introduction and Background

1.1. Introduction

The healthcare industry is experiencing a digital transformation, with an unprecedented amount of patient data being collected through Electronic Health Records (EHR). Within these records, clinical notes contain rich, detailed information about patient conditions, treatments, and outcomes. However, extracting meaningful insights from these notes presents significant challenges due to their unstructured nature, extensive use of medical jargon and acronyms, and varying writing styles across healthcare providers.

This dissertation addresses three critical challenges in applying machine learning to clinical text: (1) Understanding medical acronyms in context using a cost effective approach, (2) identifying patterns in Opioid Use Disorder (OUD) patients and opioid prescribing behavior by analyzing different modalities of patient EHR data, and (3) Developing explainable models for complex medical diagnoses. These challenges share a common thread - they all require sophisticated natural language processing techniques that can handle the unique characteristics of clinical text while providing interpretable results that healthcare providers can trust and use.

The first contribution of this work is CLASSE GATOR, a novel system for disambiguating medical acronyms in clinical notes. Unlike previous approaches that relied heavily on manually curated dictionaries or expert annotations, our distantly supervised approach uses medical research papers to learn context for clinical acronym disambiguation. This is described in greater detail in Chapter 3.

The second contribution focuses on the critical issue of opioid prescribing patterns and Opioid Use Disorder (OUD). We develop a multi-modal deep learning approach that combines structured EHR data with unstructured clinical notes to predict both opioid prescriptions and OUD risk. This work which is described in Chapter 4 is particularly significant given the ongoing opioid crisis and the need for better tools to support clinical decision-making around opioid prescribing.

Our third contribution addresses the challenge of diagnosing complex conditions like dementia, where multiple subtypes exist and early detection is crucial. In Chapter 5, we present an explainable machine learning framework that not only predicts the type of dementia but also provides clinically meaningful explanations for its predictions using concept bottleneck models.

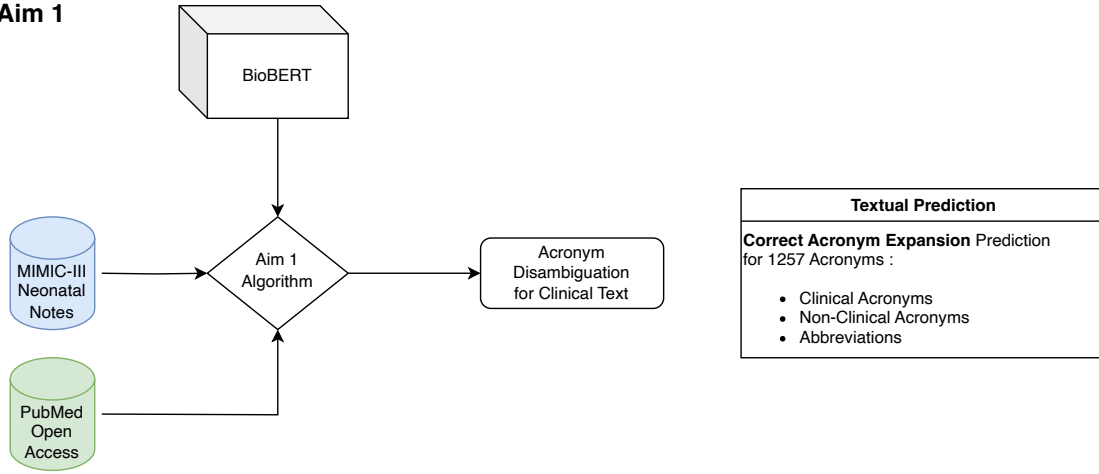
A common thread throughout this work is our focus on developing methods that require minimal manual data curation by medical experts. This approach is essential for creating scalable solutions that can be deployed across different healthcare settings and institutions. Our methods leverage recent advances in large language models while maintaining interpretability and clinical relevance. Figure 1.1 provides an overview of the goals in my dissertation. The research contributions that I have made are:

1. Developing a pipeline that can automatically discern the meaning or expansion of acronyms in clinical notes without needing expensive manually annotated training data.
2. Developing a pipeline that can analyze different EHR data modalities and multiple long clinical notes per patient by building a Hierarchical Attention Model, and using ClinicalBERT to handle ungrammatical clinical text.
3. Using Concept Bottleneck Models to help focus on relevant clinical features and reduce spurious correlations while emphasizing explainability.

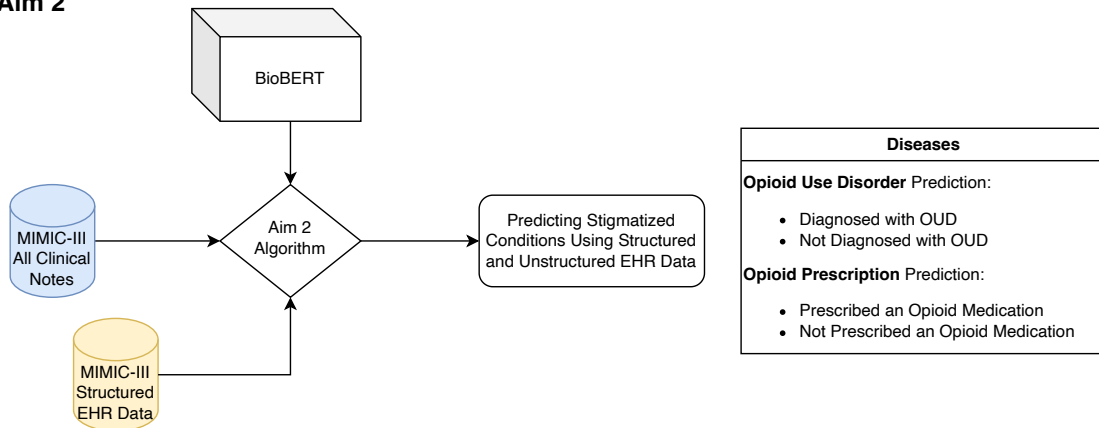
To provide context for these contributions, we begin with an overview of the current state of natural language processing in healthcare, examining both the opportunities and challenges presented by clinical text data. We then explore the evolution of language models and their increasing capability to handle complex medical text, setting the stage for our specific contributions in subsequent chapters.

Overall Theme of Aims: Enhancing Understanding of Patient Care through the Appropriate Use of Clinical Text by Reducing Annotation Costs and Explainable AI Methods

Aim 1



Aim 2



Aim 3

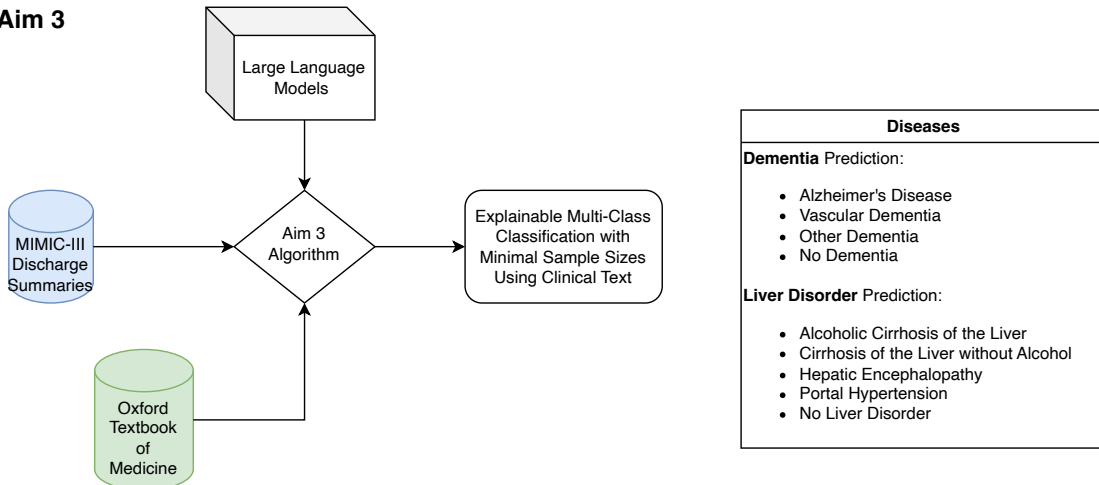


Figure 1.1: Overview of Dissertation Aims.

1.2. Background

As I aim to make my dissertation accessible to people without an NLP background, I provide a brief overview of the NLP space along with some challenges encountered in the following subsections.

1.2.1. Language Model Development History

Teaching a computer to read, write and communicate like humans has been a very challenging task, with some of the latest Natural Language Processing (NLP) models still suffering from several limitations (Zhao et al., 2023). Statistical language models (SLMs) (Rosenfeld, 2000) rose in the 1990s and were developed on the Markov Assumption where the next word was predicted based on the most recent context. These models however suffered from the “curse of dimensionality”, making it difficult to approximate high-order language models since an exponential number of transition probabilities needed to be approximated. These models were followed by Neural Language Models (NLMs), where the probability of word sequences was characterized by Neural Networks. Word2Vec (Mikolov et al., 2013b,a), a popular NLM was created to learn distributed word representations, which showed to be effective across a number of NLP tasks. These models, however, were unable to accommodate homonyms, words which are spelt alike but have different meanings based on context (e.g., “bug” can either be interpreted as an “insect” or a “computer virus”). Pre-trained language models (PLMs) like ELMo (Peters et al., 2018) and BERT (Devlin et al., 2018) were created to capture context-aware word representations and were shown to be very effective as general-purpose semantic features. These models after being finetuned on downstream tasks were shown to outperform SLMs and NLMs. Researchers further found that increasing the number of parameters of PLMs or increasing the size of the data they were trained on led to much better performance on downstream tasks. But a more surprising fact was that these larger models showed emergent abilities (Wei et al., 2022a) that helped them better solve complex tasks. The most common way of describing large pretrained language models is now Large Language Models (LLMs). GPT-3 (Brown et al., 2020) and ChatGPT are LLMs that were shown to outperform their predecessors on a wide variety of complex NLP tasks through in-context learning (Zhao et al., 2023). InstructGPT (Ouyang et al., 2022) was proposed in early 2022 and was used to improve GPT-3

by formally establishing a reinforcement learning from human feedback (RLHF) algorithm, called GPT-3.5 or chatGPT. A recent paper went so far as to argue that OpenAI's latest model GPT-4, which was trained on other modalities of data in addition to text, demonstrated some characteristics that might lead to Artificial General Intelligence (Bubeck et al., 2023).

It is clear that there is tremendous interest in large language models. Since they're relatively new, there are many emerging applications of LLMs in different fields. In this thesis, I apply LLMs to Electronic Health Record data, starting with BERT and going to GPT-4.

1.3. Introduction to Clinical Text Data Used in the NLP Domain

Clinical data is recorded when an individual interacts with the health system either through the inpatient (i.e., hospital) or outpatient (i.e., clinic) setting, and in other ancillary settings (e.g., picking up prescriptions at the pharmacy). All of these interactions with the health system generate large amounts of health data used in clinical research. This includes both structured and unstructured data elements that often contain patient demographic information, medications prescribed, diagnoses, procedures, allergies, family medical history, and so forth. Growth in the volume of clinical health data is expected to increase tremendously with the increase in adoption of mobile devices and Electronic Health Records (EHR).

Predicting risks for certain clinical outcomes is an important step that clinicians use to effectively treat patients and new methods utilize EHR data for this purpose. Researchers have worked on developing Machine Learning (ML) and Natural Language Processing (NLP) models that utilize unstructured EHR elements to identify potential risks and diagnose patients based on their medical history recorded in EHRs. Researchers working in this domain encounter several challenges, which include obtaining task specific, sufficiently sized data-sets to train models due to patient privacy issues, understanding non-standard acronyms and abbreviations, variability in note structure between physicians, departments and hospitals, and incomplete documentation by providers, and many other challenges (Xiao et al., 2018; Martínez-García and Hernández-Lemus, 2022). Some of these challenges can be seen in the clinical note excerpt shown below ([**] represents de-identified information)

Clinical Note Excerpt History of Present Illness: 75 yo male, advanced gastric ca recently started on chemotherapy presenting and hypotension and episode of unresponsiveness after diarrhea and narcotics + megace. Pt was seen on [**] by his oncologist where he was determined to be hypovolemic and received a total of 3 L IVF between We and Fri but still c/o poor po intake secondary to abdominal pain and fullness, nausea, and decreased appetite. This morning upon awakening pt was lethargic. He took MSO₄, Dilaudid and an unspecified dose of Megace. He then became flushed and pale, had copious diarrhea (non-bloody), then became unresponsive. EMS called and he was found with SBP 80. Narcan given with improvement in mental status but in [**] pt was persistently hypotensive requiring 6 L IVF and Levophed. Labs notable for ANC 500 and lactate 17. Sepsis protocol was initiated. Pt given Flagyl, Levofloxacin, and cefepime and sent to [**].

The following challenges are seen in the clinical note excerpt shown above:

1. abundant use of acronyms and abbreviations
2. non-standard use of grammar and punctuation

An additional challenge is the fact that patients have clinical notes similar to the one above whose lengths can range between 200 and 20,000 words.

1.3.1. Clinical Dataset Availability

There is limited availability of medical text with the main exception being the Medical Information Mart for Intensive Care (MIMIC) (Johnson et al., 2016b). MIMIC is a freely accessible critical care database that stores a wide range of clinical narratives, supports reproducibility of findings and enables continual research advances. MIMIC is widely used in health research with over 50 publications as of 2024.

There are not many public clinical data-sets available that contain clinical text, but there are large

private data-sets? A study carried out by Spasic et al. (2020) analyzed 110 studies that used clinical text in their experiments, and found that majority of the datasets used have cardinality in the range of hundreds and thousands, while only 10 studies used tens of thousands of documents. They concluded that the main reasons for such poor data utilization was a) the annotation bottleneck faced by supervised machine learning algorithms, coupled with the fact that b) the provenance of data is confined to a small number of contributing institutions. Additionally, privacy concerns are heightened when making clinical text publicly available as institutions are more comfortable with making structured billing code information public rather than clinical text, which can contain nuances that may reveal patient identity. These factors make the data inaccessible or unusable to researchers outside those institutions.

1.3.2. Clinical Datasets for NLP

Of the 110 studies analyzed by Spasic et al. (2020), most clinical applications of Natural Language Processing (NLP) focused on text classification using private datasets that were not publicly accessible with results supporting patient phenotyping¹ of events such as:

1. falls (Bates et al., 2016)
2. long bone fractures (Grundmeier et al.)
3. axial spondyloarthritis (Walsh et al., 2017)
4. hypertension (Teixeira et al., 2016)
5. systemic lupus erythematosus (Turner, 2017)
6. dermatitis (Gustafson et al., 2017)
7. obesity (Lingren et al., 2016)
8. celiac disease (Chen et al., 2016)

¹Phenotyping is the process of identifying and categorizing observable traits or characteristics of an organism, typically based on genetic, environmental, or clinical factors.

9. epilepsy (Goodwin and Harabagiu, 2017)
10. autism (SCHULER et al.)
11. psychiatric problems in general (J et al., 2017)
12. prognosis based on 3-month survival (I et al., 2018a)
13. likelihood of intracranial hemorrhage (I et al., 2018b)
14. development of coronary artery disease (Buchan et al., 2017; A et al., 2015; J et al., 2015)
15. cancer staging (G et al., 2016)

Most of these studies included data-sets from a few institutions and relatively small cohort sizes, thus posing problems typically associated with unrepresentative samples such as not reflecting the distribution characteristics of the target problem, bias and model over-fitting.

1.4. Approaches to Low Availability of Clinical Data

Researchers have attempted to address some of the challenges of working with limited clinical text data by providing a range of different transfer learning solutions, with pre-trained transformers (Vaswani et al., 2017) being the most popular component in these models (Laparra et al., 2021). BERT (Devlin et al., 2018) is a language representation model essentially containing stacked multi-layer bidirectional transformers that has enabled transfer learning for NLP tasks and resulted in state-of-the-art performances for a surprising number of tasks. BERT was trained on English Wikipedia and BookCorpus (Zhu et al., 2015), and therefore does not transfer well to the medical/clinical domain. Some approaches attempt to solve this problem by pretraining BERT on PubMed abstracts and PubMed Central full text articles (Lee et al., 2020), also including clinical notes from MIMIC-III v1.4 database in the pretraining stage, resulting in clinicalBERT (Alsentzer et al., 2019). Pretraining BERT on medical and clinical text achieves state of the art results across several clinical NLP tasks. Similarly, other researchers trained T5 models on Mimic-III and Mimic-IV to create Clinical-T5 (Lehman and Johnson, 2023), showing that it is an effective

way of boosting model performance on downstream clinical tasks.

1.4.1. Datasets and their Implications: General Domain

The training data distribution that these LLMs are trained on is an important indicator of which problem domains they find easy/struggle with. Most of the recent LLMs like GPT-3/4 (Brown et al., 2020), PaLM (Chowdhery et al., 2023) and LLaMA (Touvron et al., 2023) were trained on a combination of the following datasets:

- BookCorpus dataset (Zhu et al., 2015)
- CommonCrawl dataset
- Reddit data
- Wikipedia
- Open-source licensed codes from the Internet and The Pile (Gao et al., 2020)
- A text dataset consisting of over 800GB of data from multiple sources, including books, websites, code for open-source software, scientific papers and social media platforms (Zhao et al., 2023)

A model trained on this data distribution might excel at analyzing general web articles but may face significant challenges when analyzing clinical notes, which are often ungrammatical and include language that is difficult to process, including medical jargon. Even if a small amount of medical relevant data is included, because it is dwarfed by other data, it could be affected by "Catastrophic Forgetting" (Fu et al., 2022; Ye et al., 2023).

One can adapt neural network models trained on general web data to the clinical domain by either further pretraining the model on large amounts of clinical data or finetuning them on large task relevant clinical datasets. While there are huge public datasets available for social media research (Reddit (Baumgartner et al., 2020), Twitter (Yang and Leskovec, 2011)), Image Recognition (ImageNet (Russakovsky et al., 2015), MS-COCO (Lin et al.)), political research (New York

Times (Sandhaus, 2008), Australian Broadcasting Corporation (Kulkarni, 2018)), linguistic analysis (Project Gutenberg corpus (Gerlach and Font-Clos, 2020), BookCorpus (Zhu et al., 2015)), Machine translation (Opus (Tiedemann, 2012), EuroParl (Koehn, 2005)) and biomedical research (1000 Genomes project (Fairley et al., 2020), Personal Genome Project (Church, 2005)), very few datasets are publicly accessible for clinical text research due to data sensitivity and ethical issues (Tayefi et al., 2021; Management, 2017). Additionally, fragmented, duplicate and missing data can lead to inadequate insights and drive poor decisions (Demigos, September 22, 2021).

1.5. Machine Learning for Acronym Disambiguation

Early approaches to disambiguate acronyms involved relying on manually curated dictionaries and rule based approaches to map abbreviations to their expansions in clinical notes (Schwartz and Hearst, 2002; Toole, 2000; García et al., 2010; Navigli et al., 2011). A major drawback however was that these early approaches were unable to handle previously unseen acronyms. Following this, researchers used supervised learning approaches with classifiers such as Naive Bayes, Support Vector Machines and Decision Trees along with manually crafted features to predict acronym expansions (Moon et al., 2012b; Joshi et al., 2006; Finley et al., 2017). Wu et al. (2015b) and Wu et al. (2017) also showed improved prediction accuracy while using pretrained neural embeddings, investigating window size and training size on disambiguation accuracy.

As deep learning grew in popularity, neural networks were shown to outperform traditional approaches for acronym disambiguation (Kumar et al., 2020). Joopudi et al. (2018) utilized Convolutional Neural Networks to account for long-range dependencies in text for improved performance. Skreta et al. (2021) utilized biomedical ontologies as prior knowledge through pre-training to address the limited dataset size, showing strong performance gains. Kacker et al. (2022), Seneviratne et al. (2022) and Wagh and Khanna (2023) trained different models based on the BERT (Devlin et al., 2018) architecture achieving higher performance scores compared to their predecessors.

1.6. Need for Novel Methods for Prediction of a Stigmatized Disease Using Clinical Notes and Structured EHRs

Opioid Use Disorder (OUD) is a substance disorder that is characterized by a problematic pattern of opioid use that causes significant impairment, distress and/or death. OUD is a treatable, chronic disease that was estimated to effect 6.1 million people in 2022 (SAMHSA, 2023). OUD is a highly stigmatized condition with many sufferers being labeled as ‘addicts’ and often results in sub-optimal clinical care.

Alzeer et al. (2018) presented a survey of past work related to diagnosing OUD from Electronic Health Records (EHRs). They searched the OVID database for work related to designing opioid abuse predictive models, and reviewed 7 papers and 9 models with 75 distinct variables. Most of these studies relied on used ICD-9 diagnostic codes to define opioid abuse and found age and gender to be the most predictive demographic variables. Investigators have also trained different Machine Learning models on patient EHRs like Random Forest Classifiers to predict opioid dependence (Ellis et al., 2019) and Long Short-Term Memory models (Dong et al., 2021a) to predict opioid overdose risk. Findings of these studies included significantly higher correlations of White Blood Cell markers, respiratory disturbances, malnourishment and psychiatric disorders among opioid dependent patients; and pain scale scores, anti-propulsives, general anesthetics and alcohol use among patients at risk of overdose. Another group of researchers used NLP to process clinical text to identify opioid related overdoses (Hazlehurst et al., 2019). They used MediClass (Hazlehurst et al., 2005) to first extract medical concepts from the text, following which they used a manually constructed “rule tree” to classify opioid related overdoses. Dong et al. (2021b) used EHRs from Cerner’s Health Facts Database (UTHSC, 2016) to build prediction models for OUD risk based on 5 recent prior encounters before the target encounter. They compared several models including long short-term memory (LSTM), logistic regression, decision tress, random forests and BERT. They found that the LSTM model performed the best, with an F1 score of 0.802, even beating the then state-of-the-art Language model BERT.

Although machine learning has been used to predict OUD, one aspect that has not been done before

is predicting whether a patient will be prescribed an opioid or not, which I cover in Chapter 4. This would be valuable to physicians, as they could use these models in retrospect to identify factors that lead to an opioid prescription and possibly re-evaluate their approach. In addition, the prescription of an opioid or not can be also related to evidence of OUD, stigmatization and other causes indicating the need for this to be predicted by an algorithm.

1.7. Need for Explainable Methods for Multi-Class Outcome Prediction from Clinical Text

Some diseases, such as dementia, are multi-class outcomes with Alzheimer’s Disease (AD), vascular dementia and other dementias being the various ‘types’ of dementia. This presents a challenging task for Machine Learning due to the a) multi-class aspect of the outcome; b) low number of patients per class type and c) the sparsity of the clinical text with regards to that specific outcome.

1.7.1. Background on Dementia

In 2021, Kumar et al. (2021) surveyed studies using Machine Learning based analysis of Alzheimer’s Disease (AD) dementia modeling. They reviewed 64 articles between 2010 and 2020 and found that most of the research focused on predicting progression of AD dementia using publicly available datasets containing both neuroimaging and clinical data. The Alzheimer’s Disease Neuroimaging Initiative (ADNI) (Petersen et al., 2010) dataset was most widely used public dataset across the studies, with some of the other data sources including the National Alzheimer’s Coordinating Center (Beekly et al., 2007), Australian Imaging, Biomarker and Lifestyle Flagship study of Ageing (Ellis et al., 2009) and Coalition Against Major Databases (Fisher et al., 2019). Authors in (Kebets et al., 2015; Forouzaneshad et al., 2018; Lee et al., 2019) aimed to predict the development of AD dementia in individuals who were cognitively normal initially or had mild cognitive impairment. Nie et al. (2016); Kersting and Železný (2013); Zhu and Sabuncu (2018) built models for personalized forecasting of disease progression for patients showing mild symptoms of dementia using longitudinal trajectories of clinical variables. In 2022, chatGPT was released, introducing the value of Large Language Models to a wide range of researchers. Wang et al. (2023c) carried out a study to evaluate the capabilities of GPT-4 in the context of dementia diagnosis and prediction. Their experimental results on two real clinical datasets indicated that although GPT-4 has shown

remarkable performance in some professional benchmarks, it does not currently outperform traditional AI tools such as Logistic Regression and Decision Trees in dementia diagnosis and prediction tasks.

The method that we develop focuses on the utilization of automatically extracted key clinical features of dementia from medical textbooks using GPT-4, following which we use this to guide diagnostic models that analyze clinical notes (Chapter 5). This should reduce costs in the future, although current financial costs of some Large Language Models (LLMs) may be high (as of late 2024), we expect these to go down over time in accordance with Moore’s Law.

1.8. Brief Synopsis

Overall, in my dissertation, we aim to reduce the need for carefully curated data by medical experts through the development of novel Machine Learning (ML) algorithms designed for a number of different clinical use cases. These include acronym disambiguation, OUD detection and prescribing behavior detection, and multi-class prediction of dementia types of patient notes.

CHAPTER 2

Related Work

In my dissertation, I focus on creating Natural Language Processing (NLP) and Machine Learning (ML) models designed to predict outcomes in the clinical domain and attempt to address the following challenges:

1. **Acronym Disambiguation:** Acronym expansion is essential for accurately interpreting medical abbreviations, which are prevalent in clinical notes and often have multiple potential meanings. I develop a distantly supervised approach to obtain training data from medical research papers to build Logistic Regression and BioBERT based predictive models (Chapter 3). This work is based on the paper Kashyap et al. (2020) that was published in the *International Journal of Medical Informatics* in 2020.
2. **Classification of multi-modal EHRs containing lengthy clinical notes:** Patient EHRs contain a wealth of information in both structured and unstructured formats. Additionally, patients contain multiple long clinical notes associated with their hospital admissions. To analyze the different EHR components, I build a hierarchical attention model built on top of ClinicalBERT to predict Opioid Use Disorder (OUD) diagnosis and Opioid Prescription likelihood (Chapter 4). This work is based on the paper Kashyap et al. (2023) that was published in the *International Journal of Medical Informatics* in 2023.
3. **Explainability:** I develop an explainable prediction pipeline inspired by Concept Bottleneck Models. I use GPT-4 to extract relevant clinical features of patients from the *Oxford Textbook of Medicine*. Using these extracted features, I focus on the task of predicting dementia diagnosis type (if any) for patients using clinical notes from their EHRs (Chapter 5). This work is based on a paper I've submitted to the journal *BioInformatics* in November, 2024.

In this chapter, I provide a background on the utilization of EHRs in developing predictive models for healthcare. I will cover some of the challenges researchers encounter, clinical outcomes studied,

datasets used and related work.

2.1. An Introduction to EHRs

Electronic Health Records (EHRs) have become a valuable resource for developing predictive models in healthcare, offering a wealth of patient information that can be leveraged to improve clinical decision-making and patient outcomes (Swaminathan et al., 2024; He et al., 2023). These records contain both structured data, such as laboratory test results and medication information, and unstructured data, primarily in the form of free-text clinical notes (Akbulgic et al., 2019). The integration of these diverse data-types presents an opportunity to enhance the accuracy and scope of clinical outcome predictions.

In Chapters 3, 4 and 5 of my dissertation, I use data from the MIMIC-III database (Johnson et al., 2016b), which contains de-identified, comprehensive clinical data of patients admitted to the Beth Israel Deaconess Medical Center in Boston, Massachusetts. Data available in the MIMIC-III database ranges from time-stamped, nurse-verified physiological measurements made at the bedside to free-text interpretations of imaging studies provided by the radiology department. Figure 2.1 provides an overview of the MIMIC-III database and Table 2.1 shows the different classes of data available.

Class of Data	Description
Billing	Coded data recorded primarily for billing and administrative purposes. Includes Current Procedural Terminology (CPT) codes, Diagnosis-Related Group (DRG) codes, and International Classification of Diseases (ICD) codes.
Descriptive	Demographic detail, admission and discharge times, and dates of death.
Dictionary	Look-up tables for cross referencing concept identifiers (for example, International Classification of Diseases (ICD) codes) with associated labels.
Interventions	Procedures such as dialysis, imaging studies, and placement of lines.
Laboratory	Blood chemistry, hematology, urine analysis, and microbiology test results.
Medications	Administration records of intravenous medications and medication orders.
Notes	Free text notes such as provider progress notes and hospital discharge summaries.
Physiologic	Nurse-verified vital signs, approximately hourly (e.g., heart rate, blood pressure, respiratory rate).
Reports	Free text reports of electrocardiogram and imaging studies.

Table 2.1: Classes of Data Available in the MIMIC-III Critical Care Database (Courtesy of Johnson et al. (2016b))

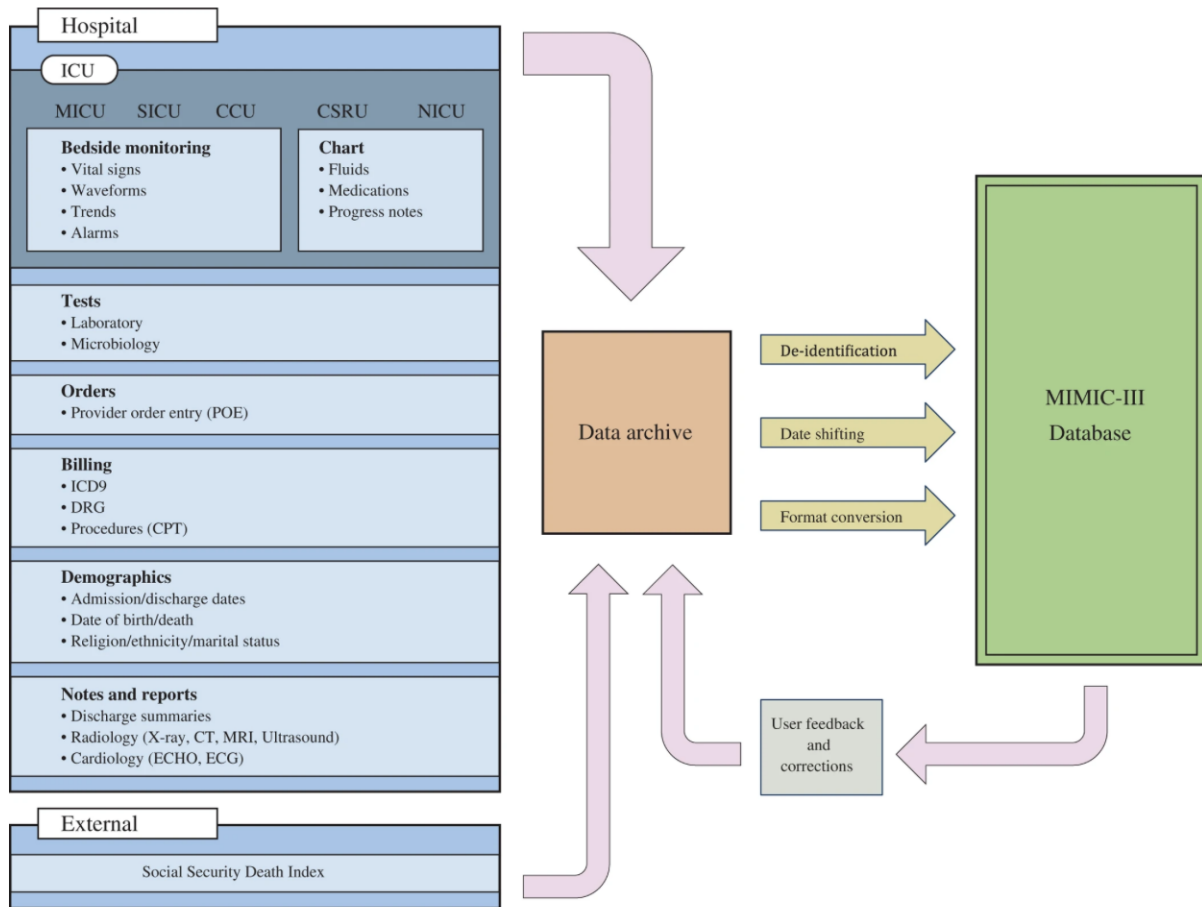


Figure 2.1: Overview of the MIMIC-III Critical Care Database (Courtesy of Johnson et al. (2016b)) and reprinted under the terms of the Creative Commons CC BY license Johnson et al. (2016a).

The development of accurate prediction models using EHR data is valuable for several reasons. Firstly, these models can assist in early intervention and detection for high-risk patients, potentially improving health outcomes and reducing healthcare costs (Glanz et al., 2023). Secondly, they can help in developing personalized treatment plans for patients by considering their comprehensive medical history and individual characteristics (Hansen et al., 2021). Finally, EHR-based models have the potential to identify previously unknown risk factors or disease patterns, contributing to advancements in medical knowledge (He et al., 2023; Akbilgic et al., 2019). Recent research has shown promising results in leveraging both structured and unstructured EHR data for clinical outcome prediction. Machine learning techniques, particularly deep learning approaches, have demonstrated superior performance compared to traditional statistical methods (Sung et al., 2022). These advanced models can capture complex relationships within the data and handle challenges associated with EHR analysis more effectively (Golmaei and Luo, 2021). Furthermore, the integration of multimodal data, including clinical notes, laboratory results, and even patient networks, has led to improvements in prediction accuracy for various outcomes, such as hospital readmissions and mortality risk (Sung et al., 2022; Sun and Zhou, 2022).

Despite the potential benefits, working with EHRs present several challenges. One of the major obstacles which can complicate model development and interpretation is the high dimensionality and sparsity of the data, i.e, a large number of features describe each patient, but many features have non-zero values for only a fraction of the patients (Ma et al., 2024). Artis et al. (2019) found that incomplete data can result in misdiagnoses and medical errors when clinicians rely on the EHR during hand-offs (Shenvi et al., 2018). Additionally, the unstructured nature of clinical notes requires sophisticated Natural Language Processing (NLP) techniques to extract meaningful information (Swaminathan et al., 2024). The temporal aspect of EHRs with variable formats across different departments and institutions adds another layer of complexity to the analysis (Ma et al., 2024). For example, the clinical events of each patient can either be represented as a sequence according to reporting chronology or an unordered set, in effect disregarding temporal information. Treating clinical events as sequences is problematic for 2 reasons: (i) Several events have the same timestamp and handling such events is unclear, and (ii) It is unclear whether reported events reflect reality,

i.e., we cannot know whether the sequence of reported events is the same as the actual sequence of events.

Additionally, using patient health data with LLMs such as OpenAI’s GPT-4 involves complex practical and legal challenges. Health data is subject to strict regulations, particularly by the Health Insurance Portability and Accountability Act (HIPAA) in the U.S., which mandates strict privacy protections for patient information. For instance, HIPAA requires that any health data used by third-party services like LLM APIs be de-identified or encrypted to prevent unauthorized access. This makes it difficult to run OpenAI APIs on sensitive health data, as these models are not typically designed to meet these specific compliance standards out of the box. Additionally, health data often includes Protected Health Information (PHI), which may require explicit patient consent for use, further complicating the process. There are also concerns about how health data is processed, stored, and shared, especially since OpenAI’s APIs are cloud-based and may involve data transfers across jurisdictions with different privacy laws.

2.2. Acronym Disambiguation (AD)

Acronyms and abbreviations are widespread in clinical documentations, serving to streamline communication among healthcare professionals. However, this prevalence often leads to ambiguity, as many acronyms have multiple possible expansions depending on the surrounding context. For instance, *RA* could stand for “rheumatoid arthritis”, “right atrium” or “room air”. This ambiguity poses significant challenges for both human readers and developing NLP models. The task of acronym disambiguation in clinical notes is crucial for various applications, including clinical decision support systems, information retrieval, and automated coding (Amosa et al., 2023).

Acronym disambiguation shares several similarities with the task of word sense disambiguation (WSD) in NLP. They both revolve around the fundamental goal of resolving ambiguity by relying on contextual clues. The goal in WSD is to predict the correct sense of a word in a specific context when a word has multiple possible meanings. For example, the word “bank” could either be a financial institution or the side of a river. Additionally, lexical resources like Wordnet (Miller, 1995) are often used to provide possible senses of words while the Acronym Finder Database (acr) among

others is used to provide possible senses of acronyms.

Early attempts at acronym disambiguation in clinical notes primarily relied on rule-based methods (Pakhomov et al., 2005). These approaches typically involved creating manually curated dictionaries of acronyms and their possible expansions, along with a set of rules for determining the correct sense based on contextual cues. While rule-based methods can be effective for well-defined acronyms, they often struggle with ambiguous cases and require significant manual effort to maintain and update (Moon et al., 2015; Wu et al., 2015a). As the field of NLP advanced, researchers began to explore machine learning techniques for acronym disambiguation. These methods leverage statistical patterns in large corpora of clinical text to learn the most likely expansion for a given acronym in context (Rubio-López et al., 2017; Moon et al., 2015). Supervised learning approaches, such as support vector machines (SVM) and random forests, have shown promising results in disambiguating clinical acronyms (Link et al., 2020; Wu et al., 2017).

The advent of deep learning has led to significant improvements in acronym disambiguation performance. Neural network architectures, particularly those based on transformers, have demonstrated the ability to capture complex contextual relationships and achieve state-of-the-art results on various clinical NLP tasks, including acronym disambiguation (Rohanian et al., 2023). Models like BERT and its clinical variants (BioBERT, ClinicalBERT) have been successfully applied to this problem, showing superior performance compared to traditional machine learning methods (Sivarajkumar et al., 2024).

BERT (Devlin et al., 2018) was introduced in 2019 and showed state-of-the-art performance on a number of tasks, primarily due to the fact that it was able to create contextualized word embeddings and was effective at capturing complex semantic dependencies. Kacker et al. (2022) created ABB-BERT, a lightweight BERT model for abbreviation and contraction disambiguation. The abbreviations in the input text were replaced by a special token [ABB] and the model was evaluated on the SciAD dataset, which included text containing abbreviations from scientific literature. ABB-BERT outperformed other models on this dataset obtaining an average rank of 1.55. However, the dataset containing scientific papers was not representative of performance on clinical text data due

to the difference in language, style and format. Seneviratne et al. (2022) used a pretrained SciBERT model (Beltagy et al., 2019) in a modified triplet network architecture for acronym disambiguation. They achieve F-scores of 87.31%, 70.67% and 75.75% on the SDU (Veyseh et al., 2020), CASI (Moon et al., 2014) and MeDAL (Wen et al., 2020) datasets respectively showcasing comparable performance to the base network but with only 12% of the parameters used. Wagh and Khanna (2023) compared different versions of the BERT model, which included SciBERT, ClinicalBERT and BioBERT for acronym disambiguation. ClinicalBERT achieved the highest F-score of 91.49% on the CASI dataset as it was additionally pretrained on large amounts of clinical text and was able to capture domain-specific vocabulary.

Given the scarcity of labeled data in the clinical domain, researchers have also explored unsupervised and semi-supervised approaches to acronym disambiguation. These methods aim to leverage large amounts of unlabeled clinical text to identify patterns and relationships that can aid in disambiguation (Rubio-López et al., 2017; Wu et al., 2017). They use clustering-based methods and profile-based word sense disambiguation, showing promise in scenarios where labeled data is limited.

The emergence of large language models (LLMs) has increased the performance bar for acronym disambiguation in clinical notes. Zero-shot and few-shot learning techniques, which allow models to perform tasks with little or no task-specific training data, have shown promising results in various clinical NLP tasks, including acronym disambiguation. These approaches leverage the vast knowledge encoded in pre-trained LLMs to infer the correct expansion of acronyms based on context, even for previously unseen acronyms or domains (Sivarajkumar et al., 2024; Sivarajkumar and Wang, 2022).

Researchers have also explored the development of real-time acronym disambiguation systems that can assist healthcare providers during the clinical documentation process. These systems aim to integrate with electronic health record (EHR) systems to provide immediate feedback and clarification of ambiguous acronyms as they are being entered, potentially reducing errors and improving the overall quality of clinical notes (Wu et al., 2015a).

Several datasets have been developed for creating and evaluating clinical acronym disambiguation systems:

- CASI Dataset: The Clinical Abbreviation Sense Inventory (CASI) contains 440 ambiguous abbreviations with 949 possible expansions (Moon et al., 2012a).
- UMLS Dataset: The Unified Medical Language System (UMLS) has been leveraged to create large-scale datasets for acronym disambiguation (Kim et al., 2011).
- MIMIC-III Dataset: The Medical Information Mart for Intensive Care (MIMIC-III) database, which contains de-identified clinical notes from intensive care unit patients, has been used to create realistic evaluation datasets for clinical acronym disambiguation (Skreta et al., 2021; Jaber and Martínez, 2022).

Many challenges remain in evaluating clinical acronym disambiguation systems. For instance, there are limited publicly available gold-standard datasets as manually annotating examples is time consuming and requires domain expertise. Additionally, assessing performance on rare or newly coined acronyms is often difficult due to the lack of sufficient examples in existing datasets. Finally, the use of real clinical data for evaluation raises privacy concerns, limiting the ability to create and share large-scale, realistic evaluation datasets.

In Chapter 3, I introduce a distantly supervised approach to predict acronym senses in clinical notes. I first create a dataset of acronyms with their possible expansions and surrounding contexts from papers published in PubMed Central using regular expressions and a set of hand written rules. We identify 1,257 acronyms and 8,287 definitions from 31,764 PubMed Central papers with the average number of expansions per acronym being 6.6 (min=2, max=50). Next, I train and evaluate Logistic Regression and BioBERT on this dataset, in addition to clinical notes from MIMIC-III. I also compare these models to other approaches in the literature, both in terms of performance metrics and cost effectiveness.

2.3. Long Document Classification

Electronic Health Records contain vast amounts of information about a patient's health history, diagnoses, treatments, lab results, medical imaging reports and more. They are often long, unstructured and complex which makes classifying and extracting useful insights from them a challenging task. Since this data must be interpreted in context to accurately diagnose patients, run risk assessments and provide treatment recommendations, it requires processing patient EHRs in its entirety to ensure that the relationships between different sections are captured. As a result, developing NLP architectures that can handle long sequences of tokens and process documents several pages in length is a crucial step for patient EHR classification.

Long document classification is a challenging task in NLP that involves categorizing extensive textual content into predefined classes. In the medical domain, this is especially hard due to the presence of several high semantic clinical themes interacting over different parts of long clinical text that need to be effectively captured and understood. This task has gained significant attention in recent years due to the increasing availability of large-scale document collections and the need for efficient information retrieval and organization (Chalkidis et al., 2022), along with the development of Large Language Models like GPT-4 with increased context windows.

In the past, traditional machine learning approaches such as Support Vector Machines (SVM), Naive Bayes, and Decision Trees were commonly used for document classification tasks (Wagh et al., 2021). These methods often relied on bag-of-words or Term Frequency-Inverse Document Frequency (TF-IDF) representations of documents, which had limitations in capturing the semantic relationships between words and the overall context of long documents (Ghumade and Deshmukh, 2019).

With the advent of deep learning, Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have shown promising results in capturing both local and global features of documents (Rao et al., 2018; Zhou et al., 2016). The introduction of transformer-based models, such as BERT and its variants, revolutionized the field of NLP, including document classification (Chalkidis et al., 2022). However, these models often

Model	Maximum Context Window Size	Release Date
BERT	512	October, 2018
Llama	2,048	February, 2023
ChatGPT	4,000	November, 2022
Llama 2	4,000	July, 2023
GPT-4 32k	32,000	March, 2023
Gemini	32,000	December, 2023
Claude-2 100k	100,000	July, 2023
GPT-4 128k	128,000	November, 2023

Table 2.2: Context Windows for Different NLP Models along with their Release Date.

face challenges when dealing with long documents due to their limited input sequence length. To address this issue, researchers proposed various approaches, including hierarchical attention networks and sparse attention mechanisms. Hierarchical models typically process documents at multiple levels, such as word, sentence, and document levels, allowing them to capture the hierarchical structure of long texts more effectively (Liu et al., 2022). To handle longer sequences, researchers developed sparse attention mechanisms that allow transformer-based models to process longer documents more efficiently. Models like Longformer have demonstrated the ability to handle documents with thousands of tokens (Beltagy et al., 2020). Table 2.2 provides a list of Language Models along with the maximum context windows that they can handle. While BERT could only process a maximum input sequence length of 512 tokens, some of the more recent models like Claude-2 and GPT-4 can handle upwards of 100,000 token sequences and are therefore better equipped to capture long range dependencies in lengthy texts.

In the clinical domain, long document classification presents unique challenges due to specialized vocabulary, complex sentence structures, and the presence of domain-specific knowledge (Crema et al., 2022). Researchers have adapted general NLP techniques to handle these challenges and developed domain-specific models for clinical document classification (Yoon et al., 2024). Pre-trained language models specifically tailored for the clinical domain, such as ClinicalBERT (Alsentzer et al., 2019) and Bio-BERT (Lee et al., 2020), have been developed to capture the nuances of medical terminology and context, showing improvements on various NLP tasks, including long document classification.

One of the major challenges of analyzing clinical documents is that they often contain a mix of structured and unstructured data, as well as templated and free-text portions. This heterogeneity poses challenges for traditional classification approaches and requires specialized techniques to effectively extract and utilize relevant information (Yoon et al., 2024).

In the clinical domain, availability of public evaluation datasets is limited due to privacy concerns and sensitive nature of medical information. However, there are some datasets that have been developed and made available for research purposes. Several include long documents suitable for testing model capabilities for long document classification:

- MIMIC-III Clinical Database: The Medical Information Mart for Intensive Care (MIMIC-III) is a large, freely available database comprising de-identified health-related data associated with over 40,000 patients who stayed in critical care units. This dataset includes various types of clinical notes, which can be used for long document classification tasks in the medical domain. The average number of words per clinical note in MIMIC-III is 406 (min = 21, max = 9,992).
- n2c2 (formerly i2b2) Datasets: The National NLP Clinical Challenges (n2c2), formerly known as Informatics for Integrating Biology and the Bedside (i2b2), provides a series of shared tasks and associated datasets for clinical NLP research. These datasets often include long clinical notes and are designed to evaluate various NLP tasks, including document classification (i2c).
- LCD (Long Clinical Document) benchmark: It is a recently proposed dataset specifically designed for long clinical document classification. This benchmark focuses on predicting 30-day out-of-hospital mortality using discharge notes from the MIMIC-IV dataset and statewide death data (Yoon et al., 2024).

In Chapter 4, I introduce an approach to predict the likelihood of an opioid prescription and an OUD diagnosis for a patient based on information present in their EHRs. I first create a dataset from MIMIC-III using a set of diagnoses codes and medications, and then train a hierarchical attention model built on top of BioBERT. Finally, the model is evaluated and compared to past approaches

and the most predictive features are analyzed.

2.4. Explainable Prediction Models

Deep Learning networks have been used in predictive tasks across several domains, but their adoption in healthcare has been slow due to their traditional lack of explainability and transparency in terms of statistical decision making (Schwartz et al., 2024). Without transparency, clinicians and other healthcare practitioners are unsure as to whether they can trust, and ultimately implement such algorithms. McCoy et al. (2019) showed that although Statistical Neural Network (SNN) models like BERT (Devlin et al., 2018) achieved high scores on different tasks, they did so by learning superficial rules that work for most training instances, rather than learning the fundamental principles of the task. Some of the more recent SNN models like ChatGPT, GPT-4 and other Large Language Models (LLMs) have been additionally shown to hallucinate when insufficient information is provided (Chen et al., 2023; Liu et al., 2023; Shen et al., 2023). They also tend to reproduce biases and factual inconsistencies in their training data (Liu et al., 2023; Shen et al., 2023). These issues are especially pronounced for the health care field where accuracy of information is crucial. In Chapter 5, I introduce an approach to diagnose patients with dementia and liver disorders based on information present in their clinical notes. The primary focus while developing this approach is explainability while ensuring that the model focuses on clinically relevant features and does not rely on unrelated superficial patterns in the data.

The importance of explainability in NLP stems from the need to build trust, ensure fairness, and comply with regulations (Luo et al., 2024). Explainable AI (XAI) techniques in NLP aim to provide human-comprehensible justifications for model predictions, enabling users to understand and potentially validate the reasoning behind the model’s outputs. This transparency is particularly crucial in high-stakes decision-making scenarios like healthcare where the consequences of errors can be severe.

Explainable NLP techniques can be broadly categorized into three main approaches: (1) interpreting the model’s predictions through related input features, (2) interpreting through natural language explanations, and (3) probing the hidden states of models and word representations (Luo et al., 2024).

Feature-based interpretation methods focus on identifying and highlighting the input features that contribute most significantly to a model’s predictions. These techniques often employ visualization methods, such as saliency heatmaps, to illustrate the importance of different words or phrases in the input text. Additionally, numerical explainability techniques like LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al., 2016) and Shapley values (Lundberg and Lee, 2017) have been adapted for NLP tasks to quantify feature importance. Natural language explanations, also known as rationalization, involves generating human-readable explanations that justify a model’s output (Gurrapu et al., 2023). Recent advancements in natural language generation have made rationalization an attractive method, as it provides explanations that are comprehensible to non-technical users. Probing techniques aim to understand the internal workings of NLP models by analyzing their hidden states and word representations. These methods involve training auxiliary models or conducting targeted experiments to reveal what information is captured in different layers or components of the model (Hewitt and Manning, 2019; Ravichander et al., 2020). Probing has been particularly useful in understanding the behavior of large language models and identifying the linguistic properties they encode.

There have also been several approaches to understanding Transformers (Vaswani, 2017), which contain encoder-decoder structures using stacked multi-head self-attention and fully connected layers, as they form the building blocks of popular language models such as BERT (Devlin et al., 2018) and GPT-2 (Radford et al., 2019). Lee et al. (2017) provides an interactive tool to visualize attention and provides an interface to dynamically adjust the search tree and attention weights. Similarly, Liu et al. (2018) proposed a flexible visualization library to visually analyze models for natural language inference and machine comprehension that relies on a perturbation-driven exploration strategy. ExBERT (Hoover et al., 2019) is another interactive visualization tool that uses linguistic annotations, masking, and nearest-neighbor search to provide insights into contextual representations learned in transformer models.

In my dissertation, I focus on building an explainable clinical decision pipeline using Concept Bottleneck Models (CBMs) (Koh et al., 2020b). CBMs are a type of interpretable neural network that

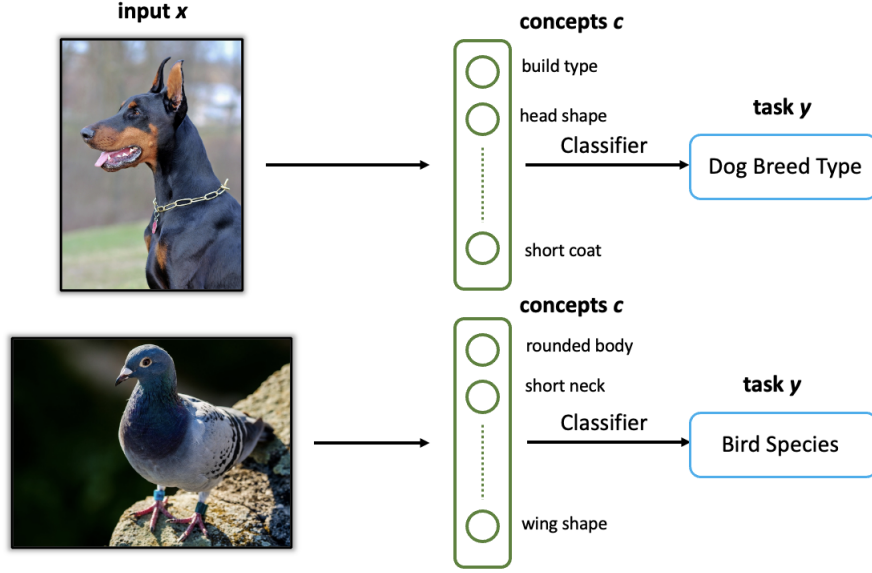


Figure 2.2: Concept Bottleneck Model for Dog Breed Type and Bird Species Identification (Inspired by Figure 1 in Koh et al. (2020a)).

operates in two stages. First, they predict labels for human-interpretable concepts relevant to the task at hand. Then, they use these concept predictions to make final classifications or predictions. This two-step process allows for a more transparent decision-making approach, as the intermediate concepts can be understood and analyzed by human experts. Figure 2.2 provides an overview of using Concept Bottleneck Models for dog breed and bird species identification (Koh et al., 2020a).

The model’s architecture enables it to predict concepts based on distinct regions of an input, supporting human interpretation when generating explanations of the model’s outputs. In healthcare, CBMs have been used for analyzing medical imaging data, such as chest X-rays, to predict diseases while providing insights into the underlying diagnostic concepts (Furby et al., 2024; Brocki and Chung, 2023). One of the primary advantages of CBMs is their inherent interpretability (Furby et al., 2023). By predicting human-defined concepts as an intermediate step, CBMs provide a level of transparency that is often lacking in traditional deep learning models. This interpretability can be crucial in fields like healthcare, where understanding the reasoning behind a model’s decisions is essential for building trust and ensuring safety. Recent developments have extended CBMs to inter-

active prediction settings, where the model can query a human collaborator for labels of certain concepts (Chauhan et al., 2023). This interactive approach allows for a collaborative decision-making process between the model and human experts, potentially improving accuracy and adaptability in complex scenarios. Yang et al. (2023) provided an approach named **L**anguage Guided **B**ottlenecks (LaBo), which leveraged a language model GPT-3 to define a large space of possible bottlenecks. Given a problem domain, LaBo used GPT-3 to first produce factual sentences about categories to form candidate concepts. LaBo then efficiently searched possible bottlenecks through a novel sub-modular utility that promoted the selection of discriminative and diverse information. Next, GPT-3’s sentential concepts were aligned to images using a pre trained alignment model CLIP, to form a bottleneck layer. Finally, they trained a linear function to predict image labels based on concept image scores. Yang et al. (2023) evaluated LaBo on 11 diverse datasets showing that it excels at few-shot classification and are 11.7% more accurate than black-box linear probes at 1 shot and comparable with more data.

Building explainable models comes with several challenges. A significant challenge is the trade-off between model performance and explainability, as more interpretable models often sacrifice some predictive power (Danilevsky et al., 2020). Additionally, the evaluation of explainability techniques remains an open problem, as there is no consensus on how to measure the quality or usefulness of explanations. Furthermore, ensuring that explanations are faithful to the model’s actual decision-making process is crucial but challenging, especially for complex models.

In Chapter 5, I provide a novel method of extracting patient clinical features related to dementia and liver disorders from the Oxford Textbook of Medicine (Warrell, 2003). Next, I develop a Concept Bottleneck Model pipeline that converts patient notes into concept vector representations, which we then use to train a linear interpretable model on top to analyze important features. We contrast the performance of our approach to a bag of words Logistic Regression baseline and a GPT-4 baseline, as well as other approaches in the literature.

2.5. Evaluation Metrics Used

In this section, I will go over metrics that I've used in Chapters 3, 4 and 5 to evaluate the models that were built. Since my work mainly deals with building classification models for different tasks, the evaluation metrics discussed are general classification metrics.

A classification task can either be binary, where the number of labels is two, or multi-class where the number of labels is more than two.

2.5.1. Confusion Matrix

The confusion matrix takes classification results for n labels and groups them into n^2 categories as shown in Figure 2.3. For Binary Classification, the 4 categories are:

- **True Positive (TP)**: Both actual and predicted values are 1.
- **True Negative (TN)**: Both the actual and predicted values are 0.
- **False Positive (FP)**: Actual value is 0 and the predicted value is 1.
- **False Negative (FN)**: Actual value is 1 and the predicted value is 0.

Similarly, for 3-way multi-class classification, there are nine groups in which three of them represent correct predictions where the actual value and predicted values coincide ($True_{00}$, $True_{11}$ and $True_{22}$). The remaining six groups represent incorrect predictions ($False_{ij}$ for actual label i and predicted label j).

Confusion matrices provide a clear, detailed view of model's performance by showing the correct and incorrect predictions across different classes, helping to assess accuracy, precision, recall and other evaluation metrics.

2.5.2. Accuracy

Accuracy measures the proportion of correctly predicted instances out of the total number of instances in the dataset. For the binary classification task, it is calculated using the following expres-

		Prediction	
		0	1
Actual	0	True Negative	False Positive
	1	False Negative	True Positive

(a)

		Prediction		
		0	1	2
Actual	0	True ₀₀	False ₀₁	False ₀₂
	1	False ₁₀	True ₁₁	False ₁₂
	2	False ₂₀	False ₂₁	True ₂₂

(b)

Figure 2.3: Confusion Matrix Format for (a) Binary Classification, and (b) 3-way Multi-class Classification.

sion:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Similarly, for 3-way multi-class classification, accuracy is calculated as follows:

$$Accuracy = \frac{True_{00}+True_{11}+True_{22}}{True_{00}+True_{11}+True_{22}+False_{01}+False_{02}+False_{10}+False_{12}+False_{20}+False_{21}}$$

While accuracy is easy to understand and provides a simple overall measure of a model's performance, it can be misleading for imbalanced datasets as the poor performance on the minority class maybe eclipsed if the majority class is predicted correctly most of the time.

2.5.3. Precision

Precision is the proportion of true positive predictions out of all positive predictions made by the model, indicating how many of the predicted positives are actually correct. For binary classification, it is calculated as:

$$Precision = \frac{TP}{TP+FP}$$

For 3-way multi-class classification, precision for a class is calculated as:

$$\begin{aligned}
Precision_0 &= \frac{True_{00}}{True_{00}+False_{10}+False_{20}} \\
Precision_1 &= \frac{True_{11}}{True_{11}+False_{01}+False_{21}} \\
Precision_2 &= \frac{True_{22}}{True_{22}+False_{02}+False_{12}}
\end{aligned}$$

Precision is particularly useful when the cost of false positives is high, as it measures the accuracy of positive predictions. However, it does not account for false negatives, which means it may ignore the model's performance on the other class.

2.5.4. Recall or Sensitivity

Recall is the proportion of true positive predictions out of all actual positive instances in the dataset, indicating how well the model identifies all relevant positive cases. For binary classification, it is calculated as:

$$Recall = \frac{TP}{TP+FN}$$

For 3-way multi-class classification, recall for a class is calculated as:

$$\begin{aligned}
Recall_0 &= \frac{True_{00}}{True_{00}+False_{01}+False_{02}} \\
Recall_1 &= \frac{True_{11}}{True_{11}+False_{10}+False_{12}} \\
Recall_2 &= \frac{True_{22}}{True_{22}+False_{20}+False_{21}}
\end{aligned}$$

Recall is valuable when it's important to minimize false negatives, such as in medical diagnoses where missing a positive case can have serious consequences. However, it can be misleading if used alone, as it may result in many false positives, reducing the model's overall reliability.

2.5.5. F1 Score

The F1 score is the harmonic mean of precision and recall, providing a balanced measure of a model's performance when both false positives and false negatives are important. For binary classification, it is calculated as:

$$F1 = \frac{2*Precision*Recall}{Precision+Recall}$$

For 3-way multi-class classification, F1 scores for a class is calculated as:

$$F1_0 = \frac{2*Precision_0*Recall_0}{Precision_0+Recall_0}$$

$$F1_1 = \frac{2*Precision_1*Recall_1}{Precision_1+Recall_1}$$

$$F1_2 = \frac{2*Precision_2*Recall_2}{Precision_2+Recall_2}$$

2.5.6. Specificity

Specificity is the proportion of true negative predictions out of all actual negative instances, indicating how well the model identifies the negative class without falsely classifying it as positive. For a binary classification task, Specificity is calculated as follows:

$$Specificity = \frac{TN}{TN+FP}$$

For 3-way multi-class classification, specificity for a class is calculated as:

$$Specificity_0 = \frac{True_{11}+True_{22}}{True_{11}+True_{22}+False_{10}+False_{20}}$$

$$Specificity_1 = \frac{True_{00}+True_{22}}{True_{00}+True_{22}+False_{01}+False_{21}}$$

$$Specificity_2 = \frac{True_{00}+True_{11}}{True_{00}+True_{11}+False_{02}+False_{12}}$$

Specificity is useful in scenarios where it's crucial to correctly identify negative cases, such as in fraud detection, where false positives can be costly.

2.5.7. AUC-ROC

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) is a metric used to evaluate the performance of binary classification models by plotting the true positive rate (recall) against the false positive rate at various threshold settings. The AUC value represents the probability that the model ranks a randomly chosen positive instance higher than a randomly chosen negative one, with values ranging from 0 to 1. A value of 1 indicates perfect performance and 0.5 indicating no discrimination ability (random guessing). The True Positive Rate (TPR) and False Positive Rate (FPR) is calculated as:

$$TPR = \frac{TP}{TP+FN}$$

$$FPR = \frac{FP}{FP+TN}$$

The curve is generated by varying the classification threshold from 0 to 1 and plotting the corresponding TPR and FPR at each threshold. However, AUC-ROC can be misleading in imbalanced datasets, as it may give an overly optimistic view of model performance by focusing more on the ability to distinguish the majority class rather than the minority class.

2.6. Large Language Models and Their Applications in Healthcare

Large Language Models (LLMs) have emerged as a transformative force in the field of healthcare and clinical research, offering unprecedented capabilities in processing and generating human-like text. These models are trained on vast amounts of textual data, have billions of parameters and show remarkable potential in addressing complex healthcare challenges, from assisting in clinical decision-making to advancing medical research (Tessler et al., 2024). In the following sections, I will first discuss the Transformer architecture (Vaswani et al., 2017), which is a core unit of LLMs. Next, I will discuss some prominent LLMs that I’ve used in my dissertation, and their applications in healthcare.

2.6.1. Transformer Architecture

The Transformer architecture was first introduced by Vaswani et al. (2017) and served as an improved approach to sequence modeling problems such as language modeling and machine translation. Previously, the state-of-the-art models were based on Recurrent neural networks, long short-term memory (LSTM) (Hochreiter, 1997) and gated recurrent (Chung et al., 2014) neural networks.

The Transformer model contains an encoder-decoder structure, where the encoder maps an input sequence of symbol representations $(x_1, x_2, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, z_2, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, y_2, \dots, y_m)$ of symbols. The encoder and decoders include stacked self-attention and point-wise full connected layers. Vaswani et al. (2017) explain self-attention as the representation of a sequence (or sentence) computed by relating different words in the same sequence. The main variables used to compute self-attention in Transformers are:

- A matrix Q that contains a set of query vectors of dimension d_k .

- A matrix K that contains a set of key vectors of dimension d_k .
- A matrix V that contains a set of value vectors of dimension d_v .

The output is computed as a weighted sum of the *values*, where the weight assigned to each *value* is computed by a compatibility function of the *query* with the corresponding *key*. The following equation is used to compute the matrix of outputs:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The dot products in the above equation are scaled by a factor of $\frac{1}{\sqrt{d_k}}$ to prevent the softmax function from operating in regions with extremely small gradients. Vaswani et al. (2017) additionally incorporates multi-head attention into the Transformer model, allowing it to jointly attend to information from different representation subspaces at different positions, thereby leading to a richer and more nuanced understanding of complex relationships in the input sequence.

In addition to the attention sub-layers, each of the layers in the encoder and decoder contains a fully connected feed-forward network which is applied to each position separately and identically. This consists of two linear transformations with a ReLU activation in between as shown below:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

To ensure that information about the order of the input sequence can be made use of by the model, positional encodings are additionally added to the bottoms of the encoder and decoder stacks. The positional encodings have the same dimension d_{model} as the embeddings so that the two can be summed. The positional encodings used in the Transformer model contain sine and cosine functions of different frequencies:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

where pos is the position and i is the dimension. Vaswani et al. (2017) mention that they chose sinusoidal versions of positional encodings as it may allow the model to extrapolate to sequence

lengths longer than the ones encountered during training.

The architecture of the Transformer model supports parallelization, speeding up training while also capturing long-range dependencies more effectively than its predecessors due to self-attention. Specifically, every token in a transformer attends to the entire sequence and captures global context. These advantages have resulted in Transformer models being an integral part of language models such as BERT and other LLMs.

2.6.2. T5 (Text-to-Text Transfer Transformer)

T5 is a language model developed by Google and released in 2019 that introduced a unified approach to various NLP tasks (Raffel et al., 2020). The model architecture is based on the original Transformer model (Vaswani et al., 2017), utilizing both encoder and decoder components which allows for handling a wide range of applications, from translation to summarization, by simply providing appropriate input prompts. Additionally, T5 uses a denoising autoencoder objective for pre-training, where it learns to reconstruct corrupted text sequences. This approach is more effective than the traditional left-to-right or masked language modeling objectives (Devlin et al., 2018) used by predecessors, leading to better generalization and performance on downstream tasks (Raffel et al., 2020). T5 was trained on a diverse dataset called C4 (Colossal Clean Crawled Corpus), which consists of cleaned web pages and comes in various sizes, with the largest version, T5-11B, containing 11 billion parameters. One of the main advantages of T5 is its versatility and ability to perform well on multiple tasks without task-specific fine-tuning (Raffel et al., 2020). However, its relatively smaller size and maximum input sequence of 512 tokens compared to more recent models may limit its performance on more complex tasks such as long document summarization and complex dialogue systems.

2.6.3. GPT (Generative Pre-trained Transformer) Models

GPT-3 was developed by OpenAI and marked a significant milestone in the field of LLMs when it was released in June, 2020 (Brown et al., 2020). With 175 billion parameters, GPT-3 was the largest language model at the time of its release, demonstrating remarkable capabilities in natural language understanding and generation. The GPT-3 architecture is based on the Transformer

decoder (Vaswani et al., 2017), utilizing a deep neural network with 96 layers. It was trained on a diverse dataset comprising web pages, books, and articles, totaling about 45 terabytes of text data. The capability of GPT-3 to perform well on new tasks with minimal task-specific examples using few shot learning was a major advantage. Additionally, it has a maximum input length of 2,048 tokens allowing it to process relatively long pieces of text compared to its predecessors. However, its large size makes it computationally expensive to run, and it sometimes generates biased or factually incorrect information (Rawte et al., 2023).

GPT-3.5 was released by OpenAI in November, 2022 and was an improved version of GPT-3. While the architecture of GPT-3.5 was similar to its predecessor (175 billion parameters), it incorporated training improvements such as Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017), which was carried out in several stages:

1. The model is initially trained on a broad dataset using unsupervised learning techniques. This stage involves learning language patterns, syntax, and a wide range of knowledge from the training data.
2. After the initial training, the model is fine-tuned using reinforcement learning. In this phase, human annotators evaluate the model’s outputs and rank responses to specific prompts based on quality, relevance, accuracy, and appropriateness.
3. The feedback from human annotators is used to create a reward model which predicts the quality of the outputs. It essentially scores the responses from the model, indicating how well they align with the desired behavior as indicated by human feedback.
4. The language model is then fine-tuned using reinforcement learning algorithms, such as Proximal Policy Optimization (Schulman et al., 2017). The model’s policy, which determines how it generates responses, is adjusted to maximize the reward as defined by the reward model. This step involves iterative training where the model learns to produce outputs that are more likely to receive higher scores from the reward model.

5. The refined model is evaluated to ensure improvements in alignment and performance. This involves further rounds of human feedback and fine-tuning to address any remaining issues and ensure the model's behavior meets the desired standards.

A major advantages of GPT-3.5 compared to GPT-3 is its enhanced ability to understand context and generate more coherent and relevant responses due to a higher maximum input length of 4096 tokens (Ye et al., 2023). But like GPT-3, it still faces challenges with occasional factual inaccuracies and potential biases. McIntosh et al. (2024) found that GPT-3.5 was more prone to hallucinations when dealing with controversial topics, subjects lacking clear scientific consensus, and brain teasers.

OpenAI released GPT-4 in March, 2023, representing the latest advancement in the GPT series (Achiam et al., 2023). Unlike the previous GPT models, GPT-4 is multimodal and capable of processing both text and image inputs. (Achiam et al., 2023) evaluated GPT-4 on a variety of exams, showing that it often outscored majority of the test takers. For example, on a simulated bar exam, GPT-4 achieved a score that falls in the top 10% of test takers while GPT-3.5 scores in the bottom 10%. GPT-4 also outperformed previous LLMs and state-of-the-art models on a suite of traditional NLP benchmarks such as MMLU (Hendrycks et al., 2020b,a). The high performance of GPT-4 can be attributed to the model size (1.76 trillion parameters), large context windows (up to 32,768 tokens) and improved data cleaning and curation techniques which have not been publicly disclosed.

2.6.4. LLaMA (Large Language Model Meta AI)

LLaMA, developed by Meta AI and initially released in February 2023, is a series of foundation language models ranging from 7 billion to 65 billion parameters (Touvron et al., 2023). A major advantage of Llama models is that they release the models publicly, allowing researchers to train/run them locally. This is especially helpful when working with patient health data that needs to be stored on secure servers and cannot be shared using APIs. This makes it more accessible for clinical and healthcare research.

2.6.5. Applications in Healthcare

One of the most notable applications of LLMs in healthcare has been their performance on standardized medical examinations (Rosol et al., 2023). In the Japanese National Nursing Examination (JNNE), GPT-4 achieved an overall accuracy rate of 79.7%, surpassing the passing standards for compulsory and general/scenario-based questions (Kaneda et al., 2023). This performance was a substantial improvement from GPT-3.5, which achieved an accuracy rate of 59.9%. In orthopedic surgery, GPT-4 demonstrated proficiency in answering questions from the Orthopaedic In-Training Examination (OITE) (Hofmann et al., 2023; Kung et al., 2023). It correctly answered 251 out of 396 attempted questions (63.4%), performing at a level comparable to an average third-year orthopedic surgery resident. In the German Medical State Examinations of 2022, GPT-4 achieved an impressive 88.1% correct answer rate, outperforming medical students who had an average correct answer rate of 74.6% (Roos et al., 2023). The model showed strength in different medical areas, with particularly high accuracy in surgery (90.00%) for GPT-4. In the Taiwanese Nuclear Medicine Specialty Exam, these models exhibited exceptional capabilities in theoretical knowledge and clinical medicine, often matching or surpassing human doctor performance (Ting et al., 2024). In a comprehensive evaluation, GPT-4 demonstrated exceptional performance on the United States Medical Licensing Examination (USMLE) (Nori et al., 2023). Without any specialized prompt crafting, GPT-4 exceeded the passing score on USMLE by over 20 points. This performance surpassed not only earlier general-purpose models like GPT-3.5 but also models specifically fine-tuned on medical knowledge, such as Med-PaLM (Singhal et al., 2022) (a prompt-tuned version of Flan-PaLM 540B (Chowdhery et al., 2023; Chung et al., 2024)). A model's ability to perform well on medical exams does not automatically imply that it's good at diagnosing patients.

Nevertheless, GPT-4 has also shown remarkable capabilities in various clinical prediction tasks. In a comparative analysis against human expertise, GPT-4 demonstrated superior accuracy in analyzing both text and images, achieving an accuracy of 73.3% for JAMA Clinical Challenge cases and 88.7% for NEJM Image Challenge cases (Han et al., 2023). Another study focused on non-small cell lung cancer (NSCLC) patients, where GPT-4 and GPT-3.5-turbo demonstrated high performance

in identifying clinical phenotypes from EHR text (Bhattarai et al., 2023, 2024). These models accurately predicted initial cancer stage, initial treatment, evidence of cancer recurrence, and affected organs during recurrence. GPT-4 achieved higher F1 scores, precision, and recall compared to other models such as Flan-T5, Llama-3-8B, and rule-based methods like medspaCy and scispaCy. A study applying GPT-4 to stage II/III localized colon cancer patients showed promising results in risk stratification. The model effectively categorized patients into low and high-risk groups for relapse, with low-risk patients having a relapse-free rate of 93% at 36 months compared to 63% in the high-risk group (Rueda-Lara et al., 2024). GPT-4 has also been used in streamlining the clinical trial screening process. In a study focused on heart failure patients, a GPT-4-based system called RECTIFIER demonstrated high accuracy in determining eligibility requirement of patients (Unlu et al., 2024). The model achieved accuracies ranging between 97.9% and 100% across various criteria, outperforming trained study staff in some aspects.

2.6.6. Different Ways of Using LLMs

Large Language Models can be used in a variety of ways depending on the task at hand and the availability of data.

Instruction Fine-Tuning: This was first introduced by Ouyang et al. (2022) to create GPT-3's InstructGPT and refers to adapting a pre-trained LLM to follow specific instructions more effectively. The model is trained on a dataset that contains examples of natural language instructions paired with the expected responses. An example use case might be to fine-tune a model to provide summaries of medical documents or respond to customer queries.

Few Shot Learning: The Few shot capabilities of LLMs was highlighted by Brown et al. (2020). In few-shot learning, a model is given a small number of examples (between 1 and 10) in the prompt to learn how to solve a particular task. The model uses these few examples to generalize and apply the learned pattern to new, unseen tasks. An example could include providing the model with a few sentence pairs for translation or question answering.

Zero-Shot Learning: This refers to using an LLM to perform tasks without any task-specific

examples in the prompt. The model is expected to generalize based on its prior knowledge and understanding of language to provide an appropriate answer or generate relevant content. It is particularly useful in scenarios where labeled data or task-specific examples are unavailable. An example could be asking the model "What are the symptoms of diabetes?" without any examples and the model would respond based on its pretraining. This significantly gained popularity with LLMs after the release of GPT-3 (Brown et al., 2020) as previous models typically required fine-tuning on specific tasks or datasets to achieve high performance in those areas.

Prompt Engineering: This involves strategically designing task-specific instructions, referred to as prompts, to guide model output without altering parameters. Sahoo et al. (2024) provides a survey of 29 distinct techniques for prompt engineering categorized by their diverse applications. Chain-of-Thought prompting (Wei et al., 2022b) is a technique to prompt LLMs in a way that facilitates coherent and step-by-step reasoning processes. For example, the prompt would show the reasoning process and final answer for a multi-step math word problem and mimic how humans break down problems into logical intermediate steps. The authors achieved state-of-the-art performance in math and commonsense reasoning benchmarks by utilizing CoT prompts for PaLM 540B (Chowdhery et al., 2023), achieving an accuracy of 90.2%. Retrieval Augmented Generation (Lewis et al., 2020) is another approach that combines information retrieval into the prompting process by analyzing user input, crafting a targeted query and searching through a pre-built knowledge base for relevant resources. The retrieved documents are incorporated into the original prompt, thereby adding contextual background. This is especially important for tasks that require up-to-date knowledge.

Fine Tuning: This involves further training a pre-trained model on a particular dataset to improve its performance on a specific task or domain. The amount of data required to fine-tune LLMs varies by task but is generally much larger than that required for few-shot learning. LLMs could be fine tuned for medical diagnosis predictions or sentiment analysis.

2.7. Legal and Privacy concerns in Clinical NLP

The use of patient information in EHRs is governed by data protection laws such as the General Data Privacy Regulation (GDPR) (Union, 2016) in the European Union and the Health Insurance Portability and Accountability Act (HIPAA) (Congress, 1996) in the United States, which imposes significant restrictions on how patient data can be collected, processed, and shared for model development and research. The sensitivity of health data creates substantial obstacles for using external LLM APIs on clinical information, such as those provided by OpenAI. As a result, healthcare institutions often require customized NLP systems that can be deployed within their secure environments, aligning with their specific practices and data governance policies (Meoni et al., 2024). Recently, Microsoft has provided access to LLMs such as GPT-4, GPT-3.5 and ChatGPT using the Azure OpenAI service (Microsoft Corporation, 2024) which conforms with HIPAA regulations, making it easier to conduct Clinical NLP research using the more recent LLMs. Another point to note is the need to mitigate legal risks and ensure patient safety by develop robust validation and testing processes for AI models used in healthcare settings (Wang et al., 2023a). This includes rigorous evaluation of model performance, assessment of potential biases, and ongoing monitoring of AI system outputs in real-world clinical environments.

CHAPTER 3

Acronym Disambiguation in Clinical Notes

Automatic acronym disambiguation is a classical NLP task that has been studied by researchers for several years (Jacobs et al., 2020; Grossman Liu et al., 2021; Menaha and Jayanthi, 2019). Acronyms are especially common in clinical notes and are difficult to understand due to the frequent use of medical jargon. In the following clinical note excerpt, a person with no medical background will find it hard to understand whether **DT** stands for *deep vein thrombosis*, *diphtheria-tetanus*, *dorsalis pedis* or *delirium tremens*, while a medical expert might identify the right expansion to be *deep vein thrombosis* based on the context .

***Clinical Note Excerpt** DISCHARGE INSTRUCTIONS: The patient was instructed that he may perform 50% weight bearing on the left lower extremity with the use of crutches. He was to be on Coumadin for **DT** prophylaxis for two weeks. The patient’s INR will be followed by the Coumadin Clinic. The patient was discharged on a regular diet. Prescriptions for Percocet for pain control were provided.*

Our goal in this chapter is to build prediction models that can accurately disambiguate acronyms in clinical notes. Training a supervised model requires labeled data containing acronym-expansion pairs along with surrounding context. At the time of our project (2020), only a few public datasets like the Medline dataset created by Pustejovsky et al. (2001), Acronym Finder (acr) and Acrophile (Larkey et al., 2000) existed that were either limited in scope (around 100 acronym-definition pairs), did not have surrounding context or were not relevant to the medical domain. Past approaches also included having data manually annotated by experts, with one study spending over \$100,000 on the annotation task (Albright et al., 2013). Therefore, as one of the initial steps, we tackle these problems by extracting acronym-expansion pairs along with context information from medical papers on PubMed Central in a distantly-supervised approach, following which we train classification models and evaluate them on clinical notes containing acronyms from MIMIC-III. To the

best of our knowledge, our approach is the first automated acronym sense disambiguation method for clinical notes which does not require an expensive manually annotated acronym-definition corpus for training.

3.1. Data for Acronym Disambiguation in the Medical Domain

3.1.1. Past Approaches

A majority of the previous acronym disambiguation approaches have relied on training classical Machine Learning models that require labeled data. Finley et al. (2017) employed the Reverse substitution (RS) method, which automatically generates training data by swapping out expansions for the corresponding abbreviations in the text. However, they found that the dispersion of terms in their full and shortened forms are usually not the same, and as a result RS leads to unbalanced training sets (Skreta et al., 2021). Wu et al. (2017) presented a system where the first module was responsible for detecting acronyms in text, while a second module which involved a clustering algorithm with manual human review of the centroids was used to obtain possible expansions. As their approach involves a manual annotation step by 2 medical students and 3 physicians, scaling up the process to thousands of acronyms with tens of thousands of possible expansions is expensive. Kirchhoff and Turner (2016) applied unsupervised approaches for expanding and disambiguating medical acronyms and abbreviations. They collected a set of acronyms with their possible expansions from different websites (Wikipedia, NIH Medline plus, etc.) and used a Statistical Machine Translation model to score possible expansions based on a log-linear model. They further used self-training, in which a systems predictions are filtered based on confidence scores, and the most confident predictions are reused to train the model thereby expanding their dataset. Other approaches such as Albright et al. (2013) have relied on medical experts manually annotating data and have shown to be very expensive.

3.1.2. Our Approach

In Kashyap et al. (2020), we present CLASSE GATOR, an algorithm for acronym sense disambiguation in clinical notes shown in Figure 3.1. While previous studies (Albright et al., 2013; Moon et al., 2012b; Savova et al., 2008) have relied on a large, annotated corpus of clinical notes to train models

which can be expensive, CLASSE GATOR does not.

CLASSE GATOR uses full-text research articles from PubMed Central, including a prenatal exposure subset (Boland et al., 2018) along with the entire open access dataset to learn features that are predictive of certain acronym ‘senses’. These feature vectors are then used to predict the meaning of acronyms from clinical notes. This can greatly reduce the burden on clinical annotators seeking to annotate the sense for a given acronym. In our dataset, we found that the number of senses (i.e., acronym definitions) in our data ranged from 2 to 50 with an average of 6.6 common senses per acronym (i.e., after introducing a threshold of >9 papers supporting the acronym-definition). This is much higher than the 2–6 senses reported in previous studies (Moon et al., 2012b) underscoring the importance of automated methods for acronym sense disambiguation (ASD).

PubMed Central (PMC) is a publicly available database maintained by the National Library of Medicine. PubMed Central differs from Pubmed in that it provides the full text for each article. Additionally, the PMC Open Access Subset includes millions of journal articles and preprints that are made available under license terms that allow reuse.

We identified a set of 31,764 prenatal-exposure papers from PubMed Central (PMC) using the method outlined previously in Boland et al. (2018). Specifically, we first provide a simple query of “prenatal” and “exposure” on PMC and retrieve 48,066 articles. However, we found that many of these retrieved articles mentioned “prenatal” and “exposure” in the reference section making the results not relevant. For example, general studies on lead, including chemical synthesis and reactivity studies would cite a reference on prenatal exposure to lead because of the media attention placed on it. Therefore, by restricting the terms “prenatal” and “exposure” to occur in the title, abstract or body of the text (i.e., not the references), we avoided many false positive literature results and ended up with 31,764 papers. We also used the PMC Open Access subset containing 2,227,674 distinct papers to increase the number of training examples for acronym-definition pairs that appeared rarely in the prenatal-exposure subset.

CLASSE GATOR: Clinical Acronym SenSE disambiGuATOR

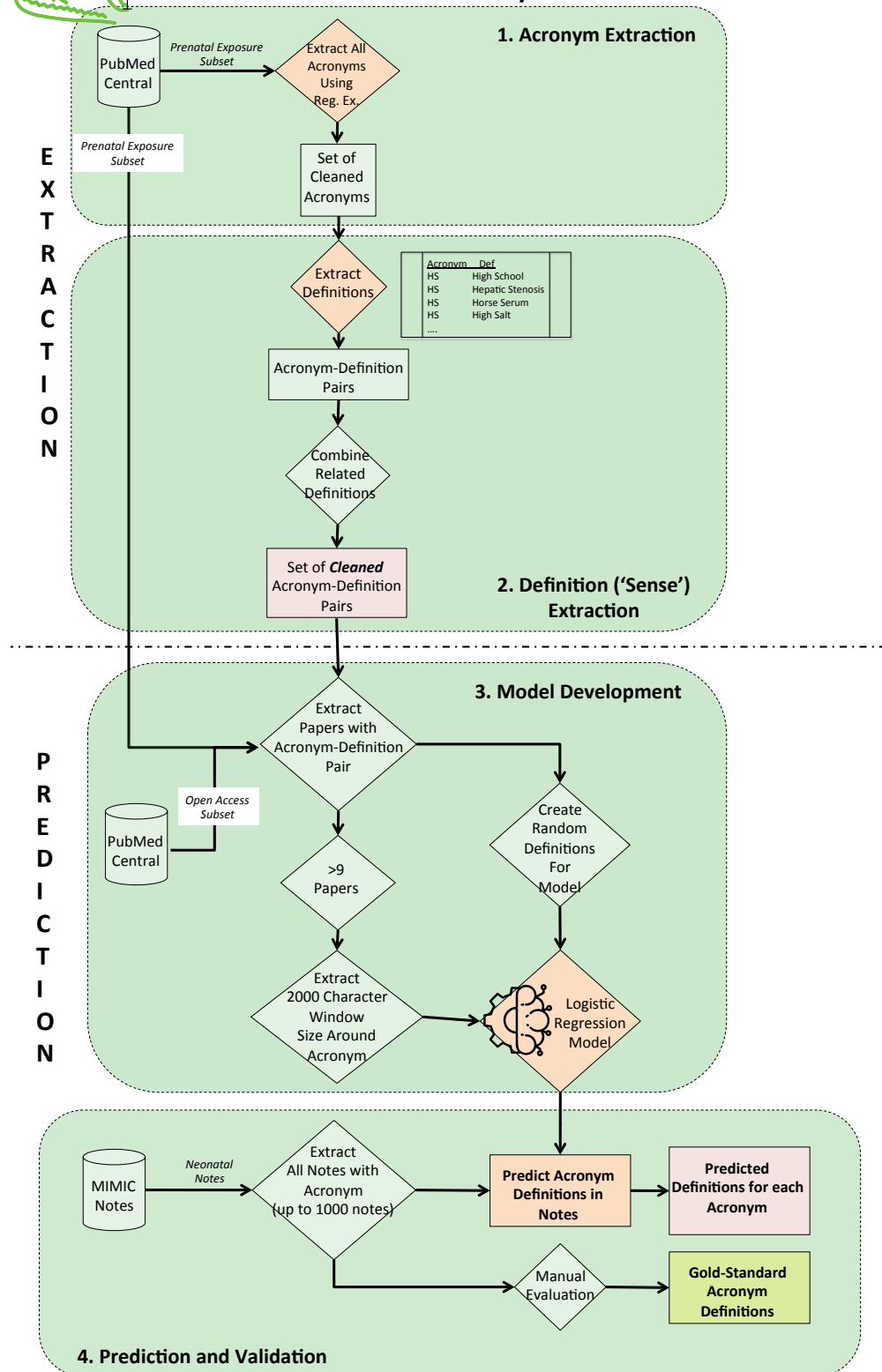


Figure 3.1: Overview of the CLASSE GATOR Algorithm.

3.2. Extraction Phase of CLASSE GATOR

3.2.1. Acronym extraction

We first removed the Reference section from 31,764 prenatal-exposure papers (Boland et al., 2018), and acronyms that occurred within parenthesis were identified and extracted using the regular expression $[([a-zA-Z]+)]$. We only used the 1500 most frequently occurring acronyms extracted that contained more than one letter. We use these 1500 acronyms to extract their definitions from the biomedical literature.

3.2.2. Extracting candidate acronym definitions or ‘Acronym expansion extraction’

We next extract candidate acronym-definition pairs, shown in 3.1 as ‘Extract Definitions’. For every mention of an acronym, CLASSE GATOR extracts two different phrases:

1. the set of $l+1$ words preceding the acronym
2. the set of l words preceding the acronym

where l is the number of letters in the acronym. We included $l+1$ words because of the frequent addition of function words in definitions (e.g., ‘United States of America’ as the meaning of USA). Therefore, for the phrase ‘*In addition, the patients’ blood pressure (bp) was alarmingly high*’, we extract both ‘patients’ blood pressure (bp)’ as $l+1$ and ‘blood pressure (bp)’ as l .

We continue collecting phrases for an acronym until one of the following conditions are met:

1. 100 non-unique phrases are obtained for that acronym
2. The entire set of prenatal-exposure papers was analyzed

For each acronym, we only consider phrases that occur more than once. This reduces the number of false positive phrases in the previous step. Additionally, we consider the remaining phrases for each acronym if they satisfy any of the following conditions:

1. If the number of letters in the acronym is equal to the number of words in the phrase, and

the acronym letters appear in the same order as the first character of each word in the phrase. For example, ‘abdominal circumference (ac)’, and ‘blood pressure (bp)’ satisfy this condition.

2. After removing stop-words from the phrase, look for the i^{th} closest word to the beginning of the phrase that begins with the first letter of the acronym. Delete all the previous words in the expansion.

We further enforced that all of the letters within the acronym need to occur in the definition and in the same order that they appear. A few examples where this condition held are: ‘**organoh**alogen (**oh**)’, and ‘**analysis of variance (anova)**’². Examples where this condition failed, and were thus removed, include, ‘**measures (mm)**’, and ‘**preterm (ptd)**’. We also excluded candidate acronym-definition pairs where none of the words in the phrase began with the first letter of the acronym, for example, ‘as abnormal shape (iii)’, and ‘conducted in Baltimore (usa)’. This rule also removed ‘lead (pb)’, which is a true acronym-definition pair, but was a rare acronym-definition pair type.

3.2.3. Combining candidate acronym-definition pairs with similar meaning

Because several expansions of a particular acronym have the same or highly related meanings, we define a method for combining expansions that are semantically similar (3.1 - ‘Combine Related Definitions’). We define *Similarity Threshold* as given below:

$$DissimilarityThreshold = \frac{2 * LevenshteinDistance(expansion_1, expansion_2)}{length(expansion_1) + length(expansion_2)}$$

where expansion refers to each candidate acronym-definition pair and $length(expansion_i)$ is the number of letters in $expansion_i$ while *LevenshteinDistance* is the Damerau-Levenshtein (D-L) distance (Bard, 2006).

Candidate acronym-definition pairs or expansions with a Dissimilarity Threshold < 0.10 were considered ‘similar’ expansions and were grouped together. The D-L distance measures the edit distance between two strings and was chosen instead of the Levenshtein distance because it allows for transpositions, which occur frequently due to human spelling errors (Damerau, 1964). For example, when

²In the example **Organohalogen (OH)**, the letters **o** and **h**, appear in the same order in the definition as they do in the acronym.

comparing two strings: ‘abcd’ and ‘badc’ the Damerau-Levenshtein distance metric considers this to be 2 differences because there are two transpositions (a and b are transposed and c and d are transposed). Using this method, we grouped the following acronym-definition pairs together (each previously distinct definition is separated by a ‘/’): ‘root mean square/ root mean squared (rms)’ and ‘human leukocyte antigen / human leucocyte antigen / human leukocyte antigens (hla)’.

3.3. Prediction Phase of CLASSE GATOR

3.3.1. Finetuning BERT and BioBERT for text classification

The BERT-base model contains an encoder with 12 Transformer blocks, 12 self-attention heads, and hidden sizes of 768. It takes in a sequence containing a maximum of 512 tokens and outputs the representation of the sequence. The input sequence has either one or two segments where the first token is always [CLS], and a [SEP] token to separate the segments. For text classification tasks, BERT takes the final hidden state $\mathbf{h}$ of the first token ([CLS]) as the representation of the entire sequence. A simple softmax classifier is added to the top of BERT to predict the probability of label $\mathbf{c}$:

$$p(\mathbf{c}|\mathbf{h}) = \text{softmax}(\mathbf{W}\mathbf{h})$$

where $\mathbf{W}$ is the task-specific parameter matrix. All the parameters from BERT are fine-tuned along with $\mathbf{W}$ jointly by maximizing the log-probability of the correct label.

To adapt BERT to the acronym disambiguation task, we first pre-process the text to ensure that the maximum input sequence length (512) of BERT is satisfied. This involves either adding extra padding tokens ([PAD]) or truncating extra context tokens. Next, we select the hidden layer that gives us the sequence representation, as different layers of a neural network can capture different levels of syntactic and semantic information (Peters et al., 2018; Howard and Ruder, 2018). The lower layers capture more basic surface features like word order and part-of-speech, while middle layers begin to encode deeper syntactic relationships, and higher layers focus on more complex semantic meaning. For our task, we chose the hidden representation from the last layer as the semantic meaning of the text is crucial to disambiguating acronyms in a specific context. Finally,

we select an optimizer (Adam (Kingma, 2014)) and an appropriate learning rate ($2e-5$) to prevent over-fitting.

3.3.2. Model construction

We used the 2,227,674 PMC Open Access Subset in addition to the 31,764 prenatal-exposure papers to increase the number of training examples for each acronym-definition pair. For each acronym under consideration, our goal was to build a model that predicted the acronym’s sense or definition based on the context (i.e. words surrounding the acronym). For every acronym, we used a set of PMC papers that contained the acronym-definition pairs to train Logistic Regression and BERT models.

We trained our model on up to 150 papers per unique acronym-definition pair. If an acronym-definition pair had more than 150 papers supporting that definition, we randomly selected 150 papers for inclusion. We also excluded all acronym-definition pairs with fewer than 10 papers to avoid skewing the model towards less frequently discussed acronym-definitions (3.1). We used words occurring within a 1000-character window on either side of the acronym-definition pair as our training examples. For each acronym, we included a ‘random expansion’ because we knew that not all acronym-definition pairs are likely to exist in our dataset and therefore we wanted to be able to capture an instance where an acronym is being used in a way that is not similar to any of our known definitions (i.e., randomly). We constructed our ‘random expansion’ set in the following manner. A random set of 150 PMC papers that did not contain the acronym were used as training examples for the ‘random expansion’ set. We randomly sampled 1000 words between the 500th and the 4500th word of every paper after removing the References section. This was done to increase the probability of choosing words from the Introduction/Methods section of the PMC paper and avoid using words contained in the abstract/author names/affiliations sections. We also removed common English stop-words and converted all the text to lower-case after lemmatization. In creating our acronym-specific random expansion set, we randomly sampled both papers and words and therefore, we performed our analyses across 10 different random seeds and report those results.

3.3.3. Equal vs. unequal class balance

Class imbalance is a major issue in machine learning models, including regression-based models. Therefore, we performed our analyses in both an equal class balance setting (ideal) and in an unequal class balance setting. For each acronym, in the equal class balance scenario, we calculated the minimum number of acronym defining instances supporting a given definition. We then randomly sampled the same number of acronym-defining instances (i.e., locations in the papers where the acronym is defined) for all classes, thus balancing the amount of information across all definitions for a given acronym. We ran our models 10 different times across different seeds to compute average results. An example of the difference between the unequal and equal class balance scenarios is as follows. If an acronym had three definitions called A, B, and C and A was supported by 20 acronym-defining instances, B by 30 acronym-defining instances and C by 40 acronym-defining instances, then we would randomly select only 20 acronym-defining instances (i.e., the minimum support among the definitions) for all three definitions in the equal class scenario. Whereas, in the unequal setting, we would chose all acronym-defining instances available - thereby resulting in class imbalance.

3.3.4. Using models to predict the sense or definition of acronyms used in clinical notes

We applied the logistic regression model of features predictive of certain definitions per acronym on neonatal clinical notes from MIMIC-III (Johnson et al., 2016b). For each acronym, we obtained up to 1000 clinical notes where that acronym was mentioned. We first cleaned the clinical note text. For example, because the clinical notes were de-identified, we removed all indicators of the censored text. These censored portions of the notes were denoted using brackets; therefore, we removed them using the regex expression `[\[\][^\[\]]*[\]]`. We also removed common English stop-words and converted all text to lower-case after lemmatization. Using this cleaned set of clinical notes, we then predicted the acronym sense or definition using our model.

In addition to logistic regression, we also compared CLASSE GATOR’s performance when using two state-of-the-art natural language processing algorithms: BERT: Bidirectional Encoder Representations from Transformers (Devlin et al., 2018) and BioBert: Bidirectional Encoder Representations from Transformers for Biomedical Text Mining (Lee et al., 2020). Bert and BioBert both utilize

the context of where a word appears in a sentence and its surrounding words to infer the probability of a sense assignment. The main difference between Bert and BioBert is that BioBert was trained on PubMed data whereas Bert was trained using BooksCorpus (800 million words) (30 and English Wikipedia 2,500 million words making Bert less specific to the biomedical domain. However, Bert has been shown to outperform traditional logistic regression for many biomedical sense disambiguation tasks due to its incorporation of context into the prediction (Si et al., 2019).

3.4. Evaluation of CLASSE GATOR

3.4.1. Five-fold cross validation for prediction on PMC articles

Before testing the prediction of CLASSE GATOR (Equal Class Distribution) on the clinical notes, the acronym expansion predictions were evaluated against PMC. For each acronym, the acronym dataset was randomly partitioned into a training subset (80%) on which the model was trained, and a validation subset (20%, on which the model was evaluated. This was carried out across 5 trials 5-fold cross validation, and an average of the validation accuracies were considered as the performance metric for that acronym model. We then report the average across 1,256 acronyms.

3.4.2. Evaluation of cross-translation accuracy for prediction on clinical notes

We evaluated the performance of CLASSE GATOR using a manually annotated set of MIMIC-III notes. These notes were annotated with the ‘sense’ (i.e., acronym meaning/definition). We selected 9 different acronyms to manually review, namely: “PEG”, “HS”, “IVF”, “GA”, “ELBW”, “MRSA”, “REM”, “PROM”, and “PSA”. We annotated up to 30 notes per acronym (if fewer than 30 notes for a given acronym existed, we annotated all notes for that acronym). Eight of the acronyms had more than 30 notes and one acronym (“PSA”) only had 5 neonatal clinical notes therefore, we annotated all 5 notes. Our final evaluation corpus consisted of 9 acronyms from 245 notes. Table 3.1 shows the acronyms, the number of ‘senses’, and the random expected accuracy given the number of definitions. The number of senses for a given acronym in our evaluation corpus ranged from 2 to 18 (Table 3.1). We manually merged highly related acronym definitions that CLASSE GATOR kept as distinct definitions. We then calculated an adjusted random accuracy expected by chance given the total number of unique definitions included (Table 3.1). For example, ‘gestation age /

gestation’ and ‘gestational / gestational ages / gestational age’ were merged. All merged acronym definitions are given in the legend (Table 3.2).

Acronym	Number of Potential Definitions	Random Accuracy	Number of Potential Definitions after Merging Related Definitions	Random Accuracy	Number of Notes Annotated
PEG	5	20	4	25	30
HS	18	5.56	18	5.56	30
IVF	5	20	3	33.33	30
GA	15	6.67	14	7.14	30
ELBW	2	50	2	50	30
MRSA	2	50	2	50	30
REM	4	25	4	25	30
PROM	6	16.67	2	50	30
PSA	10	10	10	10	5
Total					245

Table 3.1: Acronyms in the Evaluation Corpus.

Some acronym senses (definitions) were found by the annotator that were not captured by CLASSE GATOR because CLASSE GATOR did not have access to those definitions (i.e., they were not contained within our PMC subset). In those cases, the gold standard for CLASSE GATOR was ‘random’ (Table 3.2). For example, one sense for ‘HS’ was ‘heart sounds’, however this sense was missing from the 18 potential definitions identified by CLASSE GATOR (Table 3.1) and therefore our gold-standard in that case would be ‘random’.

3.4.3. Evaluation of cross-translation accuracy for prediction on UMN dataset

Several researchers at the University of Minnesota (UMN) have provided a dataset of 75 acronyms along with their appropriate sense and a clinical textual excerpt delineating its usage (Moon et al., 2012a). They have used this dataset in several projects (Moon et al., 2012b). We will evaluate CLASSE GATOR’s performance on this set of 75 acronyms, in addition to the 9 acronyms that we annotated above from MIMIC. We found that 52 of the 75 acronyms were found in our PMC dataset and therefore, we focus this section of our evaluation on the 52 found in PMC.

Sense or Definition	In CLASSE GATOR?	ELBW	GA	HS	IVF	MRSA	PEG	PROM	PSA	REM
Extremely low birth weight	Y	30	0	0	0	0	0	0	0	0
Gestational age*	Y	0	30	0	0	0	0	0	0	0
Health services	N	0	0	2	0	0	0	0	0	0
Heart sounds	N	0	0	14	0	0	0	0	0	0
Hepatosplenomegaly	N	0	0	2	0	0	0	0	0	0
High school	Y	0	0	1	0	0	0	0	0	0
History	N	0	0	1	0	0	0	0	0	0
Hour of sleep	N	0	0	1	0	0	0	0	0	0
In vitro fertilization†	Y	0	0	0	1	0	0	0	0	0
Intravenous fluids	Y	0	0	0	29	0	0	0	0	0
Methicillin resistant staphylococcus aureus	Y	0	0	0	0	30	0	0	0	0
Misspelling of has	N	0	0	4	0	0	0	0	0	0
Misspelling of he	N	0	0	1	0	0	0	0	0	0
Misspelling of his	N	0	0	2	0	0	0	0	0	0
Percutaneous endoscopic gastrostomy‡	Y	0	0	0	0	0	28	0	0	0
Polyethylene glycol	Y	0	0	0	0	0	1	0	0	0
Premature rupture of membrane	Y	0	0	0	0	0	0	28	0	0
Proper name	N	0	0	0	0	0	1	0	0	0
pseudomonas aeruginosa (psa)	Y	0	0	0	0	0	0	0	4	0
Random words+	N	0	0	0	0	0	0	2	0	0
Rapid eye movement	Y	0	0	0	0	0	0	0	0	29
Remarkable	N	0	0	0	0	0	0	0	0	1
Unknown	N	0	0	2	0	0	0	0	0	0
Unknown – social	N	0	0	0	0	0	0	0	1	0

Table 3.2: Acronyms and their Senses as Annotated in the Evaluation Corpus.

3.5. Results

3.5.1. CLASSE GATOR differences between equal and unequal class distribution

We compared CLASSE GATOR in equal and unequal settings. In the equal class scenario, we set the number of defining acronym instances to be constant across all definitions for a given acronym. Whereas, in the unequal setting, we would chose all acronym-defining instances available - thereby resulting in class imbalance. Overall, the number of acronym defining instances included in the equal class distribution model was smaller than in the unequal setting. Because of this difference in size, the overall dictionary size was lower in the equal class setting (Figure 3.2, $p < 0.001$).

CLASSE GATOR identified 1,257 acronyms and 8,287 definitions if 1-random definition per acronym is included. In the equal class setting, the average number of senses definitions per acronym was 6.6 range: 2–50) (Figure 3.3) and the number of papers per definition ranged from 10 to the max of 150 (Figure 3.3).

3.5.2. CLASSE GATOR evaluation results

Five-fold cross validation results for prediction on PMC articles

We performed 5-fold cross validation for acronym sense prediction on PMC articles. We did this to determine if CLASSE GATOR could predict acronym sense within PMC articles (i.e., the dataset it was trained on), split between 80 % training and 20 % testing. CLASSE GATOR achieved an average accuracy of 87.9 % across 1,256 acronyms range: 44.0 %–100 % accuracy).

Evaluation results for cross-translation accuracy on clinical notes

We compared our prediction results from CLASSE GATOR (Figure 3.1) to our annotated evaluation set. We investigated both the unequal class distribution and the equal class distribution scenarios (Figure 3.4). Equal class distribution is more robust because it does not depend on the number of papers per definition. The most frequent sense (definition) for ‘IVF’ in our evaluation set was ‘intravenous fluids’ because we are predicting on neonatal notes. However, there were more PMC articles with ‘in vitro fertilization’. Therefore, in the unequal class distribution scenario ‘in vitro fertilization’ was incorrectly predicted as the definition for most IVF occurrences because there are

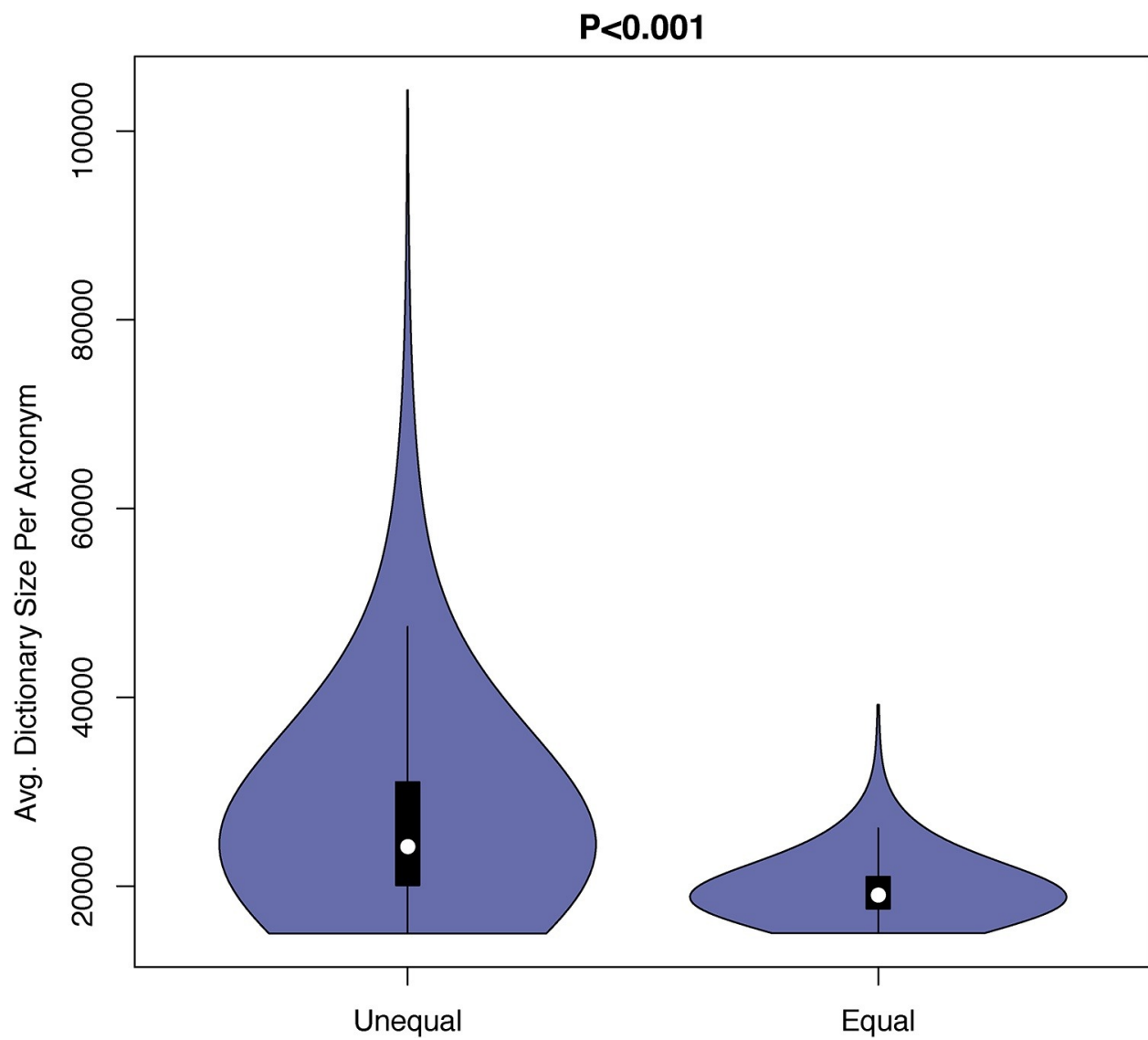


Figure 3.2: Average Dictionary Size Unequal vs. Equal Class Distribution.

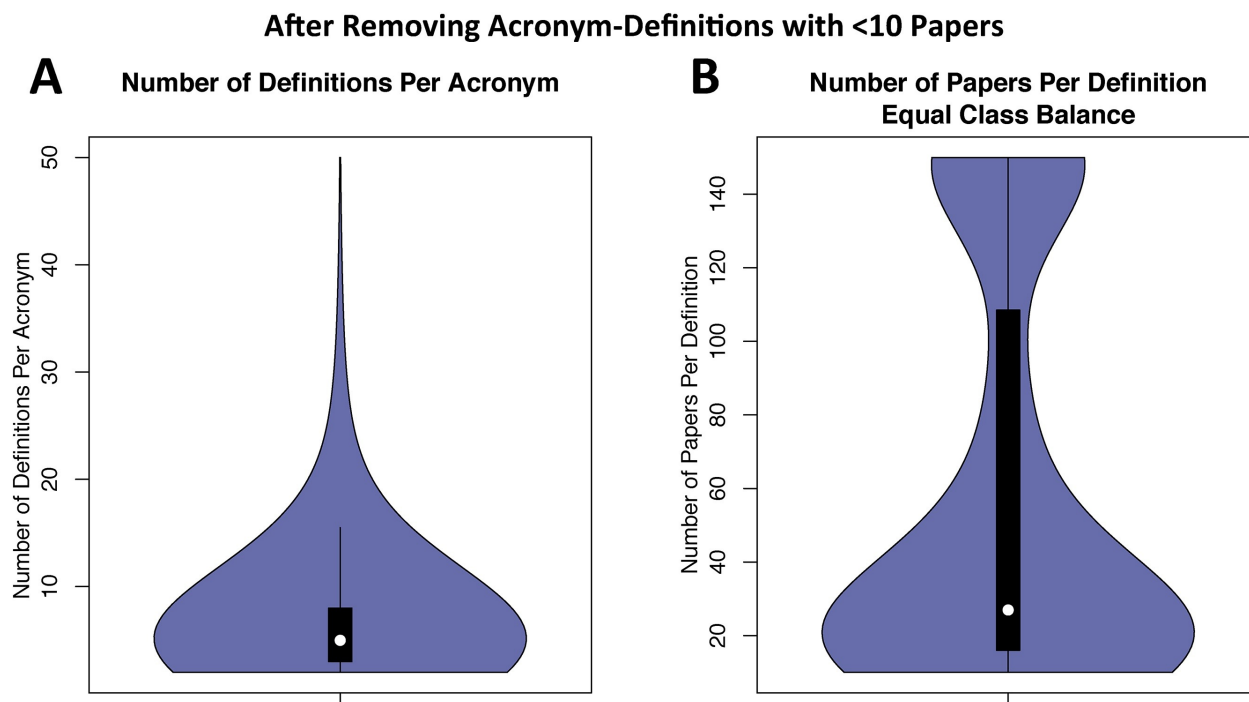


Figure 3.3: The Number of Definitions per Acronym and the Number of Papers per Definition.

more articles studying ‘in vitro fertilization’ versus ‘intravenous fluids’. Whereas, in the equal class distribution scenario CLASSE GATOR was able to assign certain instances of IVF to ‘intravenous fluids’ and others to ‘in vitro fertilization’ resulting in a significantly higher accuracy level in the equal class distribution setting ($p = 0.002$). Some acronyms did not differ greatly between the unequal and equal class distribution settings, such as MRSA and PROM (Figure 3.4). However, REM and PEG performed worse in the equal class distribution setting ($p = 0.002$ and $p = 0.008$, respectively) (Figure 3.4). In these cases, the most common usage of the acronym was also the most common in the PMC articles and therefore having more training samples (as in the unequal class balance scenario) resulted in improved algorithm prediction.

The acronym ‘HS’ had 18 potential senses identified from PMC having at least 10 papers (Table 3.1). However, in our manual review set, we identified a sense that was not found in our PMC subset, ‘Heart sounds’ that accounted for most of the usages among neonatal clinical notes (14/30 or 46.67 %). In the evaluation set, we also found that misspellings were common for ‘hs’ (7/30 or 23.33 %). These included a misspelling of the word ‘has’ (4/30 or 13.33 %), his (2/30 or 6.67 %) and

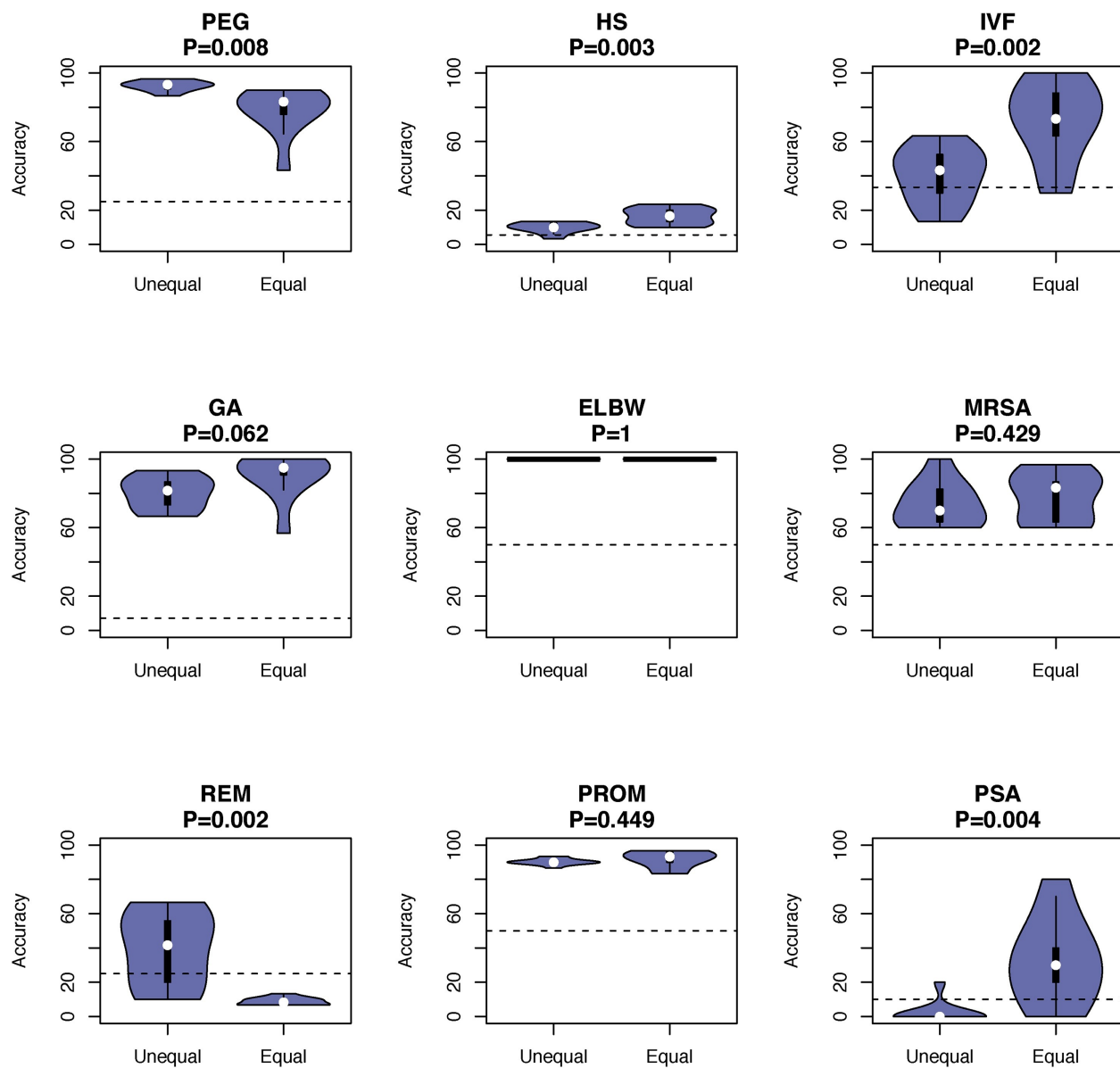


Figure 3.4: Accuracy of Algorithm by Class Distribution.

he (1/30 or 3.33 %). Other usages of the acronym ‘HS’ included health services (2/30 or 6.67 %), hepatosplenomegaly (2/30 or 6.67 %), high school (1/30 or 3.33 %), history (1/30 or 3.33 %), hour of sleep (1/30 or 3.33 %) and unknown (2/30 or 6.67 %). Table 3.2 contains the breakdown of acronyms and their senses in the 245-note evaluation corpus.

Because we were investigating neonatal clinical notes, the sense that was manually annotated for ‘PSA’ was *pseudomonas aeruginosa*, which is not the most popularly discussed meaning in the literature of ‘prostate specific antigen’. CLASSE GATOR did have *pseudomonas aeruginosa* as a potential ‘sense’ for PSA. Because *pseudomonas aeruginosa* is a rarer sense of PSA in the literature, CLASSE GATOR significantly outperformed the unequal class scenario for disambiguating PSA ($P = 0.004$) in the equal class balance scenario (Figure 3.4).

Evaluation results for cross-translation accuracy for prediction on UMN dataset

We used the UMN 75 acronym corpus (Moon et al., 2012a) along with the clinical text they provide to evaluate the performance of CLASSE GATOR. CLASSE GATOR found 52 of the 75 acronyms in PMC. Therefore, we focus our evaluation on these 52 acronyms. Table 3.3 shows the prediction results for CLASSE GATOR. We calculated the number of acronyms that outperformed random by prediction algorithm type: logistic regression with equal class distribution, Bert and BioBert. We found that CLASSE GATOR outperformed random for 63.46 % of the UMN acronyms when using logistic regression with equal class distribution, 75.00 % when using Bert, and 76.92 % when using BioBert.

One acronym - IVF was found in both the UMN corpus and our manually validated MIMIC corpus. We compared the prediction performance between these two datasets and by prediction algorithm type for CLASSE GATOR. Overall, the UMN dataset appeared to perform slightly higher than our manually evaluated MIMIC corpus (Figure 3.5). The equal class distribution for logistic regression is more robust to corpora than the unequal class distribution (Figure 3.5). This is intuitive given that unequal class distribution tends to bias the algorithm towards the more frequently occurring acronym definition in a particular corpus and therefore is less robust.

	Number of Acronyms that Outperformed Random given total 'senses' from Pubmed (N = 52)
CLASSE GATOR - logistic regression with equal class distribution as prediction algorithm	33/52 (63.46 %)
CLASSE GATOR-Bert as prediction algorithm	39/52 (75.00 %)
CLASSE GATOR-BioBert as prediction algorithm	40/52 (76.92 %)

Table 3.3: Comparison Between CLASSE GATOR Prediction Algorithm: Logistic Regression, BioBert, and Bert for the UMN Acronym Corpus.

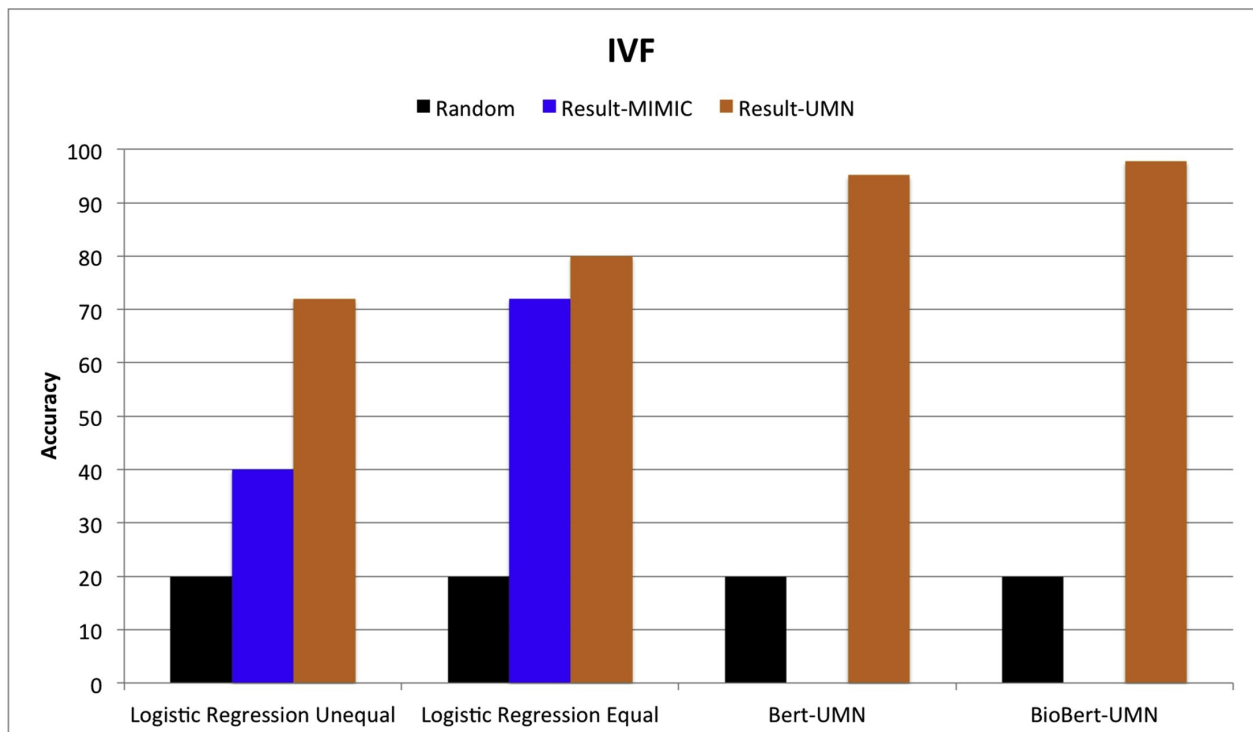


Figure 3.5: Mean Accuracy of CLASSE GATOR on UMN Corpus and MIMIC Corpus by Prediction Algorithm Type for IVF Acronym.

3.5.3. CLASSE GATOR features predictive of acronym sense

We also investigated the word features that were predictive of acronym sense in PMC articles and their presence in clinical notes. Figure 3.6 displays a word cloud for two acronyms, ‘GA’ and ‘IVF’. Many features that were predictive of acronym sense were not found in clinical notes (orange/brown in Figure 3.6). These features define what the acronym means, but the definitions are often not provided in clinical notes because the acronym definition is often implicit. Importantly, some key features were present. For example, ‘muscle’ for the ‘GA’ sense related to Gastrocnemius muscle and ‘procedure’ for the ‘GA’ sense general anesthesia.

3.6. Discussion

CLASSE GATOR incorporates knowledge from the biomedical literature on acronym sense to learn the appropriate sense used in clinical notes. This is a first, and very important, step towards disambiguating the meaning of clinical notes in a fast and cost-effective manner.

All prior studies have required carefully annotated and curated labeled datasets. For example, a prior study of only 50 clinical acronyms found that 500 manual annotations were required for their training algorithms – in their scenario the number of senses per acronym was between 2 and 6 (Moon et al., 2012b). We had 1,257 acronyms in our study and therefore 12,570 annotations would have been required, assuming that we had a similar number of senses. However, we had between 2–50 senses per acronym (versus 2–6 in (Moon et al., 2012b) with an average of 6.6 per acronym. Moon et al. reduced the number of required annotations from 500 to 125, but even with that reduction, at least 3,142.5 annotations would be required for 1,257 acronyms.

Annotating clinical text is extremely costly because of the experts required to perform the annotations. One study annotating 13,091 clinical sentences reported a cost of upwards of \$100,000 to annotate (Albright et al., 2013). Therefore, methods that require extensive manual annotations are not scalable. CLASSE GATOR provides a way to utilize already available information from the literature to understand how these acronyms are used in clinical text.

Importantly, biomedical literature is very different from clinical notes. One prior study found that

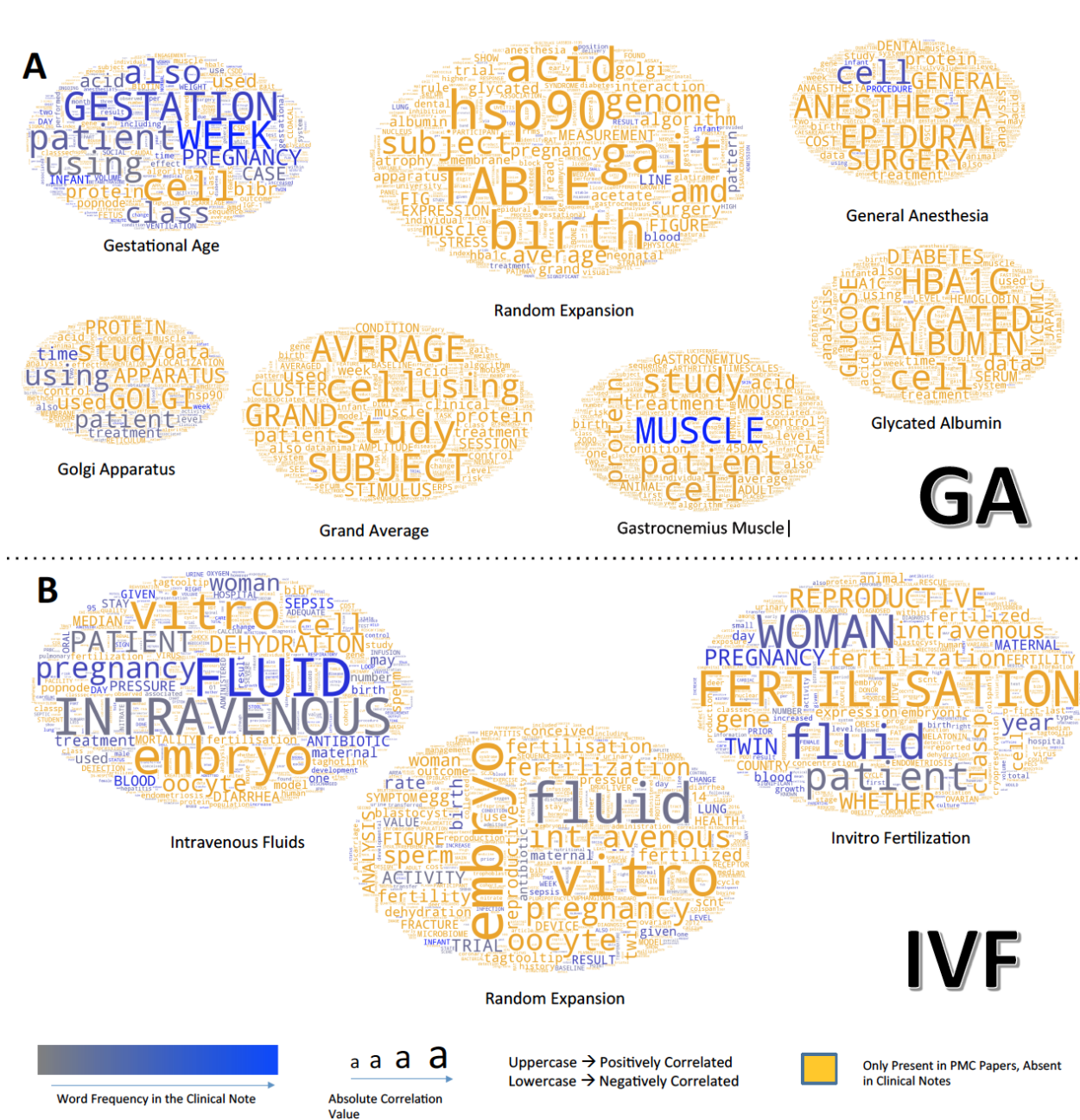


Figure 3.6: Word Clouds for ‘GA’ and ‘IVF’ Acronym Sense Predictions. Each word cloud is based on the acronym – definition predicted by the algorithm and then colors are assigned brown if either a.) the acronym-definition pair was not predicted in the clinical notes or b.) the acronym – definition pair was predicted in the clinical notes, but that particular word was not present in the notes assigned with that particular definition (or ‘sense’). Colors are assigned a color gradient from grey (rarely occurring in clinical notes) to dark blue (frequently occurring in clinical notes predicted as belonging to that acronym-definition pair). Capitalization indicates that the words are positively correlated with that definition assignment and lower-case indicates that the words are negatively correlated with that definition assignment. The size of the word indicates the absolute value of the correlation coefficient. In the interest of space, not all definitions or senses are provided in the figure, but only those that are informative or illustrative.

the Unified Medical Language System (UMLS) concepts used in clinical notes and Medline (another biomedical literature source) differed with only 57.81 % (43122/74598) of concepts being found in both with some unique to clinical notes (16963/74598) and others unique to the literature (14513/74598) (Wu and Liu, 2011). Interestingly, we found that 727 unique acronyms found by CLASSE GATOR were present in the clinical notes, which amounts to 57.29 % (727/1269). Therefore our findings on the degree of overlap across sources are consistent with prior work (Wu and Liu, 2011). The purpose of CLASSE GATOR is to utilize this overlap in information to effectively disambiguate clinical acronym usage. We included the ‘random’ definition to capture situations where an acronym was used in a manner not consistent with any of our identified definitions. In this way, our goal is to automatically identify when the acronym definition is missing from our corpus. We can then use these identified instances to cost-effectively have clinical annotators annotate these instances. Overall, CLASSE GATOR showed improved accuracy for the equal class distribution setting, outperforming random chance (Figure 3.4).

A limitation of CLASSE GATOR is the lack of misspelling adjustment. We did not correct clinical notes for misspellings prior to running our algorithm. Certain acronyms (e.g., ‘HS’) had more misspellings than other acronyms, and typically misspellings involved common short-length words (‘his’, ‘has’, ‘he’). In fact, of the 30 manually reviewed instances of ‘HS’, 7 (23.33%) were misspellings. Future work would include modifying CLASSE GATOR to account for misspellings. This is non-trivial because of the large number of acronyms in clinical notes and the difficulty of acronym detection in clinical text (Wu et al., 2011). This is further complicated by the difficulty of handling misspellings in clinical notes (Lai et al., 2015). Therefore, we leave this task to future work. Another limitation of CLASSE-GATOR is that we did not link to the Unified Medical Language System (UMLS) to combine acronyms or senses and to identify synonyms meanings similar to Moon et al. (Moon et al., 2014). Incorporating look ups to these other available dictionaries and corpora would be useful for future work as a way to expand CLASSE-GATOR.

CLASSE GATOR is the first method that does not require an automated corpus of training data and instead harnesses publically available data from PMC. Our method achieved an overall accuracy

of 63.0% and outperformed the random baseline for 8/9 acronyms (88.9%) in our evaluation set. However, there is room for improvement in terms of accuracy through use of additional machine learning methods that are actively being developed. We leave further refinements and potential increases in accuracy that those refinements may achieve to future work. A method by Savova et al. performed word sense disambiguation (WSD) in both biomedical literature and clinical notes (Savova et al., 2008) while this method was robust for prediction across domains, it still required manual annotations for training (Savova et al., 2008). The main advantage of CLASSE-GATOR is that we learned the annotations from PMC and then predicted usage on the clinical notes. Therefore, CLASSE-GATOR does not require a large annotated corpus of clinical notes (this is also very costly). Another method by Moon et al. used 50 clinical acronyms and they found that 500 manual annotations were required for their training algorithms – in their scenario the number of senses per acronym was between 2 and 6 (Moon et al., 2012b). We had 1,257 acronyms in our study and therefore 12,570 annotations would have been required, assuming that we had a similar number of senses. However, we had between 2–50 senses per acronym (versus 2–6 in (Moon et al., 2012b)) with an average of 6.6 per acronym. While Moon et al.’s algorithm (Albright et al., 2013) was able to reduce the number of required annotations from 500 to 125 at least 3,142.5 annotations would still be required for 1,257 acronyms. It is hard to predict how expensive 3,142.5 annotations would be, however one study spent upwards of \$100,000 to annotate a corpus of 13,091 sentences \$7.64 per sentence (Albright et al., 2013). If we assume that one acronym with corresponding usage takes a similar amount of time to annotate as one sentence tagging, then this corpus would cost \$24,009 to annotate even when optimized algorithms (Albright et al., 2013) are employed to reduce the annotation burden and cost. Note that our method improved this process dramatically given the state of the art at the time. Therefore, using publicly available data from PMC and learning abbreviations and their meanings from such freely available corpora is a cost-saving approach.

CHAPTER 4

A Multi-modal Deep Learning Method to Detect Opioid Prescription and Opioid Use Disorder from Electronic Health Records

In the last chapter, we introduced the acronym disambiguation task in patient clinical notes, why it is important to understand them (acronyms), and the challenges encountered when building automatic tools for this problem. We also provided a novel distantly supervised approach to disambiguating acronyms in clinical notes using BioBERT fine-tuned on a dataset from PubMed Central research papers. Training examples used to fine tune BioBERT is limited to 512 input tokens, and while this limitation did not impact our acronym disambiguation approach, it poses a significant challenge when analyzing long clinical texts of several thousands of words. This is one of the problems we attempt to address in this chapter.

We focus on the development of diagnostic models that analyze patient Electronic Health Records (EHRs) to predict a clinical outcome. Specifically, we focus on the likelihood of an opioid prescription and an Opioid Use Disorder (OUD) diagnosis for a patient given information present in their Electronic Health Records. There are 2 major challenges that we attempt to address: (i) Multi-modal Classification: Developing a Deep Learning architecture that can simultaneously process different types of data present in patient EHRs (such as X and Y), and (ii) Designing a model that can handle clinical notes of variable sizes - from a few words to thousands of words.

In Kashyap et al. (2023), we develop deep learning models using the MIMIC-III data-set for two different tasks;

1. predicting likelihood of an opioid medication prescription for patients
2. predicting likelihood of Opioid Use Disorder (OUD) diagnoses for patients

One of the challenges that we addressed in this work, was to build a model architecture that could analyze clinical notes of high variability in length (some clinical notes contained 200 tokens while

others contained as many as 20,000 tokens). In addition, we also created different modules in the prediction pipeline, each handling a different data modality.

Training data for these models was identified using a set of ICD-9 codes for the OUD task and the occurrence of an opioid medication in the patient EHR for the Opioid Prescription task. ICD-9 codes (International Classification of Diseases, 9th Revision) are a set of medical codes used to classify diseases, conditions, and other health-related problems in order to standardize diagnosis and billing procedures across healthcare settings. These codes were part of the ICD coding system developed by the World Health Organization (WHO) for use worldwide.

The model that we developed had 3 different modules, each analyzing a particular type of input data as shown in Figure 4.1, with a hierarchical attention model (Yang et al., 2016) (Figure 4.2) built on top of ClinicalBERT used to analyze the clinical notes. The following were the 3 different data-types used as input to the model.

1. **Structured Static Data:** This data includes features of a patient that remain constant during a single admission. Example features include patients' gender, ethnicity, marital status, religion, and so forth which could influence prevalence of chronic conditions and disease outcomes for certain conditions.
2. **Structured Time Varying Data:** This data includes information about a patient that typically changes over the course of an admission. These include events such as procedures and lab tests performed, medications provided and fluids removed. For example, a patient provided with the medication Acamprosate likely has alcohol dependence, which in turn could lead to a higher risk of developing OUD due to a history of dependence.
3. **Unstructured Clinical Text:** This data includes notes written by healthcare professionals while treating patients, which often includes richer data that cannot be categorized or entered into the structured parts of the EHR. Examples include patients family history such as genetic factors, social history (e.g. "Patient works as a construction laborer, lives with spouse and two children."), Psychosocial Factors (e.g. "Patient reports increased stress at work due to long

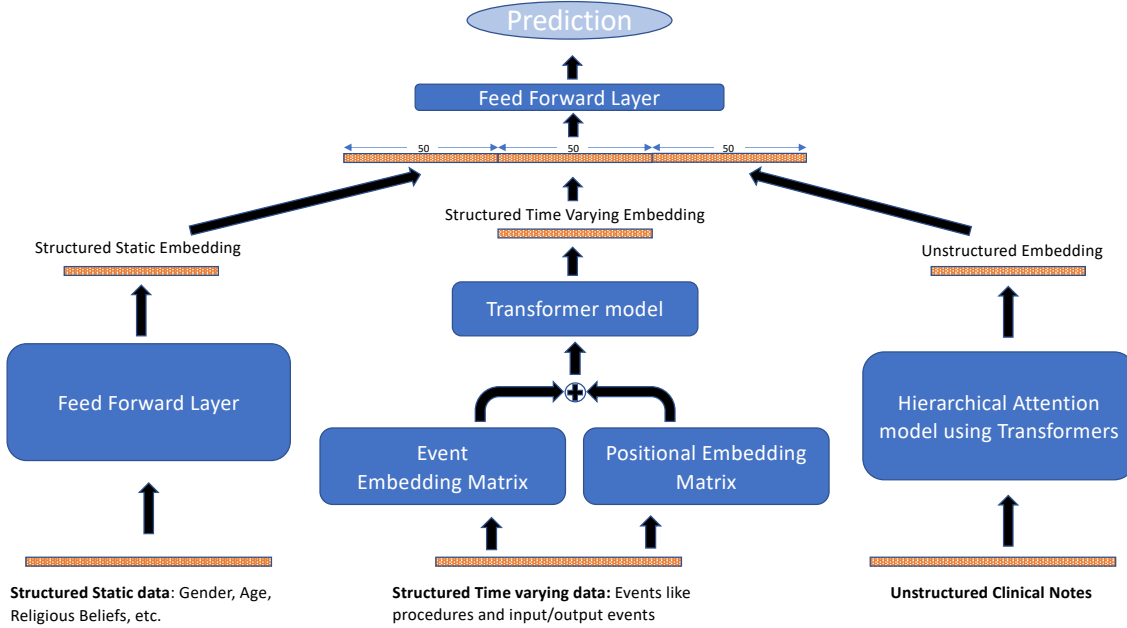


Figure 4.1: Deep Learning Model Architecture that takes in Patient EHR Data and Predicts a Binary Outcome.

hours and financial concerns." and physicians thought process during the treatment among others. This additional information could highly influence clinical outcomes being predicted.

As a preview, the Opioid Prescription model achieved an F1 score of 0.88 and an AUC of 0.93 while the OUD model achieved an F1 score of 0.82 and an AUC of 0.94 on a held-out test set. An analysis of the predictive features shows that patient age, admission type and certain medications are informative to the model while unstructured note terms related to pain, addiction and vital signs were also predictive. The opioid prescription model fills a gap as no prior methods were found for this task. The OUD model improves on past work in accuracy and F1 scores due to the deep learning approach and the utilization of multi-modal data. A comparison of various test metrics between our model and other models in the literature (Ellis et al., 2019; Dong et al., 2021a; Butler et al., 2008, 2009; Finkelman et al., 2017) is shown in Figure 4.3.

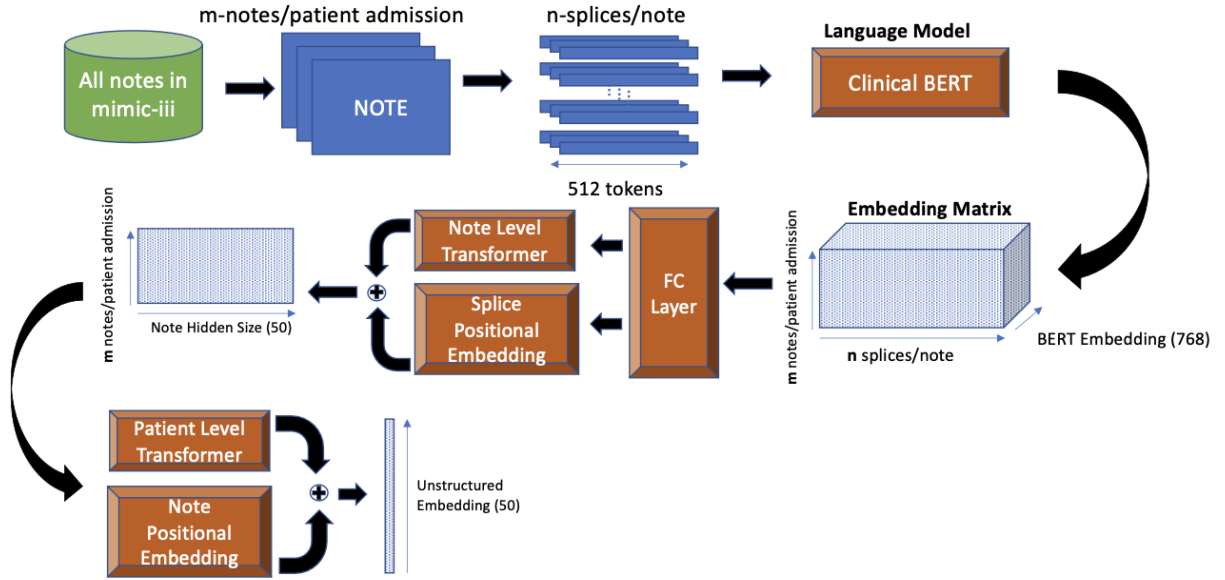


Figure 4.2: The Architecture of a Hierarchical Attention Model Built on top of ClinicalBERT to Analyze Clinical Notes.

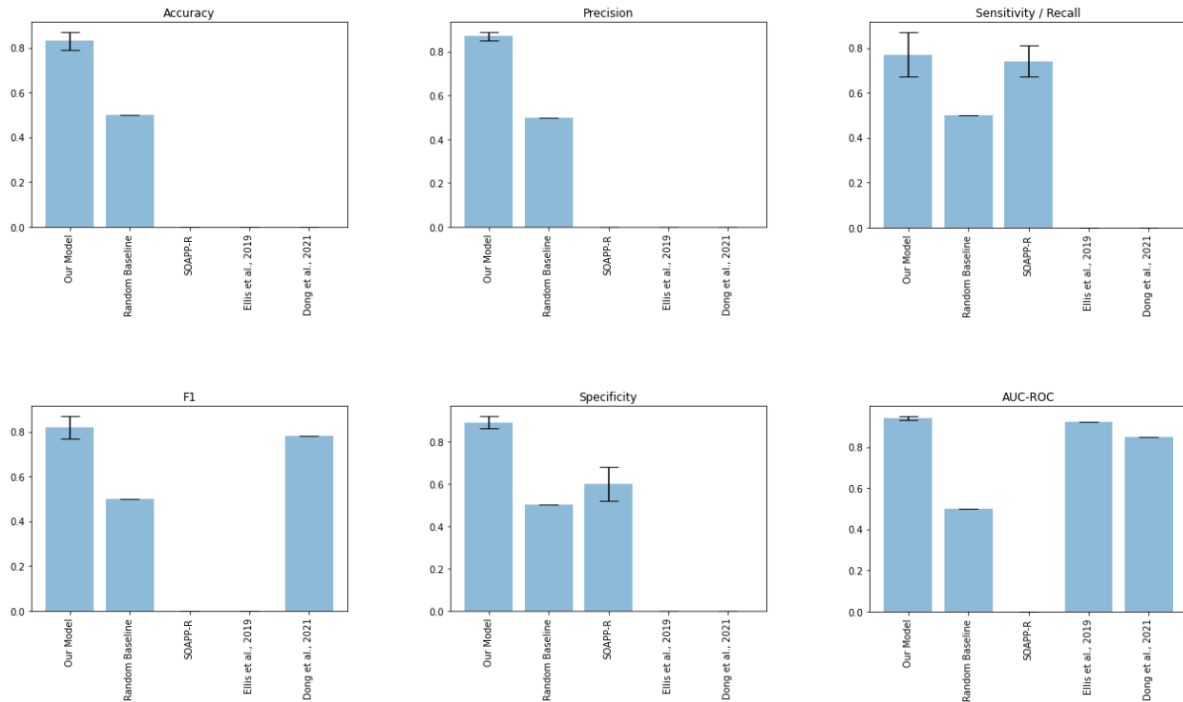


Figure 4.3: Performance Metrics of our Deep Learning Model Compared to a Random Baseline and other Models in Literature.

The main research contributions are:

1. Developing a Deep Learning Model that can analyze multiple long clinical notes using Hierarchical Transformer Models
2. Developing a Deep Learning Model that can simultaneously analyze different parts of the EHR (demographic data, time series data and clinical notes)
3. Filling a research gap by building a model that predicts the likelihood of an opioid prescription for a patient.

4.1. Obtaining a Dataset for the Opioid Prescription Prediction Model

We constructed a binary classification model that can predict whether a patient is likely to be prescribed an opioid using their EHR data. Using a list of 30 opioids (Table 4.1), we parse through the MIMIC-III dataset to identify all Hospital Admission IDs (HADM_IDs) where patients have been prescribed any of these drugs. These are taken as positive examples while the remaining set of HADM_IDs (for which patients had not been prescribed an opioid) are considered as negative examples.

Codeine	Hydromorphone	MS Contin
Percocet	Ultracet	Guiatuss AC
Dilaudid	Lorcet	Nalbuphine
Roxicet	Ultram	Robitussin DM
Duragesic	Lortab	Norco
Roxicodone	Vicodin	Safetussin DM
Endocet	Meperidine	Oxycodone
Tramadol	Delsym	Tussionex
Fentanyl	Morphine	Oxycontin
Tylenol with Codeine	Dextromethorphan	Wal Tussin Cough

Table 4.1: List of Opioids used in Obtaining Positive Training Examples for the Opioid Prescription Prediction Model.

For each HADM_ID, we retrieve 3 types of data (if available for that patient during that admission) from MIMIC-III:

1. **Structured static data:** This data includes features of a patient that remain constant

during a single admission. Example features include patients' gender, ethnicity, marital status, religion, and so forth.

2. **Structured Time Varying data:** This data includes information about a patient that typically changes over the course of an admission. These include events such as procedures and lab tests performed, and input/output events.
3. **Unstructured Clinical Notes:** This data includes notes written by healthcare workers while treating patients. The average number of words contained across all the notes for a particular patient admission in MIMIC-III is 1,900 (min = 0, max = 284,897).

For each patient admission belonging to the positive set (i.e., received an opioid prescription), we identify the date and time at which they were first prescribed an opioid. The input features that we consider include all information recorded about the patient in the MIMIC-III EHR before that point in time. This is illustrated in Figure 4.4. Prior to tokenization, we remove any mentions of opioids in the clinical notes and the Input Events of the positive examples using regular expressions containing entire medication names (Table 4.1) to ensure that information leakage does not occur in our feature space. This would, however, not account for medication misspellings and abbreviations, which through manual inspection occurred very rarely in the clinical notes.

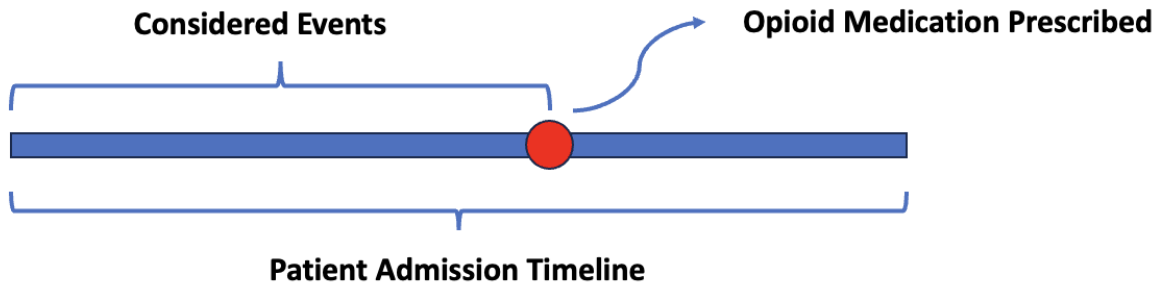


Figure 4.4: An Illustration of a Patient Admission Timeline.

For the negative examples, we consider all the information contained about that patient admission

in the MIMIC-III EHR. At test time, given an unknown patient, we consider all the information present in their EHR to make the relevant prediction. We ensure that only one hospital admission per patient is present across both the positive and negative training sets. This allows the opioid prescription prediction to occur within the course of a single hospital admission time-frame as this was deemed to be clinically more informative. An example of data for a patient used is shown in Figure 4.5.

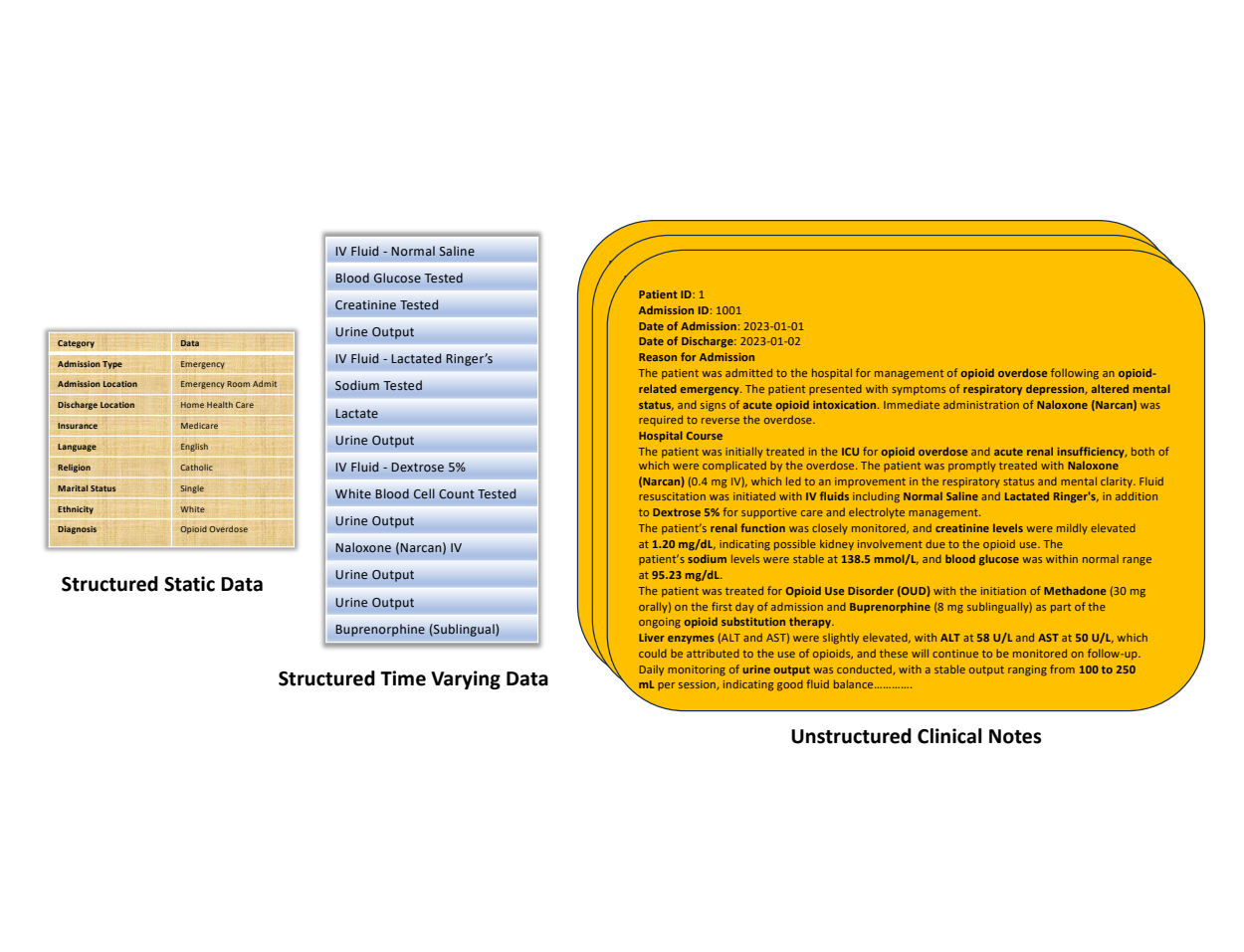


Figure 4.5: An Example of Different Data Types Analyzed for a Patient for both Tasks. We do not use Specific Timestamps of the Events, but rather their Relative Ordering.

4.2. Obtaining a Dataset for the Opioid Use Disorder (OUD) Prediction Model

In addition to seeing if a patient is prescribed an Opioid, we also look at patients that are dependent on it. We aim to build a binary classification model that takes in patient EHR data from MIMIC-III

and predicts how likely they are to be diagnosed with OUD. We identify a set of 12 ICD-9 diagnoses codes (Table 4.2) that are given to patients who were either currently showing or who have a history of OUD.

ICD9 Code	Description
30,400	Opioid type dependence, unspecified
30,401	Opioid type dependence, continuous
30,402	Opioid type dependence, episodic
30,403	Opioid type dependence, in remission
30,470	Combinations of opioid type drug with any other drug dependence, unspecified
30,471	Combinations of opioid type drug with any other drug dependence, continuous
30,472	Combinations of opioid type drug with any other drug dependence, episodic
30,473	Combinations of opioid type drug with any other drug dependence, in remission
30,550	Opioid abuse, unspecified
30,551	Opioid abuse, continuous
30,552	Opioid abuse, episodic
30,553	Opioid abuse, in remission

Table 4.2: A List of ICD9 Codes used to Identify OUD Patients.

The HADM_IDs of patients in MIMIC-III who have been diagnosed with any of these ICD9 codes were taken as positive examples and the remaining set of HADM_IDs were taken as negative examples. Structured static data, structured time varying data and unstructured clinical notes were used as input to the model. As the 12 ICD-9 codes were not present in the input features of each patient, we expect the model to learn features that are important for diagnosing OUDs.

A Venn Diagram showing the overlap between patients who have been diagnosed with OUD and have been prescribed an opioid during their hospital admission in our dataset is shown in Figure 4.6.

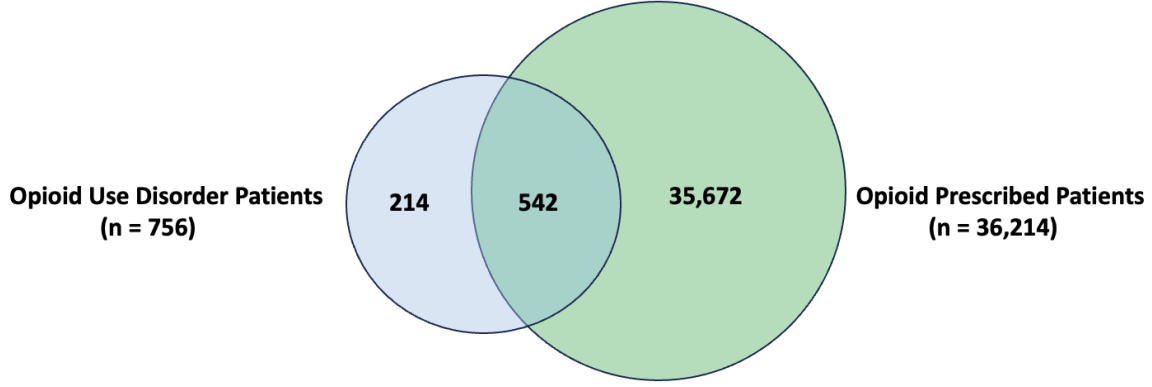


Figure 4.6: Venn Diagram showing the Overlap between Patients who have been Prescribed an Opioid Medication and who have been Diagnosed with OUD in the MIMIC-III Dataset.

4.3. Model Construction and Training

While constructing the model, two main challenges were considered: (i) Developing a Deep Learning architecture that can simultaneously process different types of data present in patient EHRs, and (ii) Designing a model that can handle long clinical notes of variable sizes. The average number of words contained across all the notes for a particular patient admission in MIMIC-III is 1,900 (min = 0³, max = 284,897). The distribution of the number of words in clinical text for each person in the Opioid Use Disorder dataset is shown in Figure 4.7. For challenge (i), we decided on having different modules to handle each data type. For the structured static data, we represent them as categorical one hot vectors and use a simple feed forward network. For the structured time varying data, using a transformer (Vaswani et al., 2017) layer with positional embeddings was the best choice at the time as they were shown to outperform RNNs and LSTMs (Sherstinsky, 2020). This was due to the self-attention mechanism that allows transformers to capture both short and long-range dependencies more effectively in addition to promoting parallelization and faster training. For the module that deals with clinical text and challenge (ii), a baseline approach was to use a bag-of-words representation with a classical machine learning model like Logistic Regression. However, the bag-of-words approach loses positional information of the text and cannot capture deeper semantic meaning of the text. BERT was the state-of-the-art model at the time we conducted the research

³There was one patient in our dataset who did not have any clinical notes associated with their EHR data.

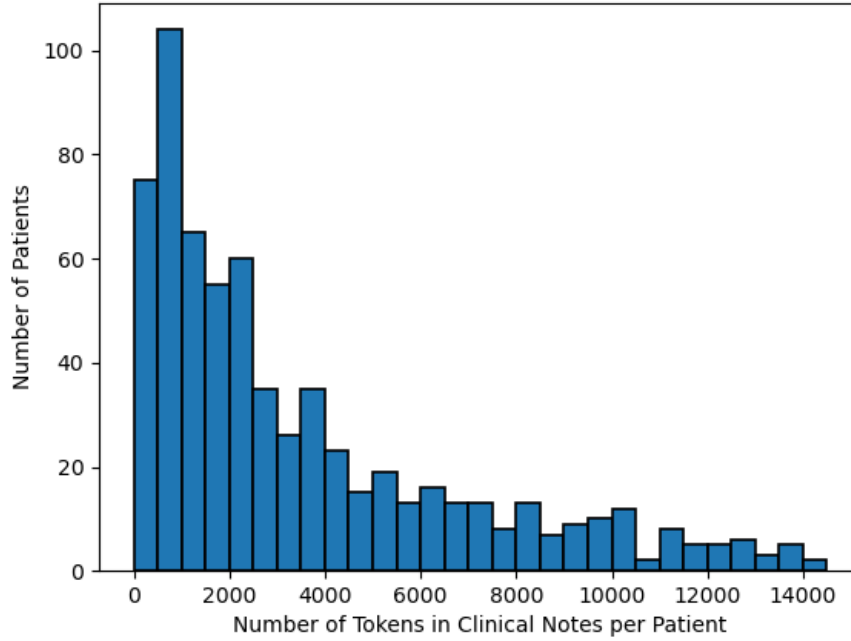


Figure 4.7: Histogram Depicting the Number of Tokens in the Clinical Notes per Patient across the Opioid Use Disorder Dataset.

presented in this chapter, but had a maximum input length of 512 tokens. With the average number of words per patient admission in our dataset being 1,900, we looked at other models like the Longformer (Beltagy et al., 2020) which could process input sizes of up to 16,000 tokens. However, a drawback with using it was the amount of GPU memory required - as the number of input tokens increases, the amount of memory required grows quadratically. We finally decided on Hierarchical attention models (Yang et al., 2016) as they were shown to perform among the best for long document classification while not requiring huge amounts of GPU memory. The basic idea was to break down long texts into 512 token chunks, obtain embeddings across each chunk from a Language Model, and aggregate information using a Transformer Layer. In this way, even though the long document was broken down and each part was embedded separately, the transformer layer would aggregate information across different chunks, thereby minimizing loss of information.

The binary classification model architectures inspired by Yang et al. (2016) for both prediction

tasks are shown in Figure 4.1. There are 4 components of the model, one for each datatype and a final one that collectively processes information from the first three components. The embeddings computed for each of the 3 data types were equal in size (1×50) to ensure an equal representation of information across the different modalities. The 4 components of the model are described below:

4.3.1. Structured static component

The structured static data containing categorical features was converted to indicator variables. A vector containing 45 features for each patient admission was then converted to a “structured static embedding” of size 50 by passing it through a feed forward layer.

4.3.2. Structured time varying component

The events in the structured time varying data were first sorted chronologically for every patient admission, following which each event was converted into a one hot encoding. We selected a maximum of 150 most recent events for every patient and appended a start and end token to each sequence. This sequence of events was then passed through the first embedding layer (event embeddings) and the second embedding layer (positional embeddings). The two embeddings were added and subsequently fed into a Transformer layer to obtain contextual event embeddings. The contextual embedding of the start token was taken as the “structured time varying embedding”.

4.3.3. Unstructured component

The notes for each patient admission were first sorted chronologically and up to 40 of their most recent notes (just prior to the opioid prescription or OUD diagnoses if a positive example or the last 40 notes if negative) were fed into a Hierarchical Transformer Model. Note embeddings were obtained using ClinicalBERT and a Note Level Transformer. The note embeddings were then passed through a Patient Level Transformer Layer that aggregated information across the 40 notes of a single patient admission as shown in Figure 4.2. The output was taken as the “Unstructured Embedding”.

4.3.4. Aggregation component

The embeddings of the first three components are finally concatenated together and passed through a Feed Forward Layer to obtain a prediction for the binary variable of interest.

We first create a balanced dataset containing an equal number of examples per class for each task by under sampling the negative examples. The reason we chose to use a balanced dataset for training was to prevent bias towards the majority class and improve minority class performance. Our earlier experiments with an imbalanced training set showed poor performance on predicting OUD diagnosis for a patient because of heavy class imbalance (the ratio between positive and negative classes was approximately 1:66). A balanced dataset allows us to do investigations into the machine learning of long document classification, but it creates the limitation that it is not directly applicable in the clinical setting.

Next, we set aside a random subset of the data (10 %) as the test set and use the remaining data for training and validating the classification models. In addition to the balanced test set, we also create an unbalanced test set that reflects the distribution of total positive to negative examples in MIMIC-III for each task. Pre-processing of the data including one hot encoding and missing value imputations were performed after the dataset was split into train, validation and test sets in order to avoid data-leakage. We trained 10 different models for each task and reported the performance metrics on the test sets as the mean $\pm$ standard deviation (Table 4.3).

4.4. Model and Data Insights

Attention maps of transformers have been a popular way for researchers to understand the decision-making process of these black box models (Niu et al., 2021). However, the complex nature of our Deep Learning model architecture along with the consideration of multimodal data makes it difficult to connect the attention maps back to the input features/tokens. So, we instead trained Logistic Regression (LogReg) models across every combination of task (opioid prescription and OUD) and data type (Structured Static Data, Structured Time Varying Data and Unstructured Clinical Notes) to provide model and data insights. It is important to keep in mind that although these LogReg models are not necessarily faithful to the true reasoning process of the deep learning models, examining the top LogReg features could give us some insights into the kind of decision made by them. The LogReg models were trained on the presence or absence of categorical features in the Structured Static Data (Table 4.4) and the Structured Time Varying Data (Table 4.4). For

Classification Model	Task	Accuracy	Precision	Recall (Sensitivity)	F1	Specificity	AUC-ROC
Random Baseline Across Both Tasks		0.5	0.5	0.5	0.5	0.5	0.5
LogReg Baseline (Structured Data only)	Opioid Prescription	0.864	0.865	0.864	0.864	0.841	0.932
LogReg Baseline (Structured Data only)	OOD	0.753	0.754	0.753	0.753	0.742	0.808
LogReg Baseline (Structured + Unstructured Data)	Opioid Prescription	0.884	0.884	0.884	0.884	0.89	0.92
LogReg Baseline (Structured + Unstructured Data)	OOD	0.818	0.819	0.818	0.818	0.828	0.845
Deep Learning Model (Structured + Unstructured Data)	Opioid Prescription	0.89 $\pm$ 0.003	0.96 $\pm$ 0.01	0.82 $\pm$ 0.01	0.88 $\pm$ 0.003	0.96 $\pm$ 0.01	0.93 $\pm$ 0.002
Deep Learning Model (Structured + Unstructured Data)	OOD	0.83 $\pm$ 0.035	0.87 $\pm$ 0.02	0.77 $\pm$ 0.1	0.82 $\pm$ 0.05	0.89 $\pm$ 0.03	0.94 $\pm$ 0.008

Table 4.3: Performance Metrics for the Logistic Regression Model Baselines and the Deep Learning Models across the 2 Tasks using a Balanced Dataset.

the Unstructured Clinical Notes, the text data was first tokenized into unigram, bigram and trigram features, converted to lower-case, and represented as a binary vector that was then used to train a LogReg model (Table 4.5). The 20,000 most frequent ngrams in the clinical notes were considered while constructing these binary vectors. Additionally, we present results of two baseline LogReg models, one trained on only the structured part of patient EHRs, while the other also incorporating the unstructured clinical text, with results proving the importance of incorporating clinical notes in the prediction pipeline (Table 4.3).

Top Positive Features	Top Positive Features	Top Negative Features	Top Negative Features
Opioid Prescription	OOD	Opioid Prescription	OOD
ADMISSION TYPE ELECTIVE	Age 25-35	ADMISSION TYPE NEWBORN	Age 65+
ADMISSION TYPE EMERGENCY	Age 18-25	Age 0-18	Age 0-18
Age 50-65	Age 35-50	ADMISSION TYPE URGENT	ADMISSION TYPE NEWBORN
Age 18-25	ADMISSION TYPE EMERGENCY	ADMISSION LOCATION UNKNOWN	INSURANCE Private
Age 35-50	INSURANCE Medicaid	ADMISSION LOCATION EMERGENCY ROOM ADMIT	MARITAL STATUS MARRIED
ADMISSION LOCATION TRANSFER FROM HOSP EXTRAM	ETHNICITY OTHER	MARITAL STATUS UNKNOWN	ETHNICITY ASIAN
Age 25-35	GENDER M	ETHNICITY BLACK/AFRICAN AMERICAN	INSURANCE UNKNOWN
ETHNICITY PATIENT DECLINED TO ANSWER	ADMISSION TYPE URGENT	ETHNICITY UNKNOWN	ADMISSION TYPE UNKNOWN
Age 65+	INSURANCE Medicare	Age UNKNOWN	GENDER UNKNOWN
ADMISSION LOCATION PHYS REFERRAL/NORMAL DELI	Age 50-65	INSURANCE UNKNOWN	Age UNKNOWN
TPN D5.0	Cryoprecipitate	E 20 FS PO	Ostomy (output)
Blood Out Lab	Naloxone (Narcan)	S 20 Fe PO	Heparin
Ativan	Acetaminophen-IV	E 20 Fe FS PO	Esmolol
AVA Line	Epinephrine	Br Milk FS PO	Metoprolol
ed ivf	Narcan	D10W	D10W
Sterile H2O GU	Invasive Ventilation	PE 20 FS PO	Neosynephrine-k
IABP line	Sandostatin	Cardiac Arrest	E 20 FS PO
Drain Out 2 Wound Vac	Folic Acid	Epinephrine	EEG
Cisatracurium	Insulin 70-30	Extubation	Esmolol
URINE CC/KG/HR	Meperidine (Demerol)	Urine	Cefazolin

Table 4.4: Top 10 Positive and Negative Features Considered by the LogReg Models across the 2 Tasks when Considering Only Structured Static Features and Structured Time Varying Features Respectively.

Top Positive Features	Top Positive Features	Top Negative Features	Top Negative Features
Opioid Prescription	OD	Opioid Prescription	OD
tracing	methadone	transfer	ventricular
comparison technique	abuse	inferior lateral	tracing normal
pain	non	rhythm rate	ischemia
tube tip	artifact	rhythm normal tracing	late
borderline left	heroin	night	borderline
osh	rhythm normal	compared	consider
findings non	wave	paced rhythm	normal sinus rhythm
patient intubated	sinus rhythm normal	tracing sinus	ventricular hypertrophy
width	overdose	abnormal ecg	tracing normal limits
sda hospital medical	non specific	changes non specific	normal sinus

Table 4.5: Top 10 Positive and Negative Features Considered by the LogReg Models across the 2 Tasks when Considering Only Unstructured Clinical Notes.

4.5. Results

4.5.1. Dataset

Using the techniques mentioned in the previous sections, we identified 37,237 unique admissions (HADM_IDs) of patients who were prescribed at least one opioid during their hospital admission. Similarly, we identified 763 unique admissions (HADM_IDs) of patients who were diagnosed with OD. Additionally, the distribution of patient demographics across the positive and negative examples for both tasks is shown in Table 4.6. Clinical notes can often contain additional information absent in the structured parts of the EHR that is relevant to predicting certain medical outcomes. For OD prediction, Table 4.7 shows note segments that could be relevant for this task.

Static Feature		Opioid Prescribed Patients(%)		OD Patients(%)	
		Pos	Neg	Pos	Neg
Admission Location	CLINIC REFERRAL/PREMATURE	19.70	20.40	26.21	22.80

	ADMISSION LOCATION EMERGENCY ROOM ADMIT	40.49	33.00	52.94	36.56
	PHYS REFERRAL/NORMAL DELI	19.79	18.06	5.76	19.39
	TRANSFER FROM HOSP/EXTRAM	17.00	9.58	14.41	14.94
	UNKNOWN	3.00	18.93	0.65	6.29
Admission Type	ELECTIVE	16.33	3.34	3.40	11.27
	EMERGENCY	78.77	54.53	95.41	71.55
	NEWBORN	0.68	21.62	0	9.82
	UNKNOWN	2.43	17.67	0	5.24
	URGENT	1.77	2.82	1.17	2.09
Age Group	0 – 18	0.93	22.56	0.13	10.61
	18 – 25	2.52	1.72	7.73	1.57
	25 – 35	3.95	2.68	19.00	3.80
	35 – 50	13.65	7.94	36.69	11.00
	50 – 65	28.11	14.38	33.15	24.11
	65 +	48.37	33.02	3.27	43.64
	UNKNOWN	2.43	17.67	0	5.24
Ethnicity	ASIAN	1.40	2.48	0.13	1.44
	BLACK/AFRICAN AMERICAN	7.86	9.17	10.61	8.25
	HISPANIC OR LATINO	2.60	2.60	3.67	2.35

	OTHER	2.18	2.30	3.54	2.23
	PATIENT DE- CLINED TO ANSWER	0.98	0.68	0.39	1.04
	UNKNOWN	14.64	28.92	11.27	17.03
	WHITE	70.31	53.81	70.38	67.62
Gender	FEMALE	42.37	37.15	34.47	43.12
	MALE	55.19	45.17	65.53	51.63
	UNKNOWN	2.43	17.67	0	5.24
Insurance	Government	2.60	2.86	7.86	2.22
	Medicaid	8.80	8.98	41.80	8.25
	Medicare	52.07	35.59	28.83	46.26
	Private	33.16	33.93	18.21	36.69
	Self-Pay	0.92	0.93	3.27	1.31
	UNKNOWN	2.43	17.67	0	5.24
Marital Status	DIVORCED	6.33	3.26	9.04	6.42
	MARRIED	47.51	26.26	16.64	41.80
	SEPARATED	1.05	0.61	2.62	0.92
	SINGLE	24.04	16.64	58.32	19.79
	UNKNOWN	7.91	43.35	9.30	19.39
	WIDOWED	13.13	9.85	4.06	11.66
Religion	CATHOLIC	35.98	26.31	39.18	34.07
	EPISCOPALIAN	1.31	0.96	0.78	1.17
	JEWISH	8.77	8.09	2.22	7.34
	OTHER	4.28	4.00	3.54	3.54
	PROTESTANT QUAKER	12.23	9.38	10.09	10.61
	UNKNOWN	37.40	51.23	44.16	43.25

Table 4.6: Population Characteristics of the Patients who were Prescribed Opioids (Pos) and who were not (Neg), and who were OUD Patients (Pos) and who were not (Neg).

1	Brief Hospital Course: history of narcotics abuse ... hospital course complicated by suicide attempt
2	PAST MEDICAL HISTORY: Chronic pain with possible opioid dependence.
3	PAST MEDICAL HISTORY:... a history of chronic low back pain and... there is a question of past history of intravenous drug abuse.
4	EMS found her unresponsive ... with empty bottle of methadone next to her.
5	Social History: Lives with son, has multiple family members nearby ... Daily heroin use, frequent cocaine, EtOH.

Table 4.7: Subset of Note Fragments from MIMIC-III Clinical Notes that might be Relevant to the Prediction of OUD in Patients.

4.5.2. Prediction models

The performance metrics of the models on the balanced held-out test sets are shown in Table 4.3. To put these results into perspective, a random baseline accuracy for both models is 0.5 on the balanced test sets, while the two Logistic Regression baselines are shown in Table 4.3. The LogReg models trained on only the structured parts of the EHR perform relatively worse compared to the model trained on both the structured and unstructured parts of the EHR (Table 4.3), proving that some important signals exist only in the clinical notes while predicting these outcomes. This motivates the use of including clinical text in the analysis pipeline for future researchers. Our opioid prescription deep learning algorithm performed well with an F1 score of 0.88 ± 0.003 and an AUC-ROC of 0.93 ± 0.002 (Table 4.3). The OUD deep learning prediction algorithm also performed well with an F1 score of 0.82 ± 0.05 (slightly lower than our opioid prescription algorithm) and an AUCROC of 0.94 ± 0.008 (slightly higher than our opioid prescription algorithm) (Table 4.3). This indicates that deep learning can be used to accurately predict both opioid prescription behavior and OUD diagnosis. Additionally, the calibration curves of the deep learning models for both tasks are shown in Figure 4.8 after performing model calibration using Isotonic Regression (Zadrozny and Elkan, 2002), with the OUD and Opioid Prescription models getting Brier Scores of 0.146 and 0.102 respectively.

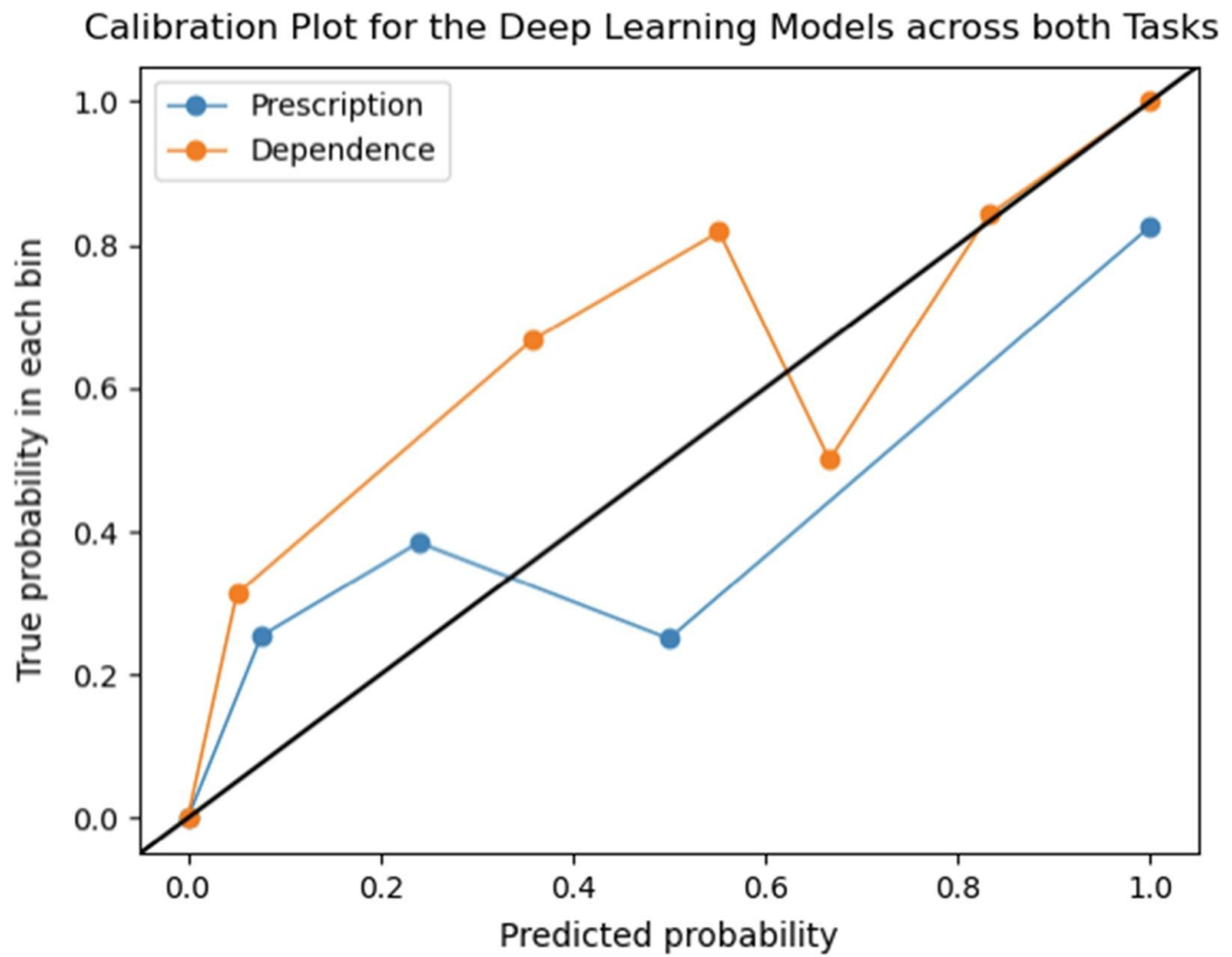


Figure 4.8: Calibration Plots of the Deep Learning Models for the Two Tasks.

4.5.3. Important features

The top positive features of each LogReg model across the 3 different data types and the 2 tasks are shown in Table 4.4 and Table 4.5 and provide some insights into the kind of patient features that the deep learning models might look for while making predictions. For example, admission types and age groups were informative at either predicting opioid prescription or non-prescribing (negative features) along with OUD diagnosis or non-diagnosis (Table 4.4). Newborn admissions were less likely to be diagnosed with either OUD or prescribed an opioid while emergency admissions were more likely to be diagnosed with an OUD or prescribed an opioid (Table 4.4). Similarly, elective admission types were more likely to be prescribed an opioid, but no relationship was found with OUD diagnosis (Table 4.4).

4.6. Discussion

4.6.1. Prediction of opioid prescribing

In the United States, the over prescription of opioids has been a major driver of the opioid abuse epidemic (Compton et al., 2015). Clinicians over prescribe opioids due to several reasons, some of them being lack of awareness about risks, limited access to non-opioid pain management, time constraints and clinical pressure. Having a automated system in place to identify patients in actual need of opioid pain medications could reduce over prescription in clinics.

We were unable to find past research that focused on predicting likelihood of opioid prescriptions based on EHRs. This work therefore makes an important contribution of filling this gap by showing that such models with high performance can be built with an F1 score of 0.88 ± 0.003 and an AUC-ROC score of 0.93 ± 0.002 (Table 4.3).

4.6.2. Prediction of Opioid Use Disorder (OUD)

We first compare the performance of our deep learning model to past research. A summary of this is shown in Figure 4.3. SOAPP-R is a self-report questionnaire used to determine which patients are high-risk for opioid misuse. It is one of the most validated tools for opioid dependence identification (Alzeer et al., 2018), and based on literature there is some variation in sensitivity and

specificity. The sensitivity of SOAPP was reported to range from 67% to 81% and the specificity ranged from 52% to 68% (Butler et al., 2008, 2009; Finkelman et al., 2017). In our OUD model, we achieved a sensitivity score of 0.77 ± 0.1 and specificity score of 0.89 ± 0.03 (Table 4.3). Therefore, our model outperformed SOAPP in terms of specificity and achieved equivalent sensitivity.

In other research (Ellis et al., 2019), a machine learning model was built on EHRs to predict opioid dependence, which obtained a mean area under the receiver operating characteristic curve (AUC-ROC) of 92%. An LSTM model with attention built in (Dong et al., 2021a) to predict overdose risk achieved an F1 score of 0.78 and an AUC-ROC of 0.8449. The OUD model built in this chapter achieved an F1 score of 0.82 ± 0.05 , and an AUC-ROC score of 0.94 ± 0.008 (Table 4.3). Although, we outperformed the prior work in this area in terms of AUC-ROC and F1 score, it is important to note that opioid dependence studies carried out often have variations in study periods, sample sizes, definitions, types of databases, and structured documentations, and therefore some of the differences in model performances built across these studies may make the models not be directly comparable.

Alzeer et al. (2018) conclude that age and gender are the most consistent demographic variables in predicting opioid abuse. The features for OUD in Table 4.4 show similar findings in our experiments, with the top three positive features being age groups, and male gender being positively correlated with OUD. These researchers also noted that certain medication variables were among the most indicative features that occurred across papers that they reviewed (Alzeer et al., 2018). We obtained similar results with medications such as naloxone, acetaminophen-IV and epinephrine (Table 4.4), and methadone and heroin (Table 4.5) being predictive features for OUD. This is also similar to the findings in (Dong et al., 2021a) where the authors mention that the medication class of opioids are among the most important features for OUD prediction. Similar to (Ellis et al., 2019), we found methadone (Table 4.5) and folic acid (Table 4.4) to be present in EHRs of OUD patients.

4.6.3. Our Deep Learning models overcome known limitations in the field

Our models overcome some limitations that researchers studying opioid dependence in EHR data have faced in the past. Previous research relies on domain knowledge from clinical experts for

feature engineering, as mentioned in Dong et al. (2021a). This typically requires a lot of effort and can be quite cumbersome and expensive. With our Deep Learning models, features are learned automatically as part of the training process, providing an inexpensive way of identifying such medical outcomes of interest. Researchers also often construct predictive models on structured parts of the EHRs, often ignoring the unstructured part which can account for 80 % of the total health data (Alzeer et al., 2018). In this paper, we build models that take in information present in both the structured and unstructured parts of the EHRs to make informed decisions by using state-of-the-art models from NLP. Hazlehurst et al. (2019) attempt to use unstructured clinical text to predict opioid related overdoses. However, their approach is expensive as it relies on domain experts to manually develop “rule trees”.

4.6.4. Limitations

The approaches developed in this chapter have several limitations that we acknowledge here. Alzeer et al. (2018) mention that ICD codes can understate the actual number of patients exhibiting the target categorization (Hylan et al., 2015; Allison, 2012) due to an intentional lack of documentation, or in other cases, prescribers trying to avoid stigmatizing patients with opioid problems. As a result, some patients who have an opioid problem may be overlooked. This could result in some False Negatives in our training data hurting model generalizability and leading to potentially more dangerous patient outcomes like overdose and death. Additionally, we only report model performance metrics on a test-set obtained from the same data distribution as the training data. This may only be partly indicative of the model’s performance in a general setting. Ideally, we would also want to evaluate the model on an out-of-distribution test dataset, but due to the difficulty in obtaining another dataset for evaluation, this was not possible for this study but remains a subject of future work. To elaborate, different hospitals and departments within the same hospital often have different formats, templates and writing styles for EHR data, in addition to using abbreviations and medical jargon unique to them (different distributions of data). Therefore, our model could perform well on clinical notes from one hospital while performing terribly on notes from another due to the shift in data distribution. Additionally, as mentioned before, our class-balanced dataset is not representative of the true distribution of OUD patient priors and our numbers are therefore

not directly interpretable for clinicians who want to apply them to a practical setting.

Also, due to population variations and procedural differences between the clinic in which this tool was created and the target clinic where this tool might be adopted, there could be some barriers to its adoption. Another important point to note is that although the Deep Learning models perform better than the respective LogReg baselines across most metrics, the Recall of the Deep Learning models is lower, which might pose a risk in certain cases where not identifying high-risk individuals is death. Further research is required to build models that overcome this drawback. Lastly, Mimic-III is skewed towards critically ill patients as the data is obtained from critical care units.

4.6.5. The Opioid Crisis in America and How Machine Learning could Help Clinicians Identify Patients at Risk

In 2017, 70,200 drug overdose deaths occurred in the USA with the sharpest increases observed among fentanyl and fentanyl analogs (Yeo et al., 2022). The number of opioid prescriptions was 81.3 per 100 people, peaking in 2012 (Chihuri and Li, 2019). Even with efforts to mitigate the steady rise in opioid prescribing, there were still 58.5 prescriptions per 100 people in 2017. The perceived overprescribing of opioids followed by drug-induced overdoses has been termed the ‘opioid epidemic’ (Ford, 2019). Methods for identifying causes of this epidemic and identifying high-risk individuals are needed to guide clinicians’ decision making (Volkow and McLellan, 2016).

Almost 50,000 people in USA have lost their lives due to an opioid related overdose in 2019 alone (NIH). This epidemic not only increases the overall national mortality rate, but also creates an “economic burden” estimated at around \$78.5 billion a year. The opioid epidemic worsened during the 2020 COVID-19 pandemic with a 29% increase in overdoses (Holland et al., 2021). The reason is multi-faceted with contributing factors including the suspension of 12 Step programs due to the requirements of social distancing that forced the suspensions of their meetings. In addition, the economic distress of the pandemic may have exacerbated the effects of the opioid epidemic on already marginalized communities. Therefore, the need to prevent and detect opioid related problems has never been greater.

Electronic Health Records are increasingly being used to perform retrospective analyses to inform clinical care - a critical component of the Learning Health System (Lowes et al., 2017; Smoyer et al., 2016). Retrospective analysis of EHRs can address some of the major challenges of the opioid epidemic by identifying high-risk patient groups and thereby provide much needed information to clinicians. EHRs can also be used to tease apart the multifactorial issues involved with both pain assessment and management, and prescribing behavior involved in the opioid epidemic (Volkow and McLellan, 2016).

CHAPTER 5

Explainable Dementia and Liver Disorder Types with Concept Bottleneck Models

Ensuring model explainability and correct reasoning is an integral part of model development to help promote trust among physicians and patients. Explainable models allow physicians to understand why a prediction was made in a clinically relevant fashion. Neural networks (NNs), which are often used for clinical outcome modeling, have been shown to have high accuracies but their predictions are often opaque and are treated as black-boxes. In this chapter, we argue that it is preferable to have a model that is not only accurate but also explainable, even if that explainability causes a tiny degradation in accuracy compared to a black-box model.

Numerous factors constrain the application of neural network approaches to domains like healthcare. For instance, McCoy et al. (2019) showed that although neural network models like BERT (Devlin et al., 2018) achieved high scores on different tasks, they did so by learning superficial rules that work for most training instances, rather than learning the fundamental principles of the task. Some of the more recent NN models like ChatGPT, GPT-4 and other Large Language Models (LLMs) have been additionally shown to hallucinate when insufficient information is provided (Chen et al., 2023; Liu et al., 2023; Shen et al., 2023). They also tend to reproduce biases and factual inconsistencies in their training data (Liu et al., 2023; Shen et al., 2023). These issues are especially pronounced for the healthcare field where accuracy of information is crucial. Sandmann et al. (2024) evaluated the clinical accuracy of GPT-3.5 and GPT-4 on 110 medical cases across diverse clinical disciplines for suggesting initial diagnosis, examination steps and treatment. They suggest not using these models without medical experts as they lack consistent high accuracies across different tasks and perform poorly on rare diseases. Another study (Casella et al., 2023) found that ChatGPT was unable to address causal relationships between acute respiratory distress syndrome and septic shock. They argue that since ChatGPT was not designed to answer medical questions, it lacks the medical expertise and context needed to fully understand the complex relationships between different conditions and treatments.

In this chapter, we keep these limitations in mind and formulate a task for predicting two clinical outcomes for a patient using clinical notes from their Electronic Health Records (EHRs): (i) Dementia type diagnosis, and (ii) Liver disorder diagnosis. We created a prediction algorithm that ensures the model looks at clinical features most relevant to the outcome of interest, thereby minimizing the likelihood of spurious correlations and hallucinations in the prediction process. We draw inspiration from Concept Bottleneck Models (CBMs) (Koh et al., 2020a) where the model is forced to focus on a set of intermediate generated concepts that are relevant to the outcome. Rather than having a Neural Network perform end to end image classification, Koh et al. (2020a) have experts decide important features and use a Neural Network to predict whether they are active in an image. Finally, they train a linear layer on these image concepts, which allows for identifying the most important features. In the past, researchers have combined deep neural networks with concept bottlenecks to solve a variety of tasks, such as retinal disease diagnosis (De Fauw et al., 2018), visual question answering (Yi et al., 2018) and content based image retrieval (Bucher et al., 2019). These models bottleneck on human-specific concepts where the model first predicts the concepts, then uses the predicted concepts to make a final prediction.

Yang et al. (2023) provided an approach named **Language Guided Bottlenecks** (LaBo), which leveraged a language model GPT-3 to define a large space of possible bottlenecks. Given a problem domain, LaBo used GPT-3 to first produce factual sentences about categories to form candidate concepts. LaBo then efficiently searched possible bottlenecks through a novel sub-modular utility that promoted the selection of discriminative and diverse information. Next, GPT-3’s sentential concepts were aligned to images using a pre trained alignment model CLIP, to form a bottleneck layer. Finally, they trained a linear function to predict image labels based on concept image scores. Yang et al. (2023) evaluated LaBo on 11 diverse datasets showing that it excels at few-shot classification and are 11.7% more accurate than black-box linear probes at 1 shot and comparable with more data.

We provide a novel method that is based on extending LaBo for extracting clinically relevant features for CBMs using the Oxford Textbook of Medicine (Warrell, 2003) and GPT-4. This allows

us to extract patient features related to dementia and liver disorders from their clinical notes. An example of an extracted feature using our approach is *“The patient exhibits poor memory.”* which is a human-readable sentence making our CBM approach interpretable. In order to predict the dementia type of patients, we first convert their clinical notes into a concept vector representation. Next, we train a linear classifier on this representation to predict patient dementia type. Finally, we analyze the weights of the linear model to provide insights into the models decisions as well as the most important clinical features for dementia type and liver order prediction prediction.

The contributions of this chapter are as follows:

1. We develop a novel method of extracting patient clinical features related to dementia and liver disorders from the Oxford Textbook of Medicine (Warrell, 2003).
2. We develop a Concept Bottleneck Model that converts patient notes into concept vector representations, which we use to train a linear interpretable model on top to analyze important features.
3. Our approach for dementia type prediction obtains an accuracy of 0.72 on a held out test-set which beats an ngram Logistic Regression baseline (0.64) and a GPT-4 baseline (0.48) while being more interpretable.
4. Our approach for liver disorder prediction only beats the GPT-4 baseline in 1 out of 4 labels and the ngram Logistic Regression baseline in 2 out of the 4 labels, but is more easily interpretable.

5.1. Background

Despite the promising performance of Deep Learning networks in various domains, their adoption in healthcare has been slow. One reason for this is due to limited insight into the inner workings of Artificial Intelligence (AI) tools is required for it to become clinically successful (Salahuddin et al., 2022). Moreover, unlike consumer applications like ChatGPT, medical applications have a higher bar for approval and are governed by strong regulatory compliance laws (Temme, 2017). While

LLMs have shown impressive performance, some of the most popular models like ChatGPT and GPT-4 are not open source, limiting transparency and the ability of researchers to mitigate biases and hallucinations (Wang et al., 2023b).

Interpretability and explainability have often been used synonymously in the field of explainable AI (Adadi and Berrada, 2018; Chakraborty et al., 2017) and have four primary dimensions along which they are defined (Luo et al., 2024): (i) Faithfulness: Measures how well an interpretation method describes the decision making process used by the underlying model. (ii) Stability: An interpretation method that provides similar explanations for similar inputs is considered stable. (iii) Comprehensibility: Measures how understandable an interpretation to an end-user is. (iv) Trustworthiness: Interpretations can be called trustworthy only when the experiment design is carefully considered, in that their results can only be relied upon if carefully controlled.

Researchers have identified important input features that significantly impact a model's prediction using different techniques. Rationale Extraction involves extracting phrases from the original textual inputs that represent critical features that influence the output (Li et al., 2018; Bastings et al., 2019; Zellers et al., 2021). Input perturbations modify or remove up to a few words in the original input to measure a performance change (Ribeiro et al., 2020; Rudin, 2018; Basaj et al., 2018; Yordanov et al., 2021). Alternatively, attention weights which are weighted sum scores of the input representation in the intermediate layers of neural networks are measured (Luong et al., 2015; Vedantam et al., 2015; Bahdanau et al., 2014; Yu, 2013). Finally, attribution methods examine the gradients of a model to identify the most important features (Tutek and Šnajder, 2018; Hendricks et al., 2018). Other approaches generate Natural Language Explanations (NLE) in addition to the outcome of interest (Camburu et al., 2018; Kumar and Talukdar, 2020; Rajani et al., 2019). Camburu et al. (2019) showed that some of the explanations with these approaches were inconsistent with the predicted outcomes, hinting at either an incorrect explanation or a flawed model decision-making process.

In this chapter, we draw inspiration from Concept Bottleneck Models (Koh et al., 2020a), which rely on the idea of generating high-level task-related concepts during training time, and then using

these concepts to predict a label. For example, a bird identification task might involve high level concepts such as *“beak length”* or *“wing color”*, while for a knee x-ray grading task, concepts could include *“bone spurs”* or *“narrow joint space”*. Finally, a simple linear model is trained on these intermediate concepts, and the most important features are obtained by examining the model weights. Concept Bottleneck models have shown competitive performance while supporting intervention and interpretation. However, they require expert annotated concepts at training time which might be expensive if manually done. Sawada and Nakamura (2022) improves on this approach by combining supervised and unsupervised concepts in the training stage, thereby increasing the number of concepts and overall performance. For an image classification task, Yang et al. (2023) use GPT-3 to generate candidate concepts followed by a sub-modular function to select concepts to construct the bottleneck, thereby not relying on manual annotation.

5.2. System Design

An illustration of the overall algorithm is shown in Figure 5.1. At a high level, we first use GPT-4 to extract patient concept features from the Oxford Textbook of Medicine. After manually annotating the concept features for quality, we obtain a set of 254 concept features related to dementia and 290 clinical features related to liver disorders. Next, we convert the patient clinical notes into a concept bottleneck vector representation using an agreement model. The agreement model is an LLM that takes an input pair consisting of a patient note and a concept feature and outputs either a “Yes”, “Uncertain” or “No” depending on whether the patient exhibits the corresponding feature in their note. Finally, we train a linear classification model on the concept vector representation of the clinical notes along with the gold labels (representing the dementia/liver disorder type) and evaluate the model on a held out test set. A more detailed description of each step is discussed in the following sections.

In the first part of this chapter, we focus on developing a concept bottleneck pipeline for predicting patient dementia type using their clinical notes. In the second part, we extend our approach to the task of predicting liver disorders.

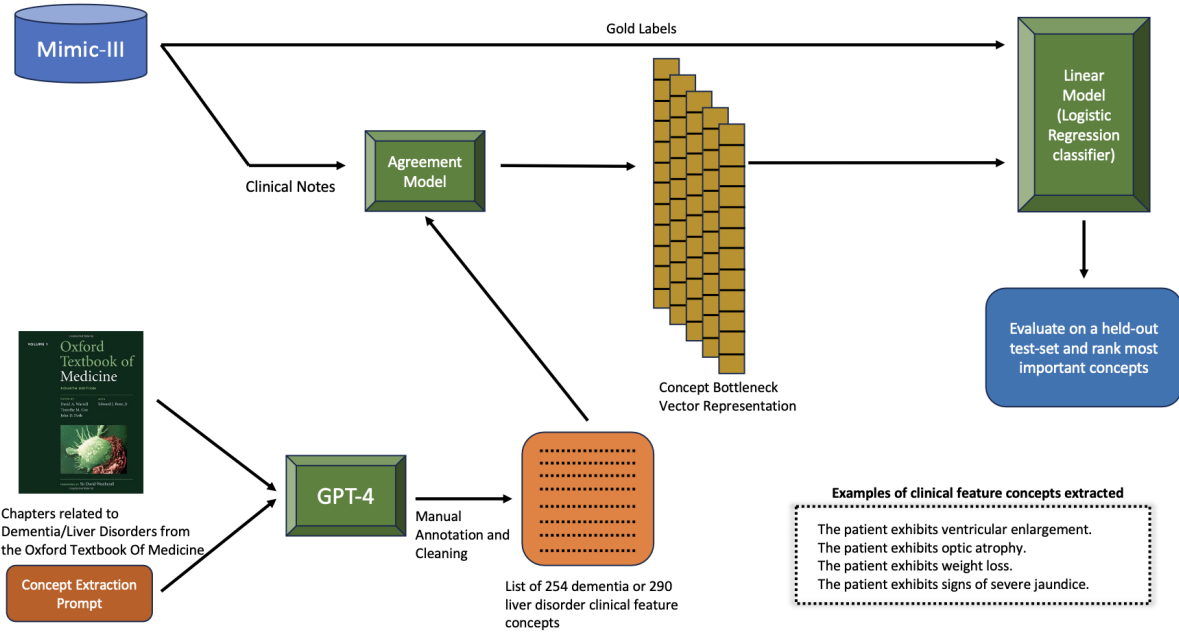


Figure 5.1: An Overview of Dementia and Liver Disorder Type Classification of Patient Clinical Notes using the Concept Bottleneck Approach.

5.2.1. Dementia Classification Datasets

We use MIMIC-III (Johnson et al., 2016b) to create a data-set of clinical notes from patient admissions containing 3 different dementia types and a control group using a set of ICD-9 codes. The distribution of this dataset is shown in Tabel 5.1. For each patient admission, we select only the discharge summary note. Additionally, we remove the phrases “*dementia*”, “*alzheimer’s*”, “*vascular dementia*” and other variants from the notes to prevent information leakage.

Dementia Label	Number of Patient Admissions
Vascular Dementia	213
Alzheimer’s	1087
Other Dementia	2650
No Dementia (Control Group)	3500

Table 5.1: Label Distribution of the Dementia Dataset Created from MIMIC-III.

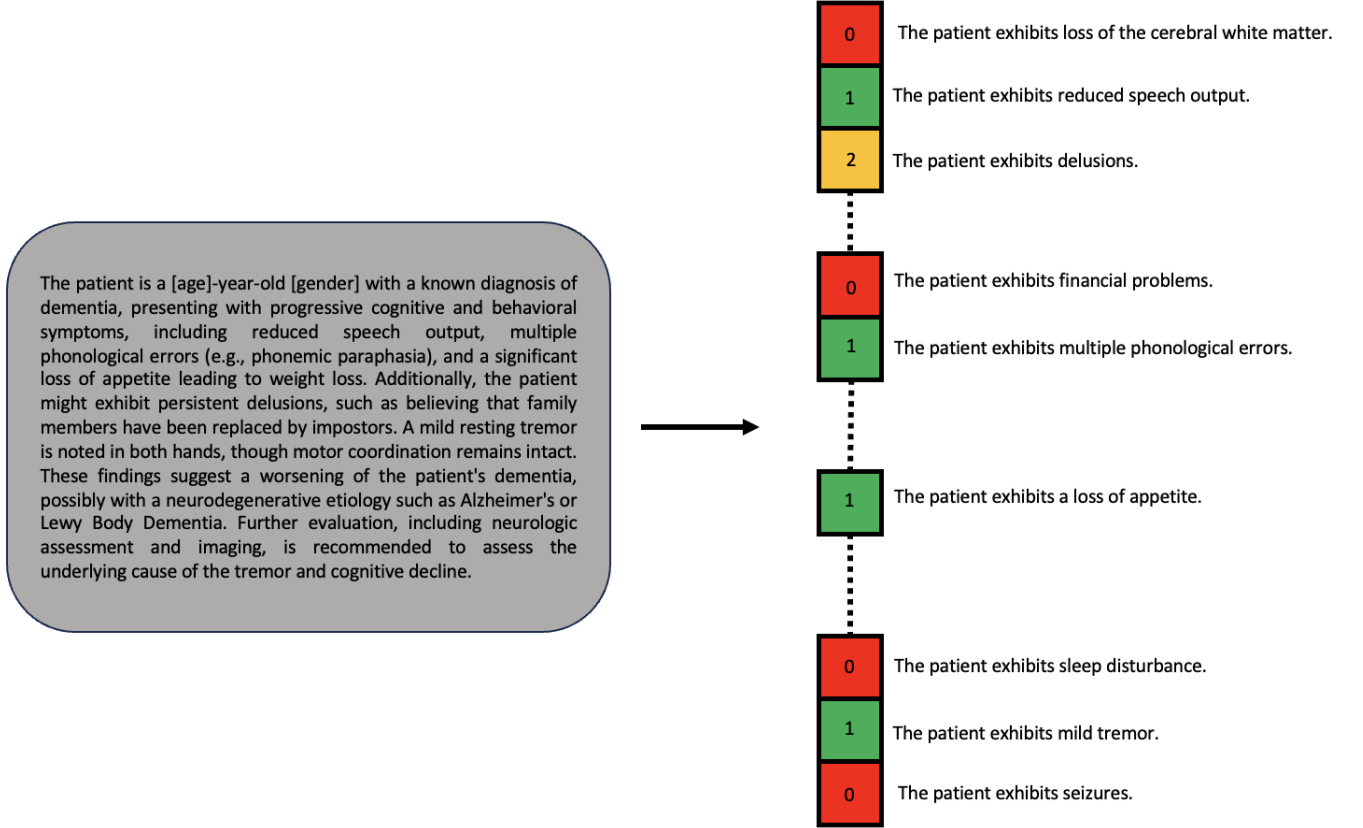


Figure 5.2: An illustrative example of the initial goal of our approach, mapping a clinical note to a concept bottleneck vector representation. *0* represents the feature is *absent*, *1* represents the feature is *present* and *2* represents the presence of the feature is *uncertain*.

5.2.2. Creating Concept Features for Dementia Type Classification

Our initial goal is to represent a patient clinical note in the form of a concept bottleneck vector where every index of the vector represents a unique atomic clinical feature related to the outcome of interest, and the values represent either a “No”, “Yes” or “Uncertain” depending on whether the feature was expressed in the clinical note or not. Figure 5.2 provides an example of a clinical note of a patient with dementia mapped onto a concept bottleneck vector.

We extract features related to Dementia from the Oxford Textbook of Medicine (Warrell, 2003) using GPT-4 (Achiam et al., 2023). For the dementia task, we first identify the chapter *Alzheimer’s Disease and Other Dementias* and extract paragraphs under the *Clinical Features* sub-heading. Next, we use GPT-4 to extract clinical features from the textbook paragraphs using the **Concept**

Extraction Prompt shown below. The **condition** in the prompt refers to the type of dementia described in the textbook (Frontotemporal Dementia, Vascular Dementia, etc.). Examples of input and outputs for this prompt are shown in the Appendix A.1.

Concept Extraction Prompt:

*Rewrite the following description about **{condition}** as multiple bullet points in the format "The patient experiences ...". Be as exhaustive as you can.'*

{textbook description}

The patient experiences:

An example of an extracted sentence using the **Concept Extraction Prompt** is “*The patient exhibits agitation, restlessness, wandering, and disinhibition.*” This example contains multiple clinical features, but we ideally want each concept feature to be an atomic clinical feature. We use GPT-4 to achieve this by using the **Concept Simplification Prompt** which contains 2 input-output examples for the task (2-shot learning). Brown et al. (2020) showed that LLMs are proficient at in-context learning and that their few-shot performance is often much higher compared to the zero-shot setting. The example containing multiple clinical features mentioned above is broken into the following concept features containing atomic clinical features. So the single sentence “*The patient exhibits agitation, restlessness, wandering, and disinhibition.*” is broken down into:

1. The patient exhibits agitation.
2. The patient exhibits restlessness.
3. The patient exhibits wandering.
4. The patient exhibits disinhibition.

Concept Simplification Prompt:

Rewrite this sentence as multiple simple sentences beginning with "The patient exhibits" one per line.

Sentence: The patient exhibits cognitive slowing plus impairment of executive (planning and organizational abilities) and visuoperceptual abilities.

Output:

The patient exhibits cognitive slowing.

The patient exhibits impairment of executive (planning and organizational) abilities.

The patient exhibits impairment of visuoperceptual abilities.

Sentence: The patient exhibits ventricular enlargement disproportionate to the degree of cortical atrophy.

Output: The patient exhibits ventricular enlargement.

The patient exhibits cortical atrophy.

The patient exhibits ventricular enlargement disproportionate to the degree of cortical atrophy.

Sentence:{sentence}

Output:

We obtained 294 unique concept features for dementia which we then manually annotated for quality. Some of the extracted concept features don't make sense due to the way we force the output to be formatted ("The patient exhibits thalamus" and "The patient exhibits perfusion MRI, MRS, and PET scans may enhance diagnostic accuracy. "). These were excluded from the final concept feature

list. We also rephrased certain concept features to exhibit proper grammar (*"The patient exhibits complaints of 'loss of memory for words'."* was changed to *"The patient complains of a loss of memory for words."*) and added additional information to reduce ambiguity (*"The patient exhibits inconsistency in test performance."* was changed to *"The patient exhibits inconsistency in MMSE test performance."*). We finally end up with 254 dementia concept features (Table A.1) that we use for the following experiments where we predict patient dementia type using clinical notes from their Electronic Health Records.

5.2.3. Converting Clinical Notes to Concept Feature Vectors using Agreement Models

The goal of this task is to convert a patient clinical note into a vector representation where the index corresponds to one of the 254 unique concept features extracted, and the value at that index corresponds to either a "Yes", "No" or "Uncertain" depending on whether that feature was present or absent (Figure 5.1). To evaluate this task, we pick 10 dementia concept features and randomly sample 20 clinical notes for each feature. We manually annotate this set containing (*patient note*, *concept feature*) pairs to evaluate the subsequent models used (refer to Table 5.6 for the selected concept features).

Using GPT-4 to extract Concept Features from Patient Notes

We provide a prompt containing (*patient note*, *concept feature*) pairs to GPT-4. An example of the prompt for a specific concept feature (Optic Atrophy) is shown in the **Patient Note Concept Activation Prompt**. The definitions for the concept features were generated by GPT-4 on a separate run.

Patient Note Concept Activation Prompt:

SYSTEM PROMPT:

You are a clinical research assistant. For the given clinical note, answer the question ("Concept Question") provided in the dictionary. Use "Definition", "Possible Responses", "Response Guide" and "Additional Response"

to help better answer the question. Your response should be of the following format: "Response": Either 'Yes', 'No' or 'Uncertain', "Supporting Information": If "Response" is either "Yes" or "Uncertain", provide supporting sentences from the clinical note. If "Response" is 'No', this should be "None"

CHAT HISTORY: *clinical_note*

USER MESSAGE:

Feature Name: *Optic atrophy*

Concept Question: *Can the following sentence be inferred from the clinical note?: 'The patient exhibits optic atrophy.'*

Definition: Optic atrophy in a clinical setting refers to the damage or degeneration of the optic nerve, which transmits visual information from the eye to the brain. This often results in a variety of visual symptoms, including blurred vision, difficulties with color vision, and reduced visual field. The causes of optic atrophy can include glaucoma, optic neuritis, stroke or trauma to the optic nerve. Optic atrophy is typically diagnosed through comprehensive eye exams and imaging studies. If left untreated, it can lead to permanent loss of vision, hence prompt diagnosis and management is essential.

Possible Responses: *Yes, No, Uncertain*

Response Guide: *Yes: The clinical note explicitly mentions that the patient has the clinical feature. No: The clinical note does not explicitly mention*

that the patient has the clinical feature and there is also no information that might suggest the patient has the clinical feature. Uncertain: The clinical note contains some information that might suggest the patient has the clinical feature (through correlations), but it is not explicitly stated

Example outputs for this prompt are shown in Table 5.2. For every patient note, we ran this for each of the 254 dementia concept features, obtaining a “yes”, “no” or “uncertain”. In this manner, we built a concept vector representation of each note which we then used to train a linear classification model on to predict dementia type for a particular patient.

Feature Name	Response	Supporting Information
Optic atrophy	No	None
Optic atrophy	Uncertain	Ophthalmology was consulted and they recommended following her decreased visual acuity in her R. eye. Bone fragment seen in R optic strut concerning for injury to the adjacent optic nerve.
Optic atrophy	Yes	28yo F w/ diabetes insipidus, diabetes mellitus, optic atrophy and deafness and a history of multiple pneumonias...

Table 5.2: GPT-4 Outputs over Different Responses for the Concept Feature “*Optic Atrophy*” using the Patient Note Concept Activation Prompt.

Reducing the Number of Calls to LLMs using Text Embeddings for Extracting Concept Features from Patient Notes

A major technical challenge that we encounter is the number of calls we have to make to GPT-4 to annotate all combinations of (patient note, concept feature) pairs. Our dementia dataset includes 7,450 patient notes with 254 concept features, having a total of 1.9 million (7,450 x 254) note-concept pairs and 7.9 billion tokens. To annotate the dataset using our access to GPT-4, it took over 50 days. Additionally, the cost of using the synchronous GPT-4 API as of June, 2024 for our dataset containing 7.9 billion input tokens and 59 million output tokens is \$240,540 (\$30/1M input tokens and \$60/1M output tokens).

In order to reduce the processing time and cost, we investigate the use of text embeddings to

minimize the number of note-concept pairs annotated by GPT-4. Rather than annotate all the pairs, we want GPT-4 to only annotate notes that are likely to have the corresponding concept activated. This is based on the hypothesis that some concepts are less common than others, occurring in only a fraction of the patient notes.

An overview of our text embedding approach is shown in Figure 5.3. We first tokenize the clinical notes into sentences using Stanza (Qi et al., 2020) and embed each note sentence and each concept feature using the OpenAI embedding models and INSTRUCTOR models (Su et al., 2022). We embed the concept features along with the corresponding definition generated by GPT-4 as mentioned in Section 5.2.3.

INSTRUCTOR (**I**nstruction-based **O**mnifarious **R**epresentations) (Su et al., 2022) is a single multitask model that generates task and domain-aware embeddings given a text input and its task instructions. Examples of task instructions mentioned in the paper include “*Represent the Medicine statement for retrieval.*”, “*Represent the Tweet post for retrieving duplicate comments.*” and “*Represent the Review sentence for classifying emotion as positive or negative.*”. We embed each patient note sentence after prepending the phrase “*Represent the clinical note sentence for patient feature identification.*” and every concept feature after prepending the phrase “*Represent the patient feature for clinical note feature identification.*”.

Next we perform pairwise cosine-similarity and rank notes containing sentences with highest similarity values for each concept feature. Finally, we select the top-k clinical notes for each concept feature and prompt GPT-4 as mentioned in Section 5.2.3. We assume that for a particular concept feature, the notes not in the top-k do not contain it.

Since smaller note chunk sizes lose information about surrounding context, we experiment with varying note chunk sizes (chunk sizes containing 1,2,3,4 and 5 sentences). We also try 2 different sampling approaches for selecting the top-k notes. In the first sampling approach, we consider a constant top 10% of clinical notes with sentences having the highest cosine similarity values for every concept feature. However, since some concept features occur at a higher frequency than

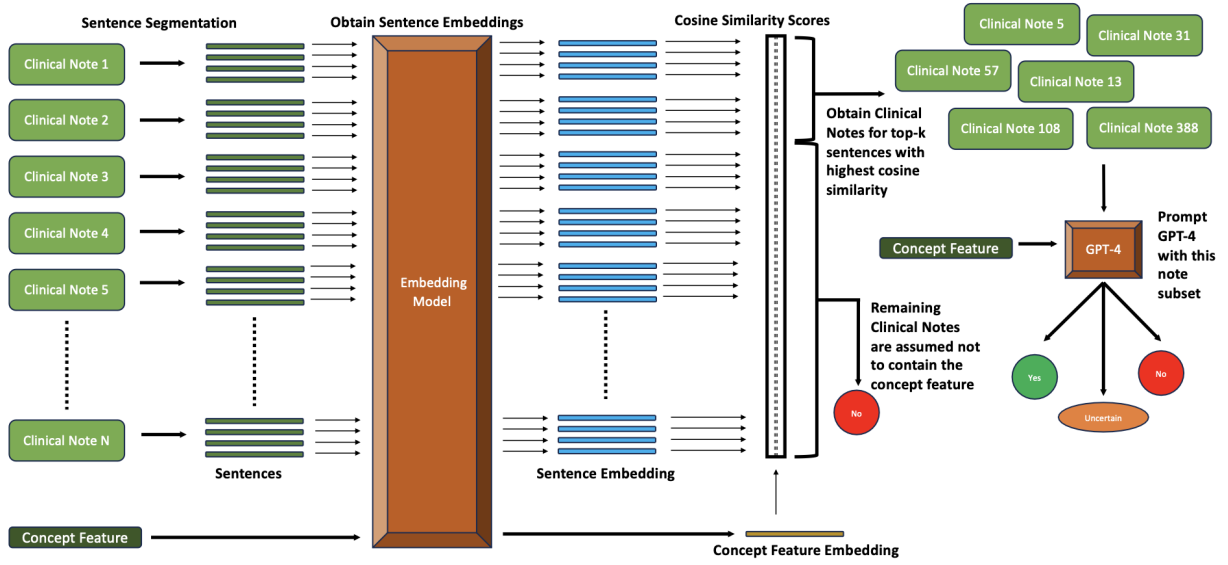


Figure 5.3: Conceptual Diagram Illustrating the Text Embedding Approach with GPT-4 for Identifying the Presence of Concepts in Patient Notes.

others, selecting a constant top-k (10%) for every concept feature might not be ideal. As a result, in the second sampling approach, we first select a random subset containing 300 patient notes and use GPT-4 to predict the presence of each concept feature. Based on the frequency of these concept features in the 300 note subset, we extrapolate to determine the approximate number of patient notes in the entire dataset that will contain a concept feature. For example, if the concept feature “*The patient exhibits multiple lesions*” occurs 60 times in the 300 note subset, then roughly 1,500 patient notes will contain it in the entire 7,450 dementia dataset.

Using Llama2 to extract Concept Features from Patient Notes

Despite GPT-4’s high performance across various tasks, their limited accessibility and high costs make them less attainable for individuals and smaller organizations. As mentioned earlier, the cost of using the synchronous GPT-4 API as of June, 2024 for our dataset containing 7.9 billion input tokens and 59 million output tokens is \$240,540 (\$30/1M input tokens and \$60/1M output tokens). OpenAI also provides an asynchronous batch API option for GPT-4 which is 50% of the synchronous API price and ideal for tasks where timing isn’t critical. Although the cost of the GPT models is

trending downwards, open source models are free to use.

Therefore, we explore finetuning Llama2 for this task through knowledge distillation using a subset of GPT-4’s outputs. Knowledge Distillation is a method for transferring advanced capabilities from leading proprietary LLMs such as GPT-4 to their open source counterparts like Llama (Xu et al., 2024). Researchers have used different distillation techniques, closing the performance gap between proprietary and open source models (Chiang et al., 2023; Xu et al., 2023).

We first use GPT-4 to create a training set of 125,000 (patient note, clinical feature) pairs with the label (Yes, No, Uncertain). Next, we utilize these examples to train a 7B and 13B Llama2 model (Touvron et al., 2023) using LoRA (Hu et al., 2021), 4-bit precision and gradient accumulation, and evaluate their performances on the manually annotated test set. As the Llama2 models used have fewer parameters (7 Billion, 13 Billion) compared to GPT-4 (over 1 Trillion), we expect the performance of these open source models to be lower.⁴

5.2.4. Dementia Type Classification

We train models on different representations of the patient notes to predict the dementia type (Table 5.1). We divide our dementia dataset containing 7,450 patient admissions into a train/val/test (0.9/0.05/0.05) split using which we train and evaluate these models.

Dementia Logistic Regression Baseline using ngrams

As a baseline, we train a Logistic Regression Model using ngram representations of the clinical notes which contain unigram, bigram and trigram features and predict the patient dementia type (Table 5.1). Each element in the ngram vector representation is binary, representing the presence or absence of a corresponding ngram feature.

Dementia GPT-4 Baseline

We provide the patient clinical notes to GPT-4 with the **GPT-4 Dementia Baseline Prompt** and expect one of the 4 label outputs: *no dementia*, *dementia*, *alzheimer’s dementia* or *vascular*

⁴Open source models have also made progress subsequent to our experiments. Llama 3 (Dubey et al., 2024) created by Meta AI is a 405B parameter model which has shown comparable performance to GPT-4 on a variety of tasks.

dementia.

GPT-4 Dementia Baseline Prompt:

For the given patient clinical note, choose one of the following diagnosis labels:

no dementia

dementia

alzheimer's dementia

vascular dementia

You should only reply with the diagnosis label and nothing else.

Our Concept Bottleneck Approach for Predicting Dementia type for Patients

Using the predictions from the agreement model across the entire Dementia dataset, we convert patient clinical notes into concept bottleneck vectors where each vector element represents a unique concept feature. Next, we train a Logistic Regression model on these feature vectors and predict the dementia label type (Table 5.1). We also evaluate a Gradient Tree Boosting model in place of Logistic Regression (for more information refer to Appendix A.3).

5.3. How well does GPT-4 Identify Clinical Features in Patient Notes?

In this section, we will first discuss the results for different approaches to build concept vectors from patient notes using agreement models. Next, we will discuss the final classification task of predicting dementia type and liver disorders using a linear interpretable model.

5.3.1. Creating Concept Feature Vectors using Agreement Models

The agreement model is an LLM that takes an input pair consisting of a patient note and a concept feature, and outputs either a “Yes”, “Uncertain” or “No” depending on whether the patient exhibits

the corresponding feature in their note. We evaluate the agreement model using the manually annotated test set mentioned in Section 5.2.3.

Using GPT-4 to extract Concept Features from Patient Notes

We first evaluate the performance of using only GPT-4 on the manually annotated test set of (patient note, concept feature) pairs. As mentioned previously, the test set contains 10 dementia concept features and 20 clinical notes for each feature randomly selected (refer to Table 5.6 for the list of concept features). We run 2 different sets of experiments, one where we prompt GPT-4 (**Patient Note Concept Activation Prompt**) without the *Definition* and the other where the *Definition* is included. The results on the manually annotated test set is shown in Table 5.3 where we also report a random baseline and a majority baseline for comparison. The fine-tuned Llama2 models (Table 5.3) perform significantly worse compared to the majority baseline and are therefore less efficient even with lower costs compared to GPT-4.

Approach	Concept Accuracy	Concept Precision	Concept Recall	Concept F1
Random Baseline	0.31	0.45	0.31	0.35
Majority Baseline	0.65	0.42	0.65	0.51
Llama2 - 7B	0.43	0.62	0.43	0.38
Llama2 - 13B	0.47	0.68	0.48	0.5
GPT-4 (prompt does not contain concept definition)	0.85	0.86	0.82	0.83
GPT-4 (prompt contains concept definition)	0.86	0.87	0.86	0.84

Table 5.3: Concept Performance Metrics of Agreement Models Compared to a Random and Majority Baseline on the Manually Annotated Test-set for the Task of Predicting Concept Presence in a Patient Note.

What is the Effect of Reducing the Number of GPT-4 Calls?

For the Text Embedding approach, we first evaluate the different embedding models for the dementia type prediction task using a note chunk size of 1-sentence and a constant top-k sampling of 10% (we select the top 10% clinical notes containing sentences with the highest cosine similarity for every concept feature). The results for these experiments on the manually annotated test set are

shown in Table 5.4. From the results, the OpenAI *text-embedding-3-large* model outperforms the other models by a significant margin. Next, we evaluate the 2 different top-k sampling approaches mentioned in the previous section (Section 5.2.3) along with different note chunk sizes, shown in Table 5.5. The top-k sampling approach using extrapolation improves the overall performance on the test-set suggesting that in a significant portion of the (note, concept) pairs, the absence of a concept in a note can be approximated by the cosine similarity of their embeddings. Additionally, we plot the cosine distance threshold against both the test-set accuracy and the dataset fraction annotated by GPT-4, which is shown in Figure 5.4. Setting a distance threshold of 0.64 achieves an accuracy of 0.86 on the test-set while requiring only 3% of the entire dataset annotated by GPT-4. With the Text Embedding approach, we can obtain a comparable performance to using GPT-4 on the entire dementia dataset, while only requiring a fraction of the GPT-4 calls. Using this approach, we reduce the cost of our overall approach from \$240,540 to \$7,276.

Approach	Embedding Model	Concept Accuracy	Concept Precision	Concept Recall	Concept F1
Majority Baseline	-	0.65	0.42	0.65	0.51
GPT-4	-	0.86	0.87	0.86	0.84
Text Embeddings + GPT-4	instructor-base	0.71	0.67	0.71	0.63
Text Embeddings + GPT-4	instructor-large	0.72	0.71	0.72	0.65
Text Embeddings + GPT-4	instructor-xl	0.71	0.68	0.71	0.63
Text Embeddings + GPT-4	text-embedding-ada-002	0.72	0.69	0.72	0.65
Text Embeddings + GPT-4	text-embedding-3-small	0.72	0.71	0.72	0.64
Text Embeddings + GPT-4	text-embedding-3-large	0.73	0.69	0.73	0.66

Table 5.4: Concept Performance Metrics of the Text Embeddings Approach with GPT-4 using Different Embedding Models.

Error Analysis for Concept Identification in Clinical Notes

Among all the approaches, using GPT-4 to annotate the entire dataset performs best on the manually annotated test-set with an accuracy of 0.86. A more detailed analysis of GPT-4’s performance across different concepts in the test-set is shown in Table 5.6. The confusion matrices of GPT-4 for the task of predicting the presence of a concept in a clinical note is shown in Figure 5.5, including

Top-k Sampling Approach	Number of sentences in note chunk	Concept Accuracy	Concept Precision	Concept Recall	Concept F1
Constant (10%)	1	0.73	0.69	0.73	0.66
Varying (using extrapolation)	1	0.81	0.83	0.81	0.76
Varying (using extrapolation)	2	0.82	0.84	0.82	0.78
Varying (using extrapolation)	3	0.78	0.80	0.78	0.73
Varying (using extrapolation)	4	0.81	0.83	0.81	0.78
Varying (using extrapolation)	5	0.78	0.72	0.78	0.74

Table 5.5: Evaluating Different Top-k Sampling Methods and Different Note Chunk Sizes for the Text Embeddings with GPT-4 Approach on the Manually Annotated Test-set.

results for (a) the entire test dataset, (b) across the single concept “*The patient exhibits forgetfulness.*”, and (c) across the single concept “*The patient exhibits substance dependence*” . For the individual concepts (b) and (c), we manually annotate an additional set of notes to ensure statistical significance in the confusion matrix.

For certain concepts like “*The patient exhibits substance dependence*”, physicians often explicitly state the presence/absence of the corresponding concept in the note (for example “*patient does not smoke and no EtOH⁵ use*” and “*patient drinks 2 beers a day....smokes marijuana daily..*”), making it easier for GPT-4 to arrive at the correct answer with little to no room for uncertainty. While GPT-4 performs well on most concept features in the test set, some features like “*The patient exhibits forgetfulness*” and “*The patient exhibits bladder dysfunction*” perform relatively poorly. Manually reviewing some of the misclassified examples (shown in Table 5.7), we find that GPT-4 fails to draw logical references and has a tendency to over-estimate or under-estimate the correlation between a concept feature and information provided in the clinical notes (also depicted by the confusion matrices in Figure 5.5 where an “*Uncertain*” gold label is often predicted as “*No*” or “*Yes*”), while in some cases completely missing vital information. For example, if a patient was stated to be

⁵EtOH is a scientific abbreviation for ethanol, the type of alcohol found in alcoholic beverages like beer, wine, and liquor

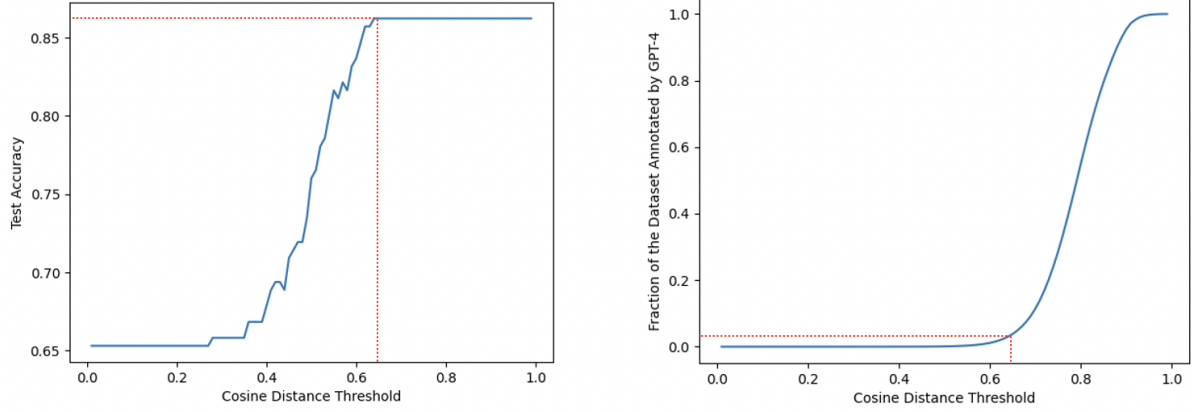


Figure 5.4: Cosine Distance Threshold versus the Held-out Test-set Accuracy and the Fraction of the Dataset Annotated by GPT-4 for the Text Embedding Approach. The Dotted Red Line indicates the Cosine Distance Threshold at which Maximum Test-set Accuracy is First Achieved and the Corresponding Dataset Fraction needed to be Annotated by GPT-4.

Concept Feature	Con- cept Accu- racy	Concept Precision	Concept Recall	Concept F1	$n/20$ pa- tients
The patient exhibits fluctuating cognitive performance	0.85	0.81	0.85	0.83	9
The patient exhibits difficulty answering questions	0.79	0.68	0.79	0.72	4
The patient exhibits reduced speech output	0.94	0.9	0.95	0.92	4
The patient exhibits forgetfulness	0.5	0.63	0.5	0.47	14
The patient exhibits bladder dysfunction	0.85	0.76	0.85	0.80	4
The patient exhibits white-matter changes	0.90	0.86	0.90	0.88	3
The patient exhibits delirium	0.89	0.85	0.89	0.87	7
The patient exhibits a loss of libido	1.0	1.0	1.0	1.0	0
The patient exhibits substance dependence	1.0	1.0	1.0	1.0	5
The patient exhibits low energy	0.89	0.92	0.89	0.88	9

Table 5.6: Performance Metrics Across Different Concept Features while using only GPT-4 on the Manually Annotated Test-set. The Last Column Represents the Number of Patients that Exhibit the Clinical Feature out of the 20 Manually Annotated Examples.

Concept Feature	Gold Label	Predicted Label	Explanation
The patient exhibits forgetfulness	Uncertain	Yes	GPT-4s explanation: <i>"The clinical note mentions: 'You were disoriented during your stay and we had to utilize soft restraints Mental status is partially oriented...' These instances suggest disorientation and potential issues with memory, which could be indicative of forgetfulness."</i>
The patient exhibits forgetfulness	Uncertain	Yes	GPT-4s explanation: <i>"Pt did not know her home medications"</i> .
The patient exhibits low energy	Uncertain	No	The clinical note includes the following information that might correlate with low energy: <i>"The patient is a 74-year-old ... who presented to the Emergency Department with a fever of 104, malaise, nausea, and vomiting..."</i> . This was missed by GPT-4
The patient exhibits difficulty answering questions	Yes	No	The clinical note includes the following information that might correlate with low energy: <i>...Upon awakening and extubation, patient was slow to answer questions and mildly inattentive (but this improved next day). Mental status was otherwise normal.</i> This was missed by GPT-4.
The patient exhibits delirium	Yes	No	The clinical note contains the following information that might suggest delirium: <i>"Pt is oriented to place 'hospital', but not year. She was not sure why she is here. Per family, pt does have some mci and this is baseline for pt."</i> . This was missed by GPT-4.

Table 5.7: Misclassified GPT-4 Examples for Concept Prediction from Patient Clinical Notes.

disoriented, GPT-4 predicted that the patient exhibited forgetfulness which is not always an outcome that co-occurs with disorientation. Similarly, a patient who does not know her home medications is not necessarily forgetful but rather, could be ignorant of the fact. There were some examples where GPT-4 missed important information in the patient notes. For instance, a 74-year-old with a fever of 104, malaise, nausea and vomiting is very likely to exhibit low energy while another patient with mild cognitive impairment (MCI) who is oriented to place but not year is very likely to exhibit delirium (Table 5.7). While the overall performance on the test set was high, the examples in Table 5.7 highlight some weaknesses in GPT-4s comprehension of clinical text.

		Predicted		
		No	Uncertain	Yes
Actual	No	122	0	6
	Uncertain	12	3	4
	Yes	5	0	44

(a)

		Predicted		
		No	Uncertain	Yes
Actual	No	19	1	3
	Uncertain	4	1	6
	Yes	1	1	12

(b)

		Predicted		
		No	Uncertain	Yes
Actual	No	43	0	0
	Uncertain	0	1	1
	Yes	2	0	3

(c)

Figure 5.5: GPT-4 Confusion Matrices for the Task of Predicting if a Concept is Present in a Clinical Note (a) Across all Concepts in the Manually Annotated Test-set, (b) For the Concept “*The patient exhibits forgetfulness.*”, (c) For the Concept “*The patient exhibits substance dependence.*”

5.4. How effective are the Extracted Clinical Features for Dementia Classification based on Patient Notes?

After we construct concept vector representations for every note, we use this representation to predict what type of dementia (if any) a patient has and compare the performance of this approach to 2 different baselines. We divide our dementia dataset containing 7,450 patient admissions into a train/val/test (0.9/0.05/0.05) split to train and evaluate the models.

We train a baseline Logistic Regression model on an ngram (unigram, bigram and trigram) representation of the patient notes and predict the type of dementia label. The top features ⁶ for each label is shown in Table 5.8 for ngram-Logistic Regression and Table 5.9 for our Concept Bottleneck approach. We additionally evaluate GPT-4 as a baseline for this task. The performance of both these baselines and our concept bottleneck approach on a held out test set is shown in Figure 5.6. Our concept bottleneck approach significantly outperforms the other baselines across both accuracy and F1 scores. Our concept bottleneck approach achieves an F1 score of 0.70 compared to 0.63 for the ngram-Logistic Regression baseline and 0.40 for the GPT-4 baseline. The confusion matrix across the different classes for our approach is shown in Figure 5.7.

⁶We multiply the standard-deviation of every feature with their respective Logistic Regression coefficients/weights. Based on this value, we sort features and rank their importance.

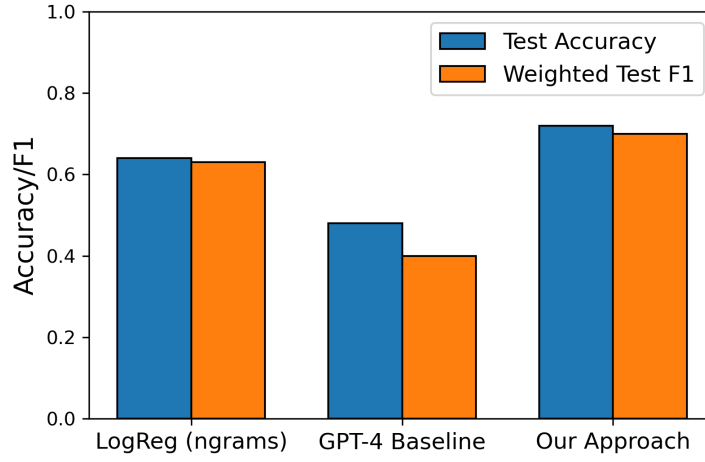


Figure 5.6: Accuracy and F1 Scores on a Held-out Test-set for 2 Baselines and our Concept Bottleneck Model Approach for the Task of Predicting Dementia Type for a Patient.

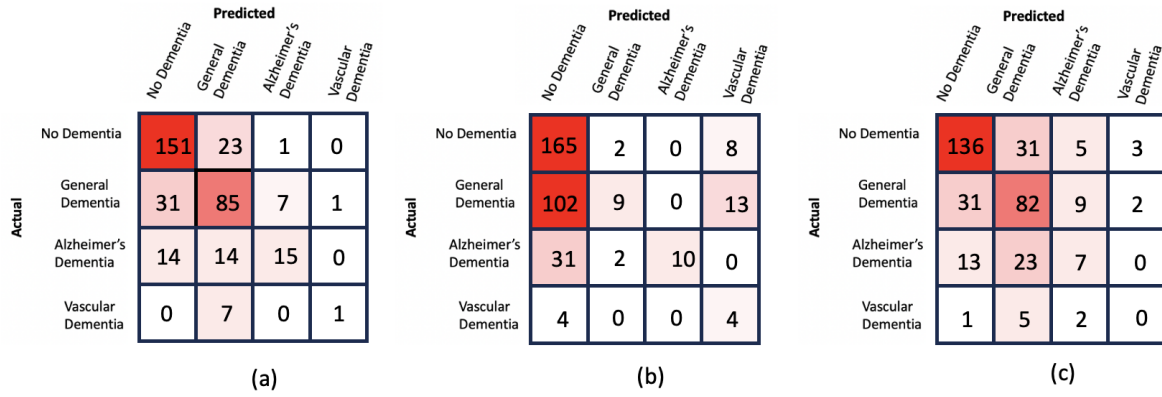


Figure 5.7: Confusion Matrix Across Different Dementia Type Labels in the Held-out Test-set for (a) our Concept Bottleneck Model Approach, (b) GPT-4 Baseline, and (c) ngram Logistic Regression Baseline.

Analyzing Most Important Features for Dementia Prediction

The ngram features from the baseline approach are not human readable full sentences, but rather words/phrases that often don't capture more complex clinical features. We rank the most important features for each dementia type label according to the weights of the trained Logistic Regression models. While some of the top ngram features such as *heparin*, *htn (hypertension)* and *aricept* can be indicative of a patient having dementia, several others are not clinically meaningful features to predict dementia type (Table 5.8). For instance, the presence of ngrams such as *cxr*, *died*, *the patient*, *he*, *iv*, *baseline* in the clinical notes shouldn't be significant contributors in diagnosing Alzheimer's Dementia for a patient. Such models should therefore be used with caution in the clinical setting as the underlying decision making process does not align with that of a healthcare professional.

The top predictive features for our concept bottleneck approach is shown in Table 5.9. For vascular dementia, the most predictive feature was "*memory loss that may be less prominent than in Alzheimer's Disease*". This is supported by Mayo Clinic (Pruthi, 2021) which states that memory loss is not a significant symptom of Vascular Dementia compared to Alzheimer's Dementia. Additionally, *infarcts involving multiple main arterial territories* (McKay and Counts, 2017) and *history of atherosclerotic risk factors* (Nordestgaard et al., 2022) have been shown to be clinical indicators of vascular dementia. (Karantzoulis and Galvin, 2011) mentions that semantic fluency is comparatively better preserved in Vascular Dementia compared to Alzheimer's Dementia, supporting our results for *poor performance on semantically based tasks* being a negative predictive feature for Vascular Dementia and *profound loss in conceptual knowledge (or semantic memory)* being a top positive predictor for Alzheimer's Disease. Looking at the top features for the control group (No Dementia), "*progressive cognitive decline*", "*cognitive impairment*" and "*poor memory*" are negatively correlated as expected.

The top features of our concept bottleneck approach involve high level clinical indicators that were extracted from the Oxford Textbook of Medicine. These might overlap with some of the more important clinical features that trained healthcare professionals focus on during the diagnosis of dementia type for a patient, thereby reducing the likelihood of spurious correlations.

No Dementia	Other Dementia	Alzheimer’s Dementia	Vascular Dementia
weeks	care	cxr	hd
date	facility	died	graft
pain	heparin	patient	stable
discharge	mild	disease	aspiration
infant	htn	aricept	hospital
birth	neck	hemorrhage	ed
good	started	appeared	vancomycin
lung	baseline	iv	hospital
tube	remained stable	baseline	new
remained	expired	risk	neurology

Table 5.8: The Top Positive ngram Features for the Baseline Logistic Regression Model for Different Dementia Class Labels.

Next, in order to validate our concept bottleneck approach in addition to the dementia type outcome, we choose a set of liver disorders prevalent in MIMIC-III to study how well our approach transfers between different clinical outcomes.

MIMIC-III contains patient data from Intensive Care Units (ICUs) and Critical Care Units (CCUs), and therefore patients are typically admitted due to acute, severe or life-threatening conditions. In this context, liver disorders are often a primary reason for a patient to be admitted as compared to dementia. Therefore, we expect a lot more clinical information to be present in the patient notes that pertain to liver disorders. This is one of the primary reasons that we chose liver disorders as a second clinical outcome to study using our concept bottleneck approach.

5.5. Liver Disorder Classification

Our concept bottleneck approach for liver disorder classification is similar to the dementia type classification previously shown (Figure 5.1). We use a set of ICD-9 codes to create a dataset of 4 different liver disorders: *Alcoholic Cirrhosis of the Liver*, *Cirrhosis of the Liver without Mention of Alcohol*, *Hepatic Encephalopathy* and *Portal Hypertension*, along with a *control group* (patients with no liver disorder). The distribution of this dataset is shown in Table 5.10. For each patient admission, we only select the discharge summary notes, and we additionally remove phrases such as “*cirrhosis*”, “*alcoholic cirrhosis*”, “*hepatic encephalopathy*”, “*portal hypertension*” and other variants from the notes to prevent information leakage. For each liver disorder label, we consider a

No Dementia	Other Dementia	Alzheimer's Dementia	Vascular Dementia
progressive cognitive decline	poor performance on semantically based tasks	memory loss that may be less prominent than in Alzheimer's disease	memory loss that may be less prominent than in Alzheimer's disease
cognitive impairment	impaired comprehension of words	decreased reflexes	narrowing of the vessel
poor memory	amnesia	profound loss in conceptual knowledge (or semantic memory)	infarcts involving multiple main arterial territories
poor concentration	myoclonus	rapid onset of memory impairment	poor performance on semantically based tasks
history of atherosclerotic risk factors	bladder dysfunction	mental inflexibility	history of atherosclerotic risk factors
rigidity	poor concentration	progressive cognitive decline	forgetfulness
profound loss in conceptual knowledge (or semantic memory)	profound loss in conceptual knowledge (or semantic memory)	fluent speech	apathy
amnesia	large infarcts	visuospatial difficulties	poor performance on neuropsychological tests
substance dependence	reduced speech output	lack of initiation	large infarcts
subcortical dementia syndrome	fluent speech	severe forgetfulness	restlessness

Table 5.9: Top Features of the Logistic Regression Model Trained on Concept Feature Vector Representations of the Patient Notes for Dementia Type Prediction. Each Cell Contains a Concept of the Form “*The patient exhibits —*”. Red Represents “*No*”, Yellow Represents “*Uncertain*” and Green Represents “*Yes*”.

Liver Disorder Label	Number of Patient Admissions
Alcoholic Cirrhosis of the Liver	1202
Cirrhosis of the liver without mention of alcohol	1287
Hepatic Encephalopathy	666
Portal Hypertension	1007
No Liver Disorder (Control Group)	1500

Table 5.10: Label Distribution of the Liver Disorder Dataset Created from MIMIC-III.

binary classification task (1 vs all). We divide the dataset containing 3,981 patient admissions into train/val/test (0.8/0.1/0.1) sets to train and evaluate our models.

We extract 292 clinical features (Table A.2) related to liver disorders from the Oxford Textbook of Medicine using GPT-4. Some examples of extracted features are: “*The patient exhibits capillary pulsation.*”, “*The patient exhibits signs of severe jaundice.*” and “*The patient exhibits skin rash.*”. For this task, we use the text embedding approach to reduce the number of calls to GPT-4. We randomly select 10 clinical features related to liver disorders and 20 notes per feature. Next, we annotate each note-feature pair with either a “*Yes*”, “*No*” or “*Uncertain*” depending on whether the clinical feature was present in the patient note. Finally, for every note-feature pair, we use GPT-4 to predict the presence of the feature in the note. Using the approach illustrated in Figure 5.3, we plot the cosine distance threshold versus accuracy on the manually annotated set, show in Figure 5.8. At a cosine distance threshold of 0.72, GPT-4 first achieves the maximum accuracy on the manually annotated test set. We use this threshold for labeling the entire set of note-feature pairs in the liver disorder dataset using the text embedding approach, only annotating note-feature pairs using GPT-4 if the cosine distance between the text-embeddings is lesser than 0.72 (44,001 out of 1,154,490 note-feature pairs). For the remaining note-feature pairs with cosine distances greater than 0.72, we assume that the clinical feature is not present in the patient note.

We construct concept vector representations for every note and use this representation to predict the liver disorder type (if any) a patient has using a linear model, and compare the performance to an ngram-Logistic Regression baseline which is trained on unigram, bigram and trigram representations of the patient notes. Additionally, we also compare our approach to a GPT-4 baseline by

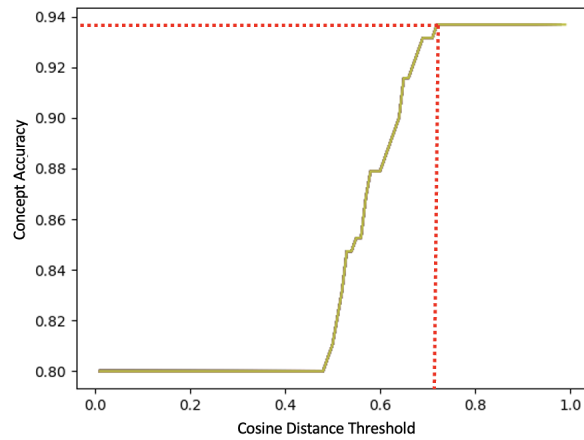


Figure 5.8: Cosine Distance Threshold versus Concept Accuracy of GPT-4 on a Manually Annotated Set of 10 Liver Disorder Clinical Features and 20 Notes per Feature for the Text Embedding Approach. The Dotted Red Line Indicates the Cosine Distance Threshold at which Maximum Concept Accuracy is First Achieved.

using the following prompt for every patient note:

GPT-4 Liver Disorder Baseline Prompt:

For the given patient clinical note, choose one or more of the following diagnosis labels:

no liver disorder

alcoholic cirrhosis of the liver

cirrhosis of the liver without alcohol

hepatic encephalopathy

portal hypertension

You should only reply with the diagnosis label (if more than one, then separated by a ",") and nothing else.

Approach	Alcoholic Cirrhosis of the Liver	Cirrhosis of the Liver without Alcohol	Hepatic Encephalopathy	Portal Hypertension
Baseline GPT-4	0.93	0.85	0.74	0.74
Baseline ngram-LogReg	0.84 $\pm$ 0.01	0.72 $\pm$ 0.01	0.86 $\pm$ 0.01	0.81 $\pm$ 0.02
Our Concept Bottleneck	0.76 $\pm$ 0.01	0.79 $\pm$ 0.02	0.81 $\pm$ 0.02	0.81 $\pm$ 0.01

Table 5.11: Accuracy of the Baseline Models vs Our Concept Bottleneck Approach on a Held-out Test-set for the Final Task of Predicting Liver Disorders in Patients.

The performance metrics of our approach and the two baseline approaches is shown in Tables 5.11 and 5.12. The GPT-4 baseline outperforms our concept bottleneck approach for the labels *Alcoholic Cirrhosis of the Liver* (F1 score of 0.86 vs 0.50 ± 0.04), *Cirrhosis of the Liver without Alcohol* (F1 score of 0.74 vs 0.46 ± 0.02) and *Portal Hypertension* (F1 score of 0.55 vs 0.52 ± 0.02) while our approach performs better for *Hepatic Encephalopathy* (F1 score of 0.43 vs 0.45 ± 0.05). The baseline ngram-logistic regression model outperforms our concept bottleneck approach for the classes *Alcoholic Cirrhosis of the Liver* (F1 score of 0.62 ± 0.03 vs 0.50 ± 0.04) and *Cirrhosis of the Liver without Alcohol* (F1 score of 0.54 ± 0.02 vs 0.46 ± 0.02) while our model achieves higher performance on the classes *Hepatic Encephalopathy* (F1 score of 0.39 ± 0.05 vs 0.45 ± 0.05) and *Portal Hypertension* (F1 score of 0.49 ± 0.04 vs 0.52 ± 0.02). The confusion matrices for the two baseline approaches and our concept bottleneck approach across the 4 liver disorder labels is shown in Figure 5.9. The top features per liver disorder label for the baseline ngram-logistic regression model is shown in Table 5.13 while the top features for our concept bottleneck approach is shown in Table 5.14.

Analyzing Most Important Features for Liver Disorder Prediction

As mentioned previously, the ngram features from the baseline logistic regression approach are not human readable but rather words/phrases that often don't capture more complex clinical features. Using the weights of the trained ngram-logistic regression model, we rank the most important features for each liver disorder label, shown in Table 5.13. While some of the top ngram features such as *etoh*, *alcoholism*, *nash* (*nonalcoholic steatohepatitis*) and *hcv* (*Hepatitis C virus*) can be indicative

Approach	Alcoholic Cirrhosis of the Liver	Cirrhosis of the Liver without Alcohol	Hepatic Encephalopathy	Portal Hypertension
Baseline GPT-4	0.86	0.74	0.43	0.55
Baseline ngram-LogReg	0.62 ± 0.03	0.54 ± 0.02	0.39 ± 0.05	0.49 ± 0.04
Our Concept Bottleneck	0.50 ± 0.04	0.46 ± 0.02	0.45 ± 0.05	0.52 ± 0.02

Table 5.12: F1 scores of the Baseline Models vs Our Concept Bottleneck Approach on a Held-out Test-set for the Final Task of Predicting Liver Disorders in Patients.

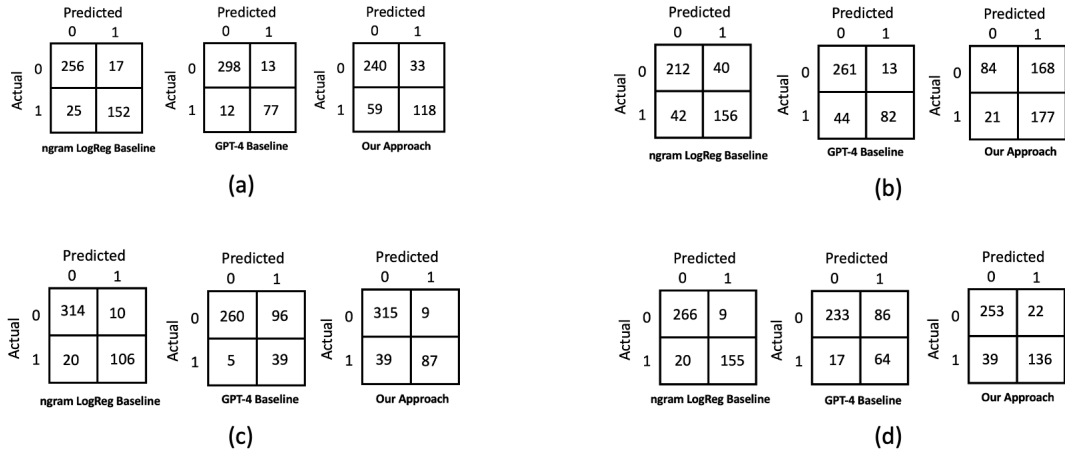


Figure 5.9: Confusion Matrices for (a) Alcoholic Cirrhosis of the Liver, (b) Cirrhosis of the Liver Without Alcohol, (c) Hepatic Encephalopathy, and (d) Portal Hypertension for the 2 Baseline Approaches and our Concept Bottleneck Approach.

Alcoholic Cirrhosis of the Liver	Cirrhosis of the Liver without Alcohol	Hepatic Encephalopathy	Portal Hypertension
alcoholic	cryptogenic	increasing	splenomegaly
etoh	nash	lactulose	vancomycin
alcohol	hcv	confusion	pt reports
admission hospital	autoimmune	alcoholic	medicine
lethargy	hep	transferred hospital	arrival
complains	unknown	nash	presented hospital
alcoholism	hiv	ams	lost
history alcoholic	hcc	liver failure	fall
medication	cirrhotic	fulminant	varices
note patient	services	services	paracentesis

Table 5.13: The Top Positive ngram Features for the Baseline Logistic Regression Model for the Liver Disorder Prediction Task.

of a patient having liver disorders, several others are not clinically meaningful features for predicting liver disorder type. For instance, the presence of ngrams such as *cryptogenic*, *unknown*, *services*, *increasing* and *arrival* in the clinical notes should not be significant contributors in diagnosing liver disorders for a patient. Hence, even though the baseline ngram-logistic regression model achieved high F1 scores, such models should be used with caution in the clinical setting as the underlying decision making process may not align with that of a healthcare professional.

The top predictive features for our concept bottleneck approach is shown in Table 5.14 and the connection between some of them with past research is highlighted below. One of the top features for *Alcoholic Cirrhosis of the Liver* is *weight loss in severe cases of alcoholic hepatitis* which is supported by Mayo Clinic (MayoClinic, 2023) which states that cirrhosis may make it more difficult for your body to process nutrients, leading to weakness and weight loss. Additionally, Li et al. (1999) found that alcoholism and impaired liver function are important predictors of the presence of spider angiomas (also referred to as spider naevus) in patients with liver cirrhosis. People with HIV and/or AIDs are more likely to develop liver cirrhosis than those without Castellares et al. (2008). This is also true for Hepatitis C infection, as NHS (2021) states that up to 1 in every 3 people infected with hepatitis C will develop cirrhosis within 10-20 years. Additionally, some HIV medicines can cause liver damage, also known as hepatotoxicity (Nunez and Soriano, 2005). Renal dysfunction

Alcoholic Cirrhosis of the Liver	Cirrhosis of the Liver without Alcohol	Hepatic Encephalopathy	Portal Hypertension
weight loss in severe cases of alcoholic hepatitis	show classical physical signs of liver disease	fluid buildup in the abdomen	signs of long-term cholestasis
spider naevi	primary biliary cirrhosis	usually tense ascites	signs of lymphoma
symptoms of hepatic granulomatous disease	symptoms of a combination of cholestasis and haemolysis	ascites	signs of sarcoidosis
intermittent jaundice	AIDS	liver failure in severe cases	typically lighter stool color
potential for developing hepatopulmonary syndrome	chance of catastrophic intraperitoneal bleeding due to tumour rupture	easily detected ascites	potential for developing hepatorenal syndrome
symptoms of cirrhosis	abdominal swelling due to liver enlargement	symptoms of Hodgkin's lymphoma	bile-duct injury
possibility of catastrophic intraperitoneal bleeding	potential chronic hepatitis C infection	hepatomegaly	exhibits possible symptoms of Zieve's syndrome
diabetes	fluid buildup in the abdomen	exhibits potential for developing hepatorenal syndrome	paraumbilical hernias
impaired synthesis of hepatocyte-derived blood clotting factors	symptoms typical of Crohn's disease	symptoms of pneumonia	severe hypoglycemia, which may exacerbate encephalopathy
weight loss in severe cases of alcoholic hepatitis	Hypoalbuminaemia	jaundice	peptic ulceration

Table 5.14: Top Features of the Logistic Regression Model Trained on Concept Feature Vector Representations of the Patient Notes for Liver Disorder Prediction. Each Cell Contains a Concept of the Form “The patient exhibits —”. Red Represents “No”, Yellow Represents “Uncertain” and Green Represents “Yes”.

with sodium and water retention is seen in patients with Hepatic Encephalopathy (Jalan and Hayes, 1997) which can manifest as fluid buildup in the affected individuals abdomen while jaundice can also be seen in affected individuals (Bleibel and Al-Osaimi, 2012).

The top features of our concept bottleneck approach involve high level clinical indicators that were extracted from the Oxford Textbook of Medicine. These might overlap with some of the more important clinical features that trained healthcare professionals focus on during the diagnosis of liver disorders for a patient, thereby reducing the likelihood of spurious correlations.

5.6. Discussion

In this chapter, we presented a novel approach for predicting dementia type from patient clinical notes using Concept Bottleneck Models (CBMs). Our key contributions were:

1. Developing a method to extract clinically relevant concept features related to dementia from the Oxford Textbook of Medicine using GPT-4. This resulted in 254 interpretable dementia concept features and 290 interpretable liver disorder concept features like “*The patient exhibits memory loss.*”, “*The patient complains of forgetfulness.*”, “*The patient exhibits signs of severe jaundice.*” and “*The patient exhibits pain in the right upper abdomen.*”
2. Converting patient clinical notes into concept vector representations using Agreement Models, which predict whether each concept feature is present, absent or uncertain based on the note text.
3. Training an interpretable linear classification model on the concept vectors to predict dementia type and liver disorder type for patients. Our concept bottleneck approach for dementia achieved 0.72 accuracy, outperforming an ngram logistic regression baseline (0.64) and GPT-4 baseline (0.48). Our liver disorder concept bottleneck model had higher F1 scores for 2 out of 4 classes compared to ngram logistic regression and 1 out of 4 classes compared to GPT-4 baseline.
4. Analyzing the important concept features and finding that top predictors aligned with clinical

Concept Feature	Sentence in Patient Note
infarcts involving multiple main arterial territories	Head CT showed multiple old infarcts but no new lesion.
infarcts involving multiple main arterial territories.	Past Medical History: Right temporal & left parietal infarcts
history of atherosclerotic risk factors	Past Medical History: Idiopathic peripheral neuropathy (per family), HTN...
history of atherosclerotic risk factors	Past Medical History: - hypertension. Social History: She is a former smoker
severe forgetfulness	Per her family, she has had confusion/dementia for quite sometime and this is stable and chronic.
severe forgetfulness	Notably, wife reported sun downing at home and issues with confusion for several months prior to admission.
decreased reflexes	Her reflexes revealed she was hyporeflexic in the lower extremities and hyperreflexic in the upper extremities.
decreased reflexes	Reflexes: Brisk in the upper extremities, normal at patellae and reduced at achilles. The plantars are mute bilaterally.

Table 5.15: Examples of Sentences in Patient Notes that Correspond to a Concept Feature. These were Obtained using Our Text Embedding Approach.

understanding, like memory loss being less prominent for vascular dementia compared to Alzheimer’s and AIDS increasing the risk of developing non-alcoholic liver cirrhosis.

Our CBM approach forced the model to focus on high-level clinically relevant concepts during prediction, potentially reducing hallucinations and spurious correlations compared to opaque black-box models. Interpreting the important concept features provided insights into the model’s reasoning process.

An additional benefit of our approach is the ability to obtain specific sentences in the clinical note that correspond to a concept feature. This is achieved by our Text Embedding approach mentioned in Section 5.2.3, where we can search for sentences in the patient notes that have high cosine similarities with the concept feature when comparing their text embeddings. Examples of these are shown in Table 5.15. This provides an additional level of insight into the section of the patient note that was responsible for a particular output label while still relying on a high level clinical feature for classification.

Overall, the CBM framework offers an interpretable and accurate paradigm for clinical prediction tasks like dementia classification. Understanding why our approach performed poorly relative to the baseline approaches for liver disorder prediction will require further analysis. Some of the possible reasons are listed in the following limitations section.

5.6.1. Limitations and Future Work

As discussed previously, when tasked with identifying whether a clinical feature is present in a patient note, GPT-4 sometimes fails to comprehend clinical details to provide an accurate answer. Some examples where this failed is shown in Table 5.7. One of the reasons this could happen is because of the way we forced the clinical concept features to be phrased (“The patient exhibits _____.”) or the format of our prompt (**Patient Note Concept Activation Prompt**). Future experiments involving prompt engineering and better standardized ways of representing clinical features might help solve this issue. Another reason could be that it is a limitation of GPT-4 itself, which might be solved by future versions of Large Language Models.

A major limitation of our work is that we only consider clinical features for predicting dementia and liver disorder type. Clinical features are observable, measurable and directly related to a disease like symptoms and signs (pain, fatigue, fever, rash, etc.) and test results (blood glucose levels, MRI findings). On the other hand, non-clinical features involve background/contextual information not directly related to the disease such as environmental exposures, demographics (age, gender), lifestyle, etc. While non-clinical features don’t directly diagnose diseases, they are critical in formulating a well rounded opinion about a patients medical trajectory. As part of future work, creating an additional set of non-clinical concept features (such as “*Is the patient Male?*”, “*Does the patient live alone?*” or “*Is the patient from a low socio-economic background?*”) and integrating it with clinical features extracted from the medical textbook would create a more well-rounded algorithm to predict clinical outcomes. Careful attention will need to be paid for potential discriminatory effects of such factors.

Additionally, we do not consider temporal information like timing, onset and progression of the clinical features in our concept bottleneck approach which may end up being critical to diagnosing

a clinical outcome. For example, the pace at which certain symptoms appear is a key feature in differentiating dementia type, as most of Alzheimer’s dementia typically develop slowly over months or years whereas Vascular dementia may develop abruptly following a stroke or other cerebrovascular event. Alzheimer’s dementia is characterized by a phase of mild memory loss that gradually worsens over years whereas Vascular Dementia often has a stepwise progression, where extreme episodes of memory loss may happen over a few days. Therefore, a natural progression for future work would be to come up with a standardized formulation of defining key temporal features of each clinical concept and integrate it with our concept bottleneck approach.

An inherent feature of our concept bottleneck approach is that it can be applied to predict any medical outcome for a patient. However, due to computational constraints, we focus only on dementia type and liver disorder prediction. This could however hurt the performance and/or generalization of our approach. To provide an analogous example, similar to how Language Models trained on larger datasets are capable of distinguishing certain nuances and formulating a more accurate representation of different words or topics, our concept bottleneck approach when built on a universal set of clinical features (related to every possible disease) might be better equipped to understand and diagnose patients. To further explain this point, when predicting dementia for a patient, we only look at a set of 254 clinical features, most of which are not expressed in the control group. Therefore, the decision boundary constructed by our concept bottleneck model is a poor approximation of the true decision boundary constructed when considering the universal set of clinical features and better representing the control group. This is a problem that might be solvable in the future given the current trends of decreasing costs and higher API calls associated with state-of-the-art language models.

CHAPTER 6

Conclusion

This thesis sets out to address challenges encountered while using Language Models to analyze patient Electronic Health Records (EHRs).

In Chapter 3, we introduced CLASSE GATOR, a distantly supervised approach to predict acronym senses in clinical notes. To the best of our knowledge, our approach was the first automated acronym sense disambiguation method for clinical notes that did not require an expensive manually annotated acronym-definition corpus for training. Given the scarcity of public datasets for clinical acronym expansion, we first created a labeled dataset of acronyms, their possible expansions and surrounding contexts from medical research papers published on PubMed Central using regular expressions and a set of hand written rules. Next, we analyzed 31,764 papers, extracting 1,257 unique acronyms and 8,287 potential expansions, with the average number of expansions per acronym being 6.6 (min=2, max=50). Next, we trained Logistic Regression and BioBERT on this dataset and evaluated them on a manually annotated set of clinical notes from MIMIC-III, achieving an overall accuracy of 63%. A major advantage of our approach was that it enabled inexpensive creation of labeled training data, as past approaches had relied on experts manually annotating examples, with some studies spending upwards of \$100,000. A limitation of CLASSE GATOR is the lack of misspelling adjustment, i.e., we did not correct clinical notes for misspellings prior to running our algorithm. This affects certain acronyms more than others (e.g., shorter acronyms that are used commonly). For example, in our manually annotated test-set, ‘HS’ was misspelled 23.33% of the time (7 out of 30) as a substitute for common short-length words (‘his’, ‘has’, ‘he’). Other acronyms (ones that we cared about more from a clinical perspective), such as ‘IVF’ which could mean ‘in vitro fertilization’ or ‘intravenous fluids’ was less likely to be misspelled.

In Chapter 4, we developed a novel algorithm to predict opioid prescription likelihood and Opioid Use disorder (OUD) diagnosis for a patient based on information present in their EHR data. While we were not the first to develop an OUD diagnosis algorithm, we were the first at the time to

predict prescribing behavior (which can also be important in stigmatization research). From an NLP perspective, there were 2 main challenges that we attempted to address: (i) Developing a deep learning architecture that can simultaneously process different data types present in patient EHRs, and (ii) Designing a model that can handle long clinical notes of variable sizes - the average number of words per patient in our dataset was 1,900 (min = 0, max = 284,897). We developed a deep learning model with different modules, each handling a particular datatype (a simple feed forward layer for structured static data, a transformer layer with positional embeddings for structured time varying data, and a Hierarchical Attention Model built on top of BioBERT for unstructured text). To the best of our knowledge, our approach was the first to predict opioid prescription likelihood for a patient based on their EHR data, thereby filling an important research gap and showing that such models with high performance can be built (F1 score of 0.88). We also compared our OUD prediction model to previous work, outperforming them on metrics such as sensitivity (0.77), specificity (0.89) and F1 scores(0.82). However, it is important to note that OUD studies carried out often have variations in study periods, sample sizes, definitions, types of databases, and structured documentation, and therefore comparing differences in model performances built across these studies may not be ideal. We also used local surrogate models to identify the most predictive features of our hierarchical attention model for both tasks, showing a large overlap with those studied in the literature.

In Chapter 5, we focused on the challenge of explainability while building predictive models for patient dementia diagnoses using their EHR information. Inspired by Concept Bottleneck Models (CBMs), we created a prediction pipeline that ensures the model looks at clinical features most relevant to the outcome of interest, thereby minimizing the likelihood of spurious correlations and hallucinations. We provided a novel method for extracting 254 clinically relevant features for CBMs using the Oxford Textbook of Medicine and GPT-4 (an example of an extracted feature is *“The patient exhibits poor memory.”*). Next we trained a linear classifier on this representation to predict patient dementia type and finally analyzed the model weights to provide insights into the most important clinical features for dementia type prediction. Our CBM approach obtained an accuracy of 0.72 on a held out test-set, beating an ngram Logistic Regression baseline (0.64) and a GPT-

4 baseline (0.48) while being more interpretable and utilizing clinically relevant features. We also extended our concept bottleneck approach to predict liver disorder types, with our approach beating a GPT-4 baseline in 1 out of 4 labels and beating an ngram-Logistic Regression baseline in 2 out of 4 labels.

In conclusion, this thesis has made significant strides in addressing three significant challenges of applying Language Models to patient EHR data. Through the introduction of innovative frameworks and the demonstration of practical applications, we have added to the expanding body of knowledge in this area. However, our findings also emphasize the complexity of these challenges and the importance of further research to fully grasp and utilize the potential of Language Models in the clinical field. In addition, while clinical annotation can be expensive, we also found that utilization of LLMs can also be expensive with the cost of using the synchronous GPT-4 API as of June, 2024 for our dataset containing 7.9 billion input tokens and 59 million output tokens being almost a quarter of a million dollars. With the Text Embedding approach, we can obtain a comparable performance to using GPT-4 on the entire dementia dataset, while only requiring a fraction of the GPT-4 calls. Using this approach, we reduce the cost of our overall approach from \$240,540 to \$7,276. However, these cost constraints are likely only due to the novelty of the LLMs. The cost of LLMs will likely go down over time, in accordance with Moore’s Law while the cost of clinical annotators will likely only increase (due to inflation, and other economic factors) with the passage of time. Therefore, utilization of the algorithms that we developed as a part of this dissertation will likely reduce costs over the next decade.

6.1. Future Work

6.1.1. Acronym Disambiguation

Automatic acronym disambiguation methods have progressed significantly with the development of LLMs since my work that I describe in Chapter 3 (Kashyap et al., 2020). Kugic et al. (2024) evaluated GPT-4 for zero-shot acronym disambiguation in clinical narratives and found that GPT-4 achieved an accuracy of 0.98 on the Clinical Abbreviation Sense Inventory dataset (Moon et al., 2012a) containing english text. With such a high performance benchmark for zero-shot learning, I

expect the task of acronym disambiguation in English clinical notes to be completely solved with the release of larger LLMs in the future, if not already considered solved. Obtaining labeled training data in an inexpensive way to build supervised models was one of the challenges that researchers focused on in the past, but might no longer be necessary as zero-shot and few-shot approaches with LLMs have shown competitive performance.

6.1.2. Building Clinical Diagnostic Models using LLMs

Zhang et al. (2024) found that GPT-4 with chain-of-thought reasoning and few shot prompting achieved F1 scores ranging between 74% and 96% for the task of identifying patients with specific medical diagnoses using Mimic-III. In order to better understand GPT-4’s decision making process, Zhang et al. (2024) provided an input prompt that instructed the model to provide a rationale for the assigned prediction. For correct predictions (true positives and true negatives), they found that the rationales provided by GPT-4 for assigning a particular disease class were usually scientifically correct, exhaustive in context with minimal bias, demonstrating appropriate task comprehension, correct information retrieval, and good adherence to the provided clinical guideline. However, for incorrect predictions (false positives and false negatives), rationales sometimes demonstrated poor accurate comprehension of clinical guidelines, and also showed evidence of fabrication (hallucination).

Keeping these limitations in mind, a few directions for future work are listed below.

Expanding on the Concept Bottleneck Approach for Patient Diagnosis using their EHR data

Using just GPT-4 as an end-to-end model to diagnose patients for a certain clinical outcome can be risky. GPT-4 has been trained on a large part of the internet that could include reliable sources such as Mayo Clinic, Center for Disease Control and Prevention (CDC), and the National Institute of Health (NIH) as well as unreliable sources from non-medical individuals such as social media, personal blogs, news articles with sensationalized health claims and so on. As GPT-4 is a black box model, it is almost impossible to determine the extent to which its reasoning process only considers reliable sources for predicting a patient diagnosis. As a result, rather than use it as an end-to-end

model, breaking up the patient diagnosis process into intermediate steps that can be qualitatively controlled and assessed is a much better option. Hence, my future research will involve expanding on the Concept Bottleneck approach described in Chapter 5.

A limitation I stated in Chapter 5 is the fact that I only look at clinical features (obtained from the *Oxford Textbook of Medicine*) for predicting Dementia and Liver Disorder type. Non-clinical features such as environmental exposures, patient demographics and lifestyle information among others also play a critical role in influencing a patients diagnosis. As part of my future work, I aim to create an exhaustive list of non-clinical features (such as “*Is the patient Male?*”, “*Does the patient live alone?*” or “*Is the patient from a low socio-economic background?*”) . This list could be potentially generated by analyzing EHR data from several different sources and combining the benefits of GPT-4 and manual annotation to produce an exhaustive list of non-clinical features.

Including temporal information like timing, onset and progression of each clinical feature in our concept bottleneck approach is another area for future work. A significant challenge of this step is to create a framework that contains medical events, symptoms and tests, and all possible event attributes (for example, a clinical feature like “*The patient fell and broke his leg.*” does not require a *duration* argument whereas a clinical feature like “*The patient exhibits skin rash.*” would require a duration argument, i.e., was it only present for 1 day/week/month, etc.).

As mentioned in Chapter 5, having a concept bottleneck model that is built on a universal set of diseases will help the model generalize better, potentially leading to better performance for patient diagnosis. However, due to computational constraints, we were only able to construct a model for two broad outcomes (dementia and liver disorder type prediction). This is a problem that might be solvable in the future given the current trends of decreasing costs and higher API calls associated with state-of-the-art language models. Alternatively, improving our Text Embedding approach (Section 5.2.3) by further minimizing the number of note-feature pairs that GPT-4 needs to analyze may make expanding our approach to a larger set of diseases and clinical features more practically feasible and a potential area for future research.

APPENDIX A

EXPLAINABLE DEMENTIA AND LIVER DISORDER TYPES WITH CONCEPT BOTTLENECK MODELS

A.1. CONCEPT EXTRACTION PROMPT

Two examples of inputs and outputs to GPT-4 is provided below. In total, we had clinical feature text from the *Oxford Textbook of Medicine* totaling 3039 tokens related to dementia and 4206 tokens related to liver disorders.

EXAMPLE 1: TEXTBOOK EXCERPT:

Vascular Dementia: Cerebral damage may be hemorrhagic, hypoxic, or anoxic. The vascular disease can occur in larger arteries, medium or small arterioles, or capillaries. VaD may be the result of a sentinel infarct, multiple infarcts, Binswager’s disease, or vasculitis. Hypoxic damage related to systemic episodes of hypotension can also result in VaD. Given the variety of causes and manifestations, it is not surprising that there are no uniformly accepted criteria for the diagnosis of VaD. It may be overdiagnosed, given the high prevalence of subclinical cerebrovascular disease seen in elderly persons, or underdiagnosed, given the lack of clear criteria for diagnosis. VaD is the most common second diagnosis in mixed dementia, a finding that has clinical significance: AD + VaD or LBD + VaD = more cognitive and functional deficits than AD or LBD alone.

EXAMPLE 1: GPT-4 OUTPUT:

- The patient exhibits cerebral damage that may be hemorrhagic, hypoxic, or anoxic.
- The patient exhibits vascular disease involving larger arteries, medium or small arterioles, or capillaries.
- The patient exhibits vascular dementia (VaD) potentially resulting from a sentinel infarct, multiple infarcts, Binswager’s disease, or vasculitis.
- The patient exhibits hypoxic damage related to systemic episodes of hypotension that may

contribute to VaD.

- The patient exhibits difficulty in diagnosing VaD due to the variety of causes and manifestations, with no uniformly accepted criteria.
- The patient exhibits the potential for overdiagnosis of VaD, given the high prevalence of subclinical cerebrovascular disease in elderly individuals.
- The patient exhibits the possibility of underdiagnosis of VaD, due to the lack of clear diagnostic criteria.
- The patient exhibits mixed dementia, where VaD may coexist with Alzheimer's disease (AD) or Lewy body dementia (LBD), leading to more cognitive and functional deficits than either AD or LBD alone.

EXAMPLE 2: TEXTBOOK EXCERPT:

Alzheimer: Other risk factors for AD include head injury with loss of consciousness and lower educational level. Various risk factors for atherosclerosis (diabetes, hypertension) have also been shown to increase the risk of AD, as well as VaD.

Histopathologic markers of AD include extracellular senile plaques (aggregations of abnormal amyloid protein), intraneuronal neurofibrillary tangles (hyperphosphorylated tau protein), and brain cell death. Neurofibrillary tangles are preferentially distributed in the medial temporal lobe, hippocampus, and amygdala, whereas senile plaques are widely scattered throughout the cerebral cortex (Clark and Karlawish, 2003). The AD patient's brain displays shrinking gyri, widening of the sulci (particularly in the frontotemporal region), and enlargement of the ventricular system in proportion to the atrophy of the cortex. Caution should be taken in using these features exclusively for the diagnosis because normal aging occurs in much the same manner, although with more diffuse atrophy and slower progression if serial studies are performed. Decreases in hippocampal volume correlate well with deteriorating clinical status

The most prominent neurochemical defect appears to be a deficiency in the enzyme choline O-acetyltransferase, resulting in a deficit of acetylcholine. This is most evident in the nucleus basalis of Meynert and the hippocampus. In the absence of cholinergic stimulation from these lower centers of the brain, there is secondary atrophy in the cerebral cortex, leading to memory problems and other deficits in cognitive function. Neurochemical alterations probably contribute to the associated psychiatric symptoms of AD (depression, anxiety, agitation, psychosis). Early in the course of AD, impaired memory is the predominant finding. Changes in mood and personality can also be seen. Later, memory impairment becomes more dramatic, and word-finding difficulty (anomia) appears. Eventually, functional performance is affected, with deterioration in skills of driving, handling finances, meal planning, and hygiene. Memory loss becomes profound, with loss of recognition of previously familiar persons or things.

EXAMPLE 2: GPT-4 OUTPUT:

- The patient exhibits a history of head injury with loss of consciousness, which is a risk factor for Alzheimer's disease (AD).

- The patient exhibits a lower educational level, which is associated with an increased risk of AD.
- The patient exhibits risk factors for atherosclerosis, such as diabetes and hypertension, which increase the risk of AD and vascular dementia (VaD).
- The patient exhibits histopathologic markers of Alzheimer's disease, including extracellular senile plaques, which are aggregations of abnormal amyloid protein.
- The patient exhibits neurofibrillary tangles, composed of hyperphosphorylated tau protein, found predominantly in the medial temporal lobe, hippocampus, and amygdala.
- The patient exhibits brain cell death, as seen in the histopathology of AD.
- The patient's brain exhibits shrinking gyri and widening of the sulci, particularly in the frontotemporal region.
- The patient exhibits enlargement of the ventricular system, corresponding to the atrophy of the cerebral cortex.
- The patient exhibits hippocampal volume loss, which correlates with deteriorating clinical status.
- The patient exhibits a deficiency in the enzyme choline O-acetyltransferase, leading to a deficit of acetylcholine.
- The patient exhibits reduced cholinergic stimulation from lower brain centers, especially the nucleus basalis of Meynert and the hippocampus.
- The patient exhibits secondary cortical atrophy, which contributes to memory problems and other cognitive deficits.
- The patient exhibits neurochemical alterations that likely contribute to psychiatric symptoms such as depression, anxiety, agitation, and psychosis.

- The patient exhibits impaired memory as the predominant early symptom of AD.
- The patient exhibits changes in mood and personality early in the course of AD.
- The patient exhibits more dramatic memory impairment as AD progresses.
- The patient exhibits word-finding difficulty (anomia), which becomes apparent as memory impairment worsens.
- The patient exhibits functional decline, with deteriorating abilities in tasks such as driving, handling finances, meal planning, and personal hygiene.
- The patient exhibits profound memory loss, including the inability to recognize previously familiar persons or objects.

A.2. CLINICAL FEATURES FOR DEMENTIA AND LIVER DISORDERS

The 254 clinical features related to dementia and 292 clinical features related to liver disorders that were extracted using GPT-4 from the *Oxford Textbook of Medicine* and manually annotated for quality are shown in Table A.1 and Table A.2.

A.3. DEMENTIA TYPE CLASSIFICATION USING GRADIENT TREE BOOSTING

For our Concept Bottleneck approach, in addition to having Logistic Regression as the final classification model, we also evaluate a Gradient Tree Boosting model. Logistic Regression does not explicitly model interactions between features unless they are manually created. For example, if there is a reason to believe that the interaction between two features x_1 and x_2 is important, then a feature $x_1 * x_2$ needs to be manually created and added to the model. On the other hand, one of the key strengths of Gradient Tree Boosting is its ability to automatically capture feature interactions. Each decision tree in the ensemble splits the data based on feature values and these splits can represent complex, non-linear interactions between features.

We train a Gradient Tree Boosting model with 100 estimators, each with a maximum depth of 3, and a learning rate of 0.1. Our Concept Bottleneck approach with a final Gradient Tree Boosting

classifier achieves an accuracy of 0.72 and an F1 score of 0.71 for dementia type classification. The net performance improvement compared to using a final Logistic Regression classifier (Figure 5.6) is minimal (F1 score of 0.71 vs 0.70) which could be attributed to the model being able to capture complex, non-linear interactions between features. The top features ranked based on Gini importance are shown in Table A.3.

peripheral mass lesions	relatively unaffected visuospatial function	delayed verbal recall
reduced speech output	non-fluent speech	biological features of depression
pleocytosis	diminished vibration sense	recurrent cerebral infarcts
dyspraxia	change in food preference	impaired naming with semantically based errors
fluctuating cognitive performance	difficulty with complex visuospatial tasks	abnormalities in the parietotemporal regions on SPECT scans
psychomotor slowing	rapid shifts between hypoactivity and hyperactivity	disturbance of the sleep-wake cycle
apathy	multiple phonological errors	speech disorder
behavioural problems	decreased reflexes	poor performance on neuropsychological tests
worsening of semantic memory	anosmia	ventricular dilatation
mood swings	apraxia	confluent subcortical and corpus callosal demyelination
poor concentration	patchy performance on memory tasks	relatively spared memory in the early stages
a loss of appetite	impaired comprehension of faces	pout reflex
delusions	reduced category fluency	stereotyped food choices
amnesia	gross impairments in the grammatical aspects of language production	visuospatial function
mental slowing	lack of subjective feelings	perceptual distortions
impaired judgement	optic atrophy	visual hallucinations
poor understanding of complex words	sensitivity to neuroleptic medication	relatively preserved day-to-day memory (episodic memory)
indifference to professional responsibilities	lobar haemorrhages	difficulty answering questions
of a loss of memory for words	illusions	narrowing of the vessel
ventricular enlargement disproportionate to the degree of cortical atrophy	mild extrapyramidal features	marital problems
mutations on chromosome 19	Babinski reflex	small-vessel disease
disinhibition	restlessness	impairment of executive abilities
financial problems	agitation	ambulatory difficulties
involuntary movements	hallucinations	perseverations
a tendency to echolalia	snout reflex	sleep disturbance
socially inappropriate behaviour	pathological changes in the cutaneous blood vessels	loss of the cerebral white matter
periventricular high-signal changes	impaired repetition	changes in personality
sensory ataxia	white-matter changes	acute onset of attentional abnormalities
poor memory	giving bizarrely wrong answers to questions	difficulty with visuospatial tasks
a lack of interest in hobbies and activities	past psychiatric history	inconsistency in mmse test performance
memory loss that may be less prominent than in Alzheimer's disease	stroke-like episodes	cognitive slowing
mild tremor	severe forgetfulness	difficulty in managing complex day-to-day activities
pout reflexes	extensor plantar responses	abnormalities in the parietotemporal regions
ganser syndrome	history of atherosclerotic risk factors	forgetfulness
increased tone in their voice	preference towards sweet foods	oligoclonal bands
insidiously progressive changes in personality and behaviour	repetitive questioning	mild tremor with postural and action components
poorly articulated speech	gait disturbance	migraine-like headaches
cognitive impairment	gross impairments in the phonological (sound-based) aspects of language production	large infarcts
poor understanding of complex concepts	impaired repetition of multisyllabic phrases	Binswanger's disease
indifference to domestic responsibilities	progressive cognitive decline	mutations in the notch3 gene on chromosome 19
confluent subcortical demyelination	a history of theft	memory loss
visuospatial impairments	subcortical dementia syndrome	increased reflexes
muscle fasciculations	infarcts involving multiple main arterial territories	a loss of libido
impaired repetition of multisyllabic words	wasting	snout reflexes
gait apraxia	multiple lacunar lesions	bladder dysfunction
multiple lesions	diffuse white-matter disease	increased protein and oligoclonal bands
well-preserved word comprehension	postural components of tremors	difficulty understanding less frequent words
incontinence	visuospatial disturbance	intracerebral haemorrhages
grasp reflex	fluent speech	word-finding difficulties
impaired comprehension of words	substance dependence	trouble with the law
rigidity	agrammatical speech	symmetrical diffuse low-density periventricular hypodensity
impairments in job performance	obsessive-compulsive behaviour	good visuospatial skills
joint contractures	sparse language output	relatively spared memory
poor performance on semantically based tasks	unawareness of the meaning of words and phrases	macrocytic anaemia
deterioration of social skills	raised intracranial pressure	subcortical infarcts
spontaneous fluctuations in cognitive abilities	mutism	greatly increased rigidity
mental inflexibility	diminished vibration sense in the lower limbs	stereotyped daily routines
action components of tremors	well-preserved object recognition	pathological changes in the cutaneous blood vessels in a skin biopsy
leucoencephalopathy (CADASIL)	anomia	neuropsychiatric symptoms
potential concomitant organic disorder	word substitutions	severe language impairment
loss of personal identity	language difficulties	mixed reflexes
periventricular hypodensity	low energy	ataxia
rapid onset of memory impairment	asymmetries on the neurological examination	shrinking vocabulary
loss of salient personal events	small lesions	focal neurological signs
impairment of visuo-perceptual abilities	personality changes	poor recent memory
corpus callosal demyelination	impairment of episodic memory	impaired naming of low-frequency words
tendencies to hoard food	lack of feelings of depression	primitive reflexes such as snout, pout, and grasp
narrowing of the vessel to produce ischaemia	cortical atrophy	mutations in the notch3 gene
fleeting misidentification phenomena	profound loss in conceptual knowledge (or semantic memory)	semantically based errors
delirium	patchy performance on frontal tasks	lacunar infarcts
empty speech	jocularly	changes in behaviour
peripheral mass lesions of varying signal density on CT	extrapyramidal signs	hyperactivity
abnormal sexual behaviour	impairments in social function	disturbance of consciousness
seizures	unilateral visual failure	hypoactivity
lack of initiation	non-verbal problem-solving ability	fatuousness
marked fluctuations in cognitive abilities	impaired naming	myoclonus
pupillary abnormalities	cerebral amyloid angiopathy	worsening of language function
grasp reflexes	ventricular enlargement	delayed verbal recall of new material
impaired comprehension of objects	distractibility	frontal executive malfunction
impaired attention	periventricular high-signal changes on T2-weighted MRI	gradual loss of expressive abilities
rapid onset of intellectual impairment	cerebral autosomal dominant arteriopathy	ischaemia
reduced ability to maintain attention to external stimuli	marked fluctuations in behaviour	emotional lability
medial temporal lobe atrophy	visuospatial difficulties	psychomotor disturbance
poor performance on neuropsychological tests of frontal function	ophthaloplegia	wasting particularly affecting the bulbar musculature
normal repetition of words and phrases	inability to appropriately shift attention to new stimuli	

Table A.1: 254 Clinical Features of Dementia extracted by GPT-4 from the Oxford Textbook of Medicine after manually annotating for quality. Each cell represents a concept of the form *The patient exhibits* —.

capillary pulsation	a fever	increased susceptibility to bacteria
splenomegaly	metabolic acidosis	psoriasis
low levels of antinuclear antibody	moderate ascites	xanthelasma in palmar creases
symptoms of Sjögren's syndrome	symptoms of hepatocellular jaundice	lethargy
gynaecomastia	signs of severe jaundice	a possible escalation in serum bilirubin levels
right upper abdominal pain	cholestasis as indicated by serum biochemical tests	cholestasis leading to jaundice
osteoporosis	hypercalcaemia	increased susceptibility to infection
ascites	hepatic granuloma formation	symptoms of a combination of cholestasis and haemolysis
lesions on their wrists	severe hypoglycemia, which may exacerbate encephalopathy	portal hypertension in severe cases
hypokalaemic metabolic alkalosis	considerable abdominal swelling	ulcerative colitis
not exhibit any abnormalities in serum biochemistry	pleural effusions	easy bruising
an increased incidence of gallstones	right upper quadrant pain	abdominal swelling due to liver enlargement
symptoms of Q fever	symptoms of drug-induced granulomatous hepatitis	skin rash
risks of rupture due to ascites	no physical abnormalities	hepatic hydrothorax
symptoms of renal tubular acidosis	a distinct breath odor	symptoms of porphyria cutanea tarda
signs of sarcoidosis	rare attacks of acute cholangitis	respiratory acidosis
left-sided pleural effusions	paraumbilical hernias	calcinosis
tumours that are less than 3-cm in diameter	weight loss	of abnormalities could be due to primary sclerosing cholangitis
signs of Schistosoma japonicum infestation	increased susceptibility to aspergillosis	hypernatremia
persistent pain	an uncommon development of jaundice	fluctuations in levels of alkaline phosphatase
symptoms typical of Crohn's disease	the possibility of catastrophic intraperitoneal bleeding	evidence of exposure to silicone
evidence of exposure to starch	abnormal liver enzymes	haemorrhagic diathesis
an increase in serum IgM concentrations	hepatic encephalopathy	signs of urinary tract infection
moderate symmetrical distension of the abdomen	peptic ulceration	abstinence from alcohol
a general deterioration in health	Hypoalbuminaemia	thrombocytopenia
Raynaud's phenomenon	established cirrhosis	the potential to develop spider naevi
profound cholestasis	damage of bile ducts	symptoms of hepatic granulomatous disease
impaired secretion of conjugated bilirubin	hard time particularly absorbing vitamin D	low levels of smooth-muscle antibody
psychiatric and behavioural changes	pulmonary vascular dilation	anorexia
xanthomas in palmar creases	pain in the right upper abdomen	odor is often described as pungent
bile-duct injury	signs of anorexia	elevated alkaline phosphatase activity
sclerodactyly	weight loss in severe cases of alcoholic hepatitis	ascites that are only detectable by ultrasound examination
gastrointestinal bleeding from esophageal varices	advanced cholestasis	a history of smoking
disabling fatigue	is usually tense	increased susceptibility to candidiasis
hepatosplenomegaly	signs of cardiovascular changes	dilated blood vessels
possible signs of cholestasis	an increased alveolar-arterial oxygen gradient	tuberous lesions that affect nerves
fluid retention with ascites	fatigue	a low-grade fever
cholestasis leading to pruritus	leucocytosis	symptoms of granulomatous disease
signs of septicaemia	symptoms of thyroiditis	possible symptoms of Zieve's syndrome
is easily detected	clubbed fingers	evidence of pulmonary tuberculosis
prominent veins on the abdominal wall	odor is often described as sweet	gastrointestinal bleeding from an associated peptic ulcer
impaired synthesis of hepatocyte-derived blood clotting factors	abdominal swelling due to ascites	persistent upper abdominal pain
bilateral pleural effusions	symptoms of seropositive arthropathy	a bounding pulse
have AIDS	fever	malabsorption of fat-soluble vitamins
pruritus	ascites that are detected by shifting dullness	a typically lighter stool color
lesions on their elbows	hyponatremia	severe fatigue
features of primary biliary cirrhosis (PBC)	intermittent jaundice	an illness resembling viral hepatitis
signs of long-term cholestasis	evidence of tuberculosis	spontaneous bacterial peritonitis
large or gross ascites	a bruit	lesions on their pressure points
a fever	endocrine changes	signs of Schistosoma mansoni infestation
high concentrations of IgG	grade 3 ascites	primary biliary cirrhosis
changes in stool color	frequent upper gastrointestinal haemorrhage	scratch marks
mild ascites	lesions on their knees	the potential to develop palmar erythema
an enlarged and firm liver	elevated levels in serum alkaline phosphatase	signs of carcinoma
liver enlargement	obesity	hemorrhagic phenomena
lowered blood pressure	symptoms of cirrhosis	cholestasis in severe cases
lesions on their buttocks	symptoms akin to fatty liver	hypoglycaemia
fluid buildup in the abdomen	a wide range of acid-base changes	non-metastatic systemic manifestations
abdominal pain	loss of skeletal muscle mass	finger clubbing
itching	leukonychia	decreased bone density
grade 2 ascites	increased susceptibility to fungi	the possibility of tumour rupture
persistently raised serum alkaline phosphatase	small, asymptomatic tumours	oliguria
liver disease	a potential chronic hepatitis C infection	eosinophilia
symptoms of coexisting autoimmune disease	pulmonary fibrosis	prominent veins
the complications of cirrhosis	lesions on their ankles	an age of diagnosis primarily between the ages of 25 and 40
the presence of a fluid thrill on palpation	a swinging fever	systemic vasodilatation
decreased testosterone levels	bone pain	impaired hepatic venous drainage
decreased libido	the development of hepatic granulomas	a bruit over the liver
risks of strangulation due to ascites	signs of lymphoma	divarification of the rectus abdomini muscles
abdominal swelling	show classical physical signs of liver disease	abnormal protein metabolism
signs of having a variety of neoplasms	mild elevations in liver transaminases	fatigue
severe hypoglycemia	spider naevi	pruritus
cholestasis	fluctuations in levels of bilirubin	a condition named xanthomatous peripheral neuropathy
feminization	signs of having Mycobacterium avium intracellulare (MAI)	edema
symptoms of non-Hodgkin's lymphoma	potential for developing hepatopulmonary syndrome	Terry's nails
celiac disease	a risk of encephalopathy	right-sided pleural effusions
myoclonic twitching	long-standing recurrent tense ascites	a risk of bleeding
an enlarged spleen	a risk of ascites	malaise
bone pain due to osteomalacia	sinusoidal portal hypertension	nausea
tuberous lesions that affect tendon sheaths	osteomalacia	jaundice
mildly deranged liver function tests	signs of histoplasmosis	the chance of catastrophic intraperitoneal bleeding due to tumour rupture
marked abdominal distension	elevated muscle tone	reduced glomerular filtration rate
symptoms of autoimmune diseases	near-normal liver function tests	electrolyte changes
precordial heave	xanthelasma around their eyes	osteopenia
palmar erythema	a hyperdynamic circulation	evidence of exposure to suture material
notice itching first during pregnancy	steatorrhea primarily	an orange tint to their skin
rapid-onset renal failure	hypogonadism	hypoalbuminaemia
symptoms of pneumonia	xanthomas around their eyes	CREST syndrome
diabetes	bleeding from esophageal varices	hyperlipidaemia
telangiectasia	bleeding from varices	potential for developing hepatorenal syndrome
signs of lepromatous leprosy	portal hypertension	fetor hepaticus
evidence of exposure to beryllium	symptoms of Hodgkin's lymphoma	evidence of exposure to talc
a usual absence of serum mitochondrial antibodies	notice itching when on contraceptive pills	normal serum transaminase activities
liver failure in severe cases	symptoms of scleroderma	complications from cirrhosis
an enlarged liver	signs particularly of non-caseating hepatic granulomas	hepatomegaly
tuberous lesions on their extensor surfaces	experience exacerbation of encephalopathy	symptoms of seronegative arthropathy
various skin changes	a possibility of developing normocytic normocytic anaemia	symptoms of vitamin K deficiency
odor is often compared to rotting apples or corpses		

Table A.2: 292 Clinical Features of Liver Disorders extracted by GPT-4 from the Oxford Textbook of Medicine after manually annotating for quality. Each cell represents a concept of the form *The patient exhibits* —

Concept Feature	Gini Importance
poor memory	0.176
cognitive impairment	0.117
good visuospatial skills	0.078
memory loss that may be less prominent than in Alzheimer’s disease	0.042
impairment of episodic memory	0.030
amnesia	0.021
fluctuating cognitive performance	0.009
amnesia	0.009
pout reflexes	0.009
memory loss that may be less prominent than in Alzheimer’s disease	0.008
cognitive slowing	0.008
rapid onset of intellectual impairment	0.008
difficulty answering questions	0.008
pout reflexes	0.008
impairment of executive abilities	0.008
progressive cognitive decline	0.008
forgetfulness	0.007
poor performance on neuropsychological tests	0.007
history of atherosclerotic risk factors	0.007
mental slowing	0.007
subcortical dementia syndrome	0.006
cognitive impairment	0.006
memory loss	0.006
frontal executive malfunction	0.005
delayed verbal recall	0.005
loss of salient personal events	0.005
delayed verbal recall of new material	0.005
patchy performance on memory tasks	0.005
poor concentration	0.005
delayed verbal recall	0.005

Table A.3: Top 30 features and Corresponding Gini Importance for our Concept Bottleneck Approach with Gradient Tree Boosting as the Final Classification Model. Each Cell Contains a Concept of the Form “The patient exhibits —”. Red Represents “No”, Yellow Represents “Uncertain” and Green Represents “Yes”.

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